Turbo Spectrum aka CoBra-2

Specification

Sorin C. Cismas

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1 Introduction

Once upon a time, in the glorious eighties, in the medieval town of Brasov, Transylvania, a small group of young engineers and mathematicians in their late 20s and some in the early 30s, designed and built CoBra (**Co**mputer **Bra**sov), a microcomputer compatible with Sinclair ZX Spectrum, running CP/M as well. A few years later, in the late 80's/early 90's, students in Regie, Bucharest, were building CoBra using PCB boards and components from the black market. One of those students, Jesse (Cristian Savu), rebuilt CoBra in 2016 for its 30-year anniversary. He made available all CoBra technical information that he could gather at https://www.cobrasov.com.

A few years later, Bizonul (Nicolae Muntean), another CoBra fan from the late 80's/early 90's, contacted me saying he wants to rebuild Turbo Spectrum, aka CoBra2. This was a side project of mine in the late 80's. Having joined ITC Brasov after the CoBra project started, I was not part of the CoBra key team. My only contribution to Cobra was the logo addition to the boot code, right before the product release; the technical details of it are given in chapter 4 CoBra Logo.

Turbo Spectrum was built in hindsight with learnings from the main CoBra development. In the meantime, the 64Kx1 Russian DRAM chips with a single voltage became available, as well as 8-bit registers and buffers, like 74165, 74244, 74373, 74377. Using these, and a few design tricks, the total bill of materials was about half of CoBra. Not having the time to build a prototype, I introduced the schematics directly into Pix, the PCB layout software we were using at that time. 12 PCB boards were built. I populated and brought-up the first unit myself, which included a real CoBra keyboard with engraved keycaps.

The Turbo Spectrum main board was the same size as the CoBra keyboard; it was a sandwich of 6 spacers, with taller ones placed in the back to allow for more space for the connectors, while the front spacers were lower to allow for better ergonomics. The power supply was external.

After I left Romania in February 1990, the first unit ended up with an acquaintance, who tossed it after a few years when he got bored of it. I gave the remaining PCBs and the schematics to my colleague, Bogdan Morariu, who built and sold a few before he left for Canada. The remaining were handed over to Mircea Ungur who built the remaining units.

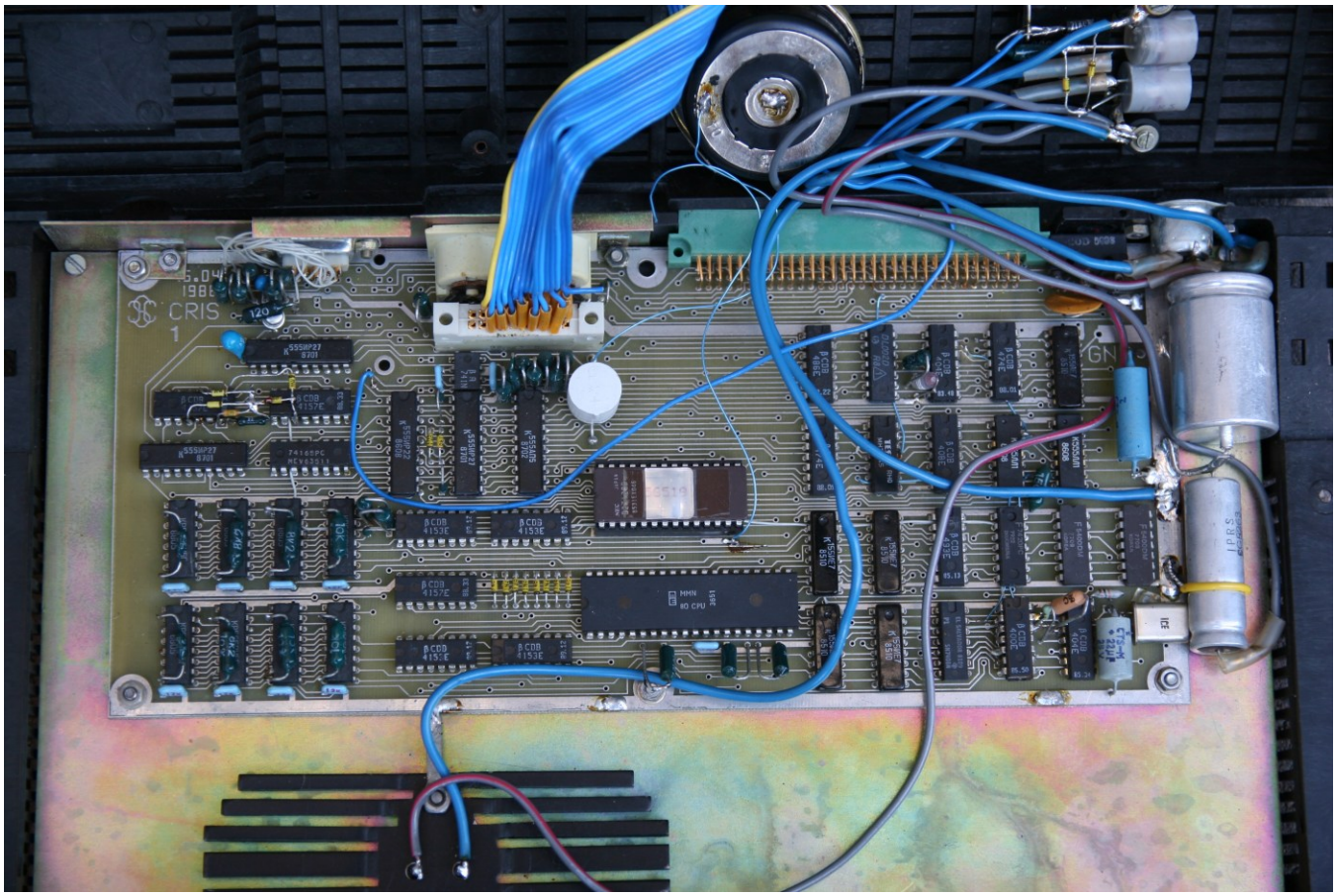


Figure 1: Last original Turbo Spectrum

After an unfortunate event in 2011, Mircea passed away; his family handed me a box with the last Turbo Spectrum PCB mounted inside a HC-85 case (see photo above). I brought the box to California and shelved it in the garage until Bizonul came up with the idea of rebuilding it. I liked the idea, but I kept postponing doing anything about it, mostly due to lack of time. Bizonul kept bugging me, so finally, in 2022, I made it my 2022 New Year's resolution to work on it. For the latest information about Turbo Spectrum after the release of this document, please visit https://www.cobrasov.org.

The EEPROM was on a socket. The source code was lost, but luckily the EEPROM had preserved all the information from the past 32 years since it was programmed. I was able to read it, but disassembling was a little challenging. It was using self-modifying code and undocumented Z80 instructions and features. I did recognize the coding style and did remember I took the approach to minimize everything; partly to reduce the code size, partly to simplify the schematics and reduce the number of components to the absolute minimum, but also as a personal challenge for my own nerdy enjoyment.

I proceeded to reconstitute the schematics, which were lost as well. I still remembered the design tricks I used. After inspecting the PCB using a Fluke Multimeter, I redrew the schematics, in part through reverse engineering, and in part from memory. The time then came to power-up the board. No sparks and no smoke came out of it. I inspected a few signals with an oscilloscope, and they looked alright to me. After some hesitation, I placed the probe on composite video out, and to my great surprise, it was looking right.

In a rush, I built an improvised circuit to convert the digital RGBI to VGA levels, then connected the circuit to a VGA display, but it could not synchronize to the 15KHz line frequency. Searching the internet, I found out that a Lenovo monitor Q24i-10 LED 24″ accepts 15KHz line frequency and 50Hz refresh rate. I bought one, and the video was synchronizing nicely.

Amazingly, this board, built 32 years ago by my friend Mircea, who passed away 11 years ago, is still running as it did on the first day. Thank you, Mircea, and rest in peace.

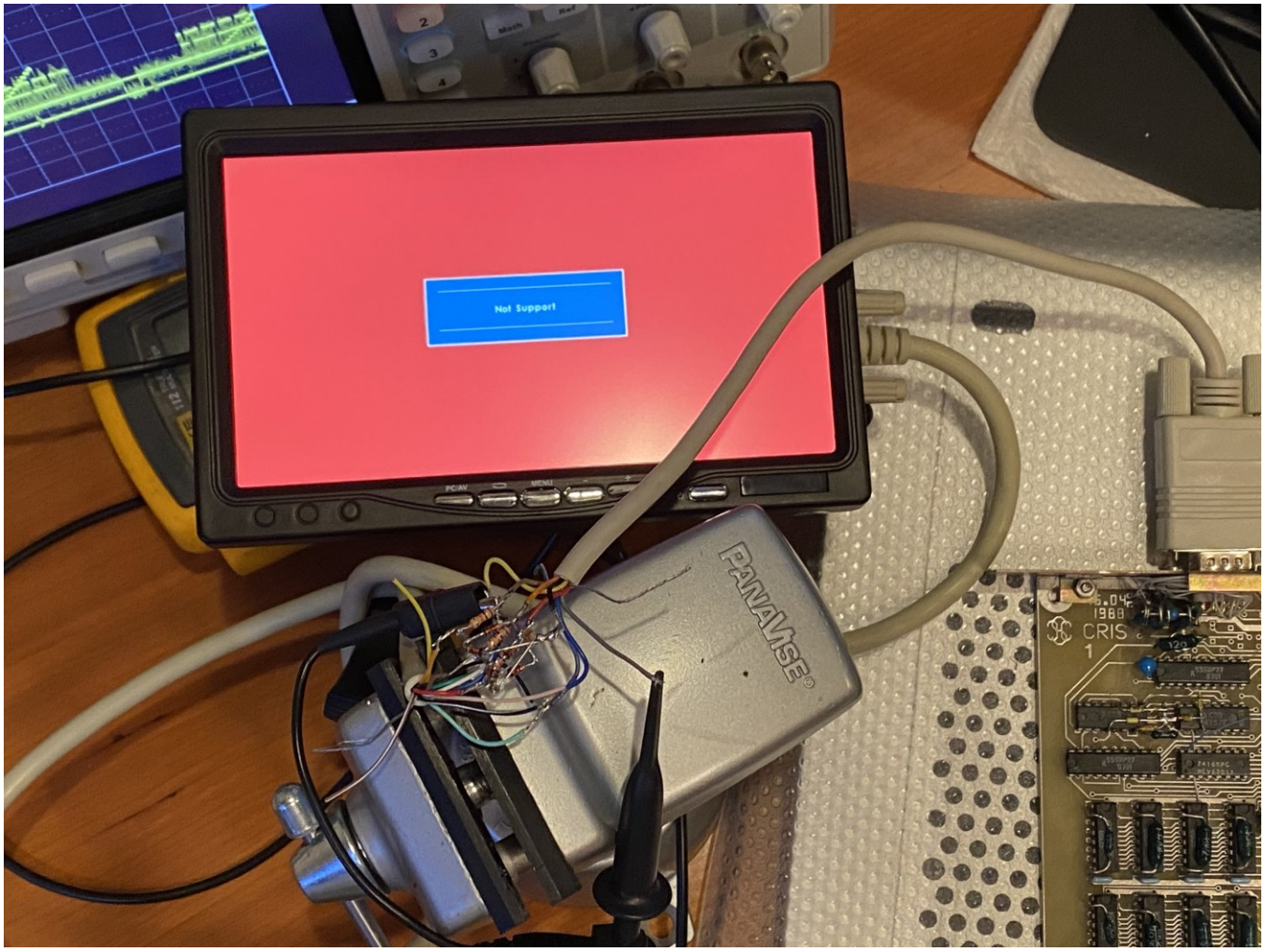


Figure 2: Improvised RGBI to VGA Conversion

2 Hardware

Turbo Spectrum (TS) includes design ideas exemplative of a minimalistic design style that are worth documenting. My goal was to rebuild TS as close as possible to the original version. I fixed a few imperfections, like PCB trace cuts and extra wires, and I added RGBI to VGA conversion. Fortunately, components from the 80's can still be found. For the digital logic circuits, I choose the 74LSxx family, as they are better, easier to find, and lower cost. I used a DE-15 VGA connector in addition to the DE-9 connector, a newer speaker instead of the original graphite phone speaker (galena), and 3.5 mm audio jacks instead of the DIN-5 audio connectors. I opted to keep the 3x32 pins DIN expansion connector, although that connector is no longer manufactured and likely no expansion boards have been built for that proprietary bus since.

Due to these changes and the desire to make sure that the Turbo Spectrum PCB could be rebuilt without any flaws, unlike in 1988, I had now decided to invest the time to prototype it. I found an Augat board on eBay and purchased a wire wrap gun. The picture below shows the working rebuilt Turbo Spectrum on the morning of August 23, 2022, before the start of the Hot Chips 34 Conference. The monitor on the right shows the DRAM Timing diagram from Figure 13 on page 13. The orange box on the left is the new model of CASduino loader purchased from Pedro Almeida in Portugal (https://retro.care).



Figure 3: The rebuilding of Turbo Spectrum

Besides Z80, the 16Kx8 EEPROM and the 8 DRAM chips, the rebuilt Turbo Spectrum has 34 74LSxx digital logic circuits, 20 1N4148 diodes, 45 resistors, 21 capacitors, a PN2222A transistor, the 14 MHz crystal, DE-15 VGA and DE-9 connectors, keyboard connector, expansion connector, 3 audio jacks, 2 small speakers, the reset switch and power jack. The complete Bill of Materials can be found in section 2.14 on page 24.

The component reference designators use the following conventions: `Z` for Z80, `M` for EEPROM, `M0-M7` for the DRAM chips, `TS` for the Trigger-Schmitt 74LS14, `G` and `G0-G9` for logic Gates, `N0-N5` for 74LSxx counters, `F` for 74LSxx Flops, `L` for 74LS373 Latch, `B` for 74LS244 Buffer, `X0-X6` for 74LS1xx multipleXers, `D*` for Diodes, `R*` for Resistors, `C*` for Capacitors, `T` for the TN2222A transistor, `J*` for connectors, `SPKR*` for Speakers, `SW` for reset Switch, and `XTAL` for the 14 MHz crystal. In the text of the following sections, underlined characters are used for signals active LOW, for example, `CK1` is `CK1` active LOW. Signals and component reference descriptors use `Courier` characters.

2.1 Power and Ground

Turbo Spectrum has a single +5V power domain. The CPWR 100µF electrolytic capacitor is used to filter out transient current spikes. In addition, the 74LS14 Trigger-Schmitt invertor TS used for clock generation and reset, Z80 Z, the final video output register F6, and all 16Kx1 DRAM chips M0-M7, have their own 0.1µF tantalum capacitors, next to the VCC pin to filter out current peaks and reduce switching noise.

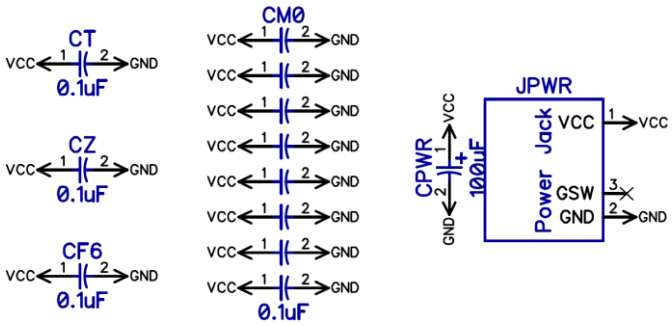


Figure 4: Power and Ground

2.2 Clocks

Figure 5 shows the Turbo Spectrum clocking circuitry. It starts with a typical clock generation used in simple digital circuits from the early 80-ties, with 2 inverters and 680Ω resistors as feedback and a 14 MHz quartz crystal. The original IPRS βCDB404E were always locking on the crystal fundamental, but the newer 74LS04 are locking on the 3rd harmonic, at 42 MHz. 74LS14 Trigger-Schmitt invertors (TS) do not have this problem. The second inverter is followed by a third to form a clean digital 14 MHz clock, used as clock for the N0 counter (pin N0.5) and the second F0 flip-flop (pin F0.11). The XTAL component has a 3rd pin connected to GND, in case a crystal with a 3rd pin connected to the crystal case is used.

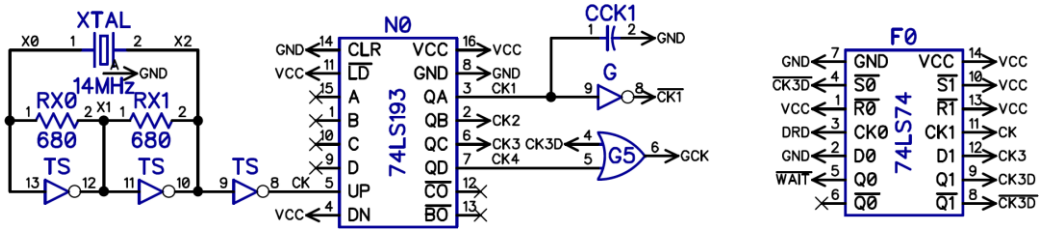


Figure 5: Clock Generation

The N0 counter generates the lower clock frequencies. CK1, the inverted version of CK1 (7 MHz), is used as the pixel clock by the F4 video bitmap shifter (pin F4.2 in Figure 27) and by F6, the final video output register (pin F6.11 in Figure 27). CK2 is the 3.5 MHz Z80 clock (pin Z.CLK in Figure 7) and it is used also as input to the G1 gate (pin G1.5 in Figure 12), that generates the DRAM RAS signal.

The 390pF CCK1 capacitor is used to delay CK1, to avoid a set-up time violation for the F6 register. This is explained in section 2.8 on page 12, and the critical timing path is shown in Figure 13 DRAM Timing Diagram.

CK3 is half the Z80 clock frequency and is used to select between Z80 and video access to the dual-port DRAM (pin G1.2 in Figure 12), as input to the second F0 flip-flop (pin F0.12) that generates delayed versions CK3D and CK3D of CK3, as input G9.1 in Figure 12 to generate the LD strobe for the F4 bitmap shifter, and as S1 select to the X1-X4 DRAM address multiplexers (pin 2 in Figure 23). CK3D is used as input G5.4 to generate the GCK graphics character clock, and as input L.11 in Figure 12 to latch the DRAM data for the Z80 memory read cycle. CK3D is used as input to the first F0 flip-flop (pin F0.4) that sets the Z80 WAIT signal, as input G7.10 in Figure 12 that enables the DRAM WE signal, and as input G9.2 to generates the LD strobe for the F4 bitmap shifter.

CK4 is 1/4th Z80 clock frequency used as input G5.5 to generate the GCK graphics character clock, as input G9.4 in Figure 12 to generate the LD strobe for the F4 bitmap shifter, as inputs G2.12 in Figure 20 and X0.1 in Figure 23 to multiplex bitmap and attributes DRAM video addresses, and as input clock N1.5 in Figure 20 for the first video timing counter.

2.3 Reset

There are two reset signals, PRST (Power-on Reset) and HR (Hard Reset). PRST has an RC circuit, RPR and CPR, with a time constant of 10KΩ * 10µF = 0.1 seconds, to keep Turbo Spectrum in the reset state until the power source stabilizes. The two Trigger-Schmitt invertors form the PR signal with standard digital TTL edges. HR has an RC circuit, RHR and CHR, with a time constant of 10KΩ * 0.1µF = 1 ms, to filter glitches from the SW reset switch. The following Trigger-Schmitt invertor forms the HR signal with standard digital TTL edges. The final RST signal is an AND of PR and HR. The SW reset switch has 2 extra pins connected to GND; these are pads that coincide with the reset switch mounting holes.

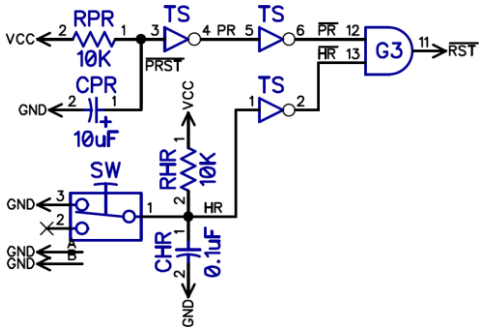


Figure 6: Reset Circuitry

2.4 Z80 CPU

Z80 has a few pull-up resistors, a 330Ω RCLK, two 10KΩ RNMI and RBR, and eight 4.7KΩ RD0-RD7. The BUSREQ, BUSACK, and HALT signals are not used, and the NMI and M1 signals are used only by the JEXT expansion connector.

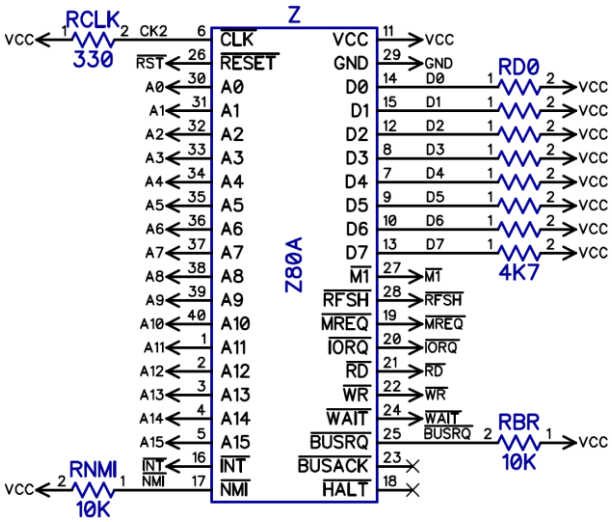


Figure 7: Z80 CPU

2.5 EEPROM

The WE pin and the not used pin 1 (A14 for the 32Kx8 EEPROMs) are connected to VCC. Turbo Spectrum supports also 32Kx8 EEPROMs, when only the upper 16K are used.

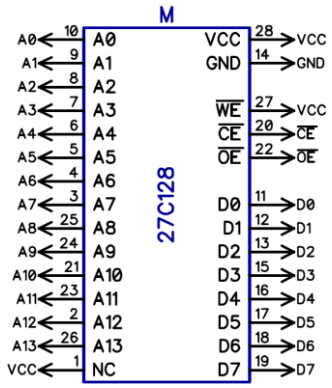


Figure 8: 16Kx8 EEPROM

2.6 DRAM

Turbo Spectrum has eight 64Kx1 DRAM chips. Pin 1 of the DRAM chips are not connected inside the package but are connected to GND in Turbo Spectrum to facilitate wider GND traces in the densely populated board.

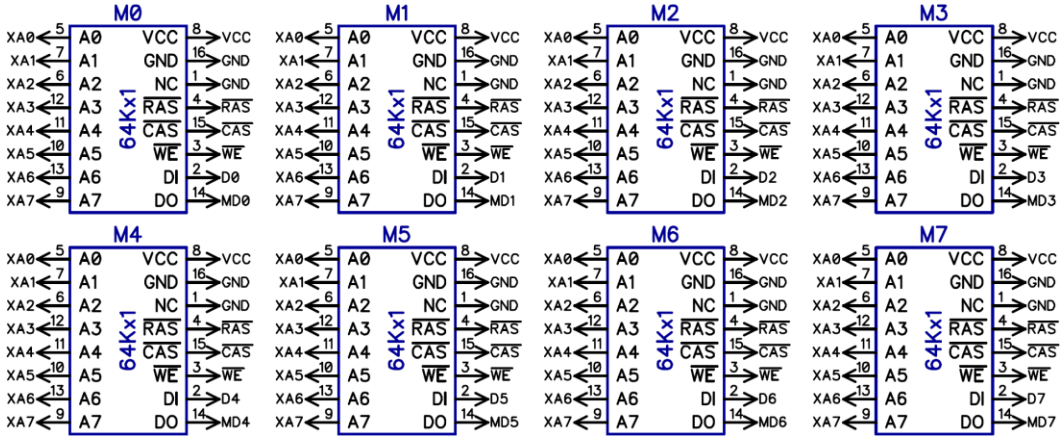


Figure 9: 64Kx8 DRAM

2.7 Operating Modes and Memory Organization

The two F2 flops drive the CE (EEPROM Chip Enable) and SM (Spectrum Mode) signals. CE is connected directly to the EEPROM CE pin. Turbo Spectrum can operate in one of 3 modes:

1. EEPROM Mode: CE=0 (enabled EEPROM) and SM=0;
2. 64K DRAM Mode: CE=1 (disabled EEPROM, 64K DRAM) and SM=0;
3. Spectrum Mode: CE=1 (disabled EEPROM, 64K DRAM) and SM=1.

RST resets both flops. When not in Spectrum Mode (SM=0 and SM=1), Z80 address bit A15 is inverted to X15 before it is used to generate the DRAM address (video memory starts at 0c000h). The IE signal is connected to the Output Enable pins B.1 and B.19 in Figure 24, the 74LS244 buffer driving the Z80 data bus when an in instruction is executed.

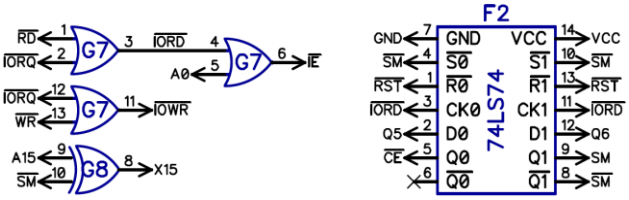


Figure 10: Operating Modes

The F5 74LS377 register in Figure 24 is used as output port. Whenever an out instruction is executed at any even address (A0=0), for example 0feh, the output data byte is stored to F5; the stored bit 5 (Q5) is used as input to the CE flop and bit 6 (Q6) as input to the SM flop; output port bits 5 and 6 are not used by Spectrum. Any in instruction loads those bits to the CE and SM flops. When SM becomes 1 (SM=0), the CE and SM flops are set to 1, from where they can come out only when RST=0.

Figure 11 shows the Turbo Spectrum memory organization for all three modes; the shaded memory areas are Read-Only. In Spectrum Mode, EEPROM is disabled, the Spectrum Basic executes out of DRAM, and writes are not permitted to the first 16K. This is necessary because some games do write to the first 16K which should behave as NOPs without any side effects.

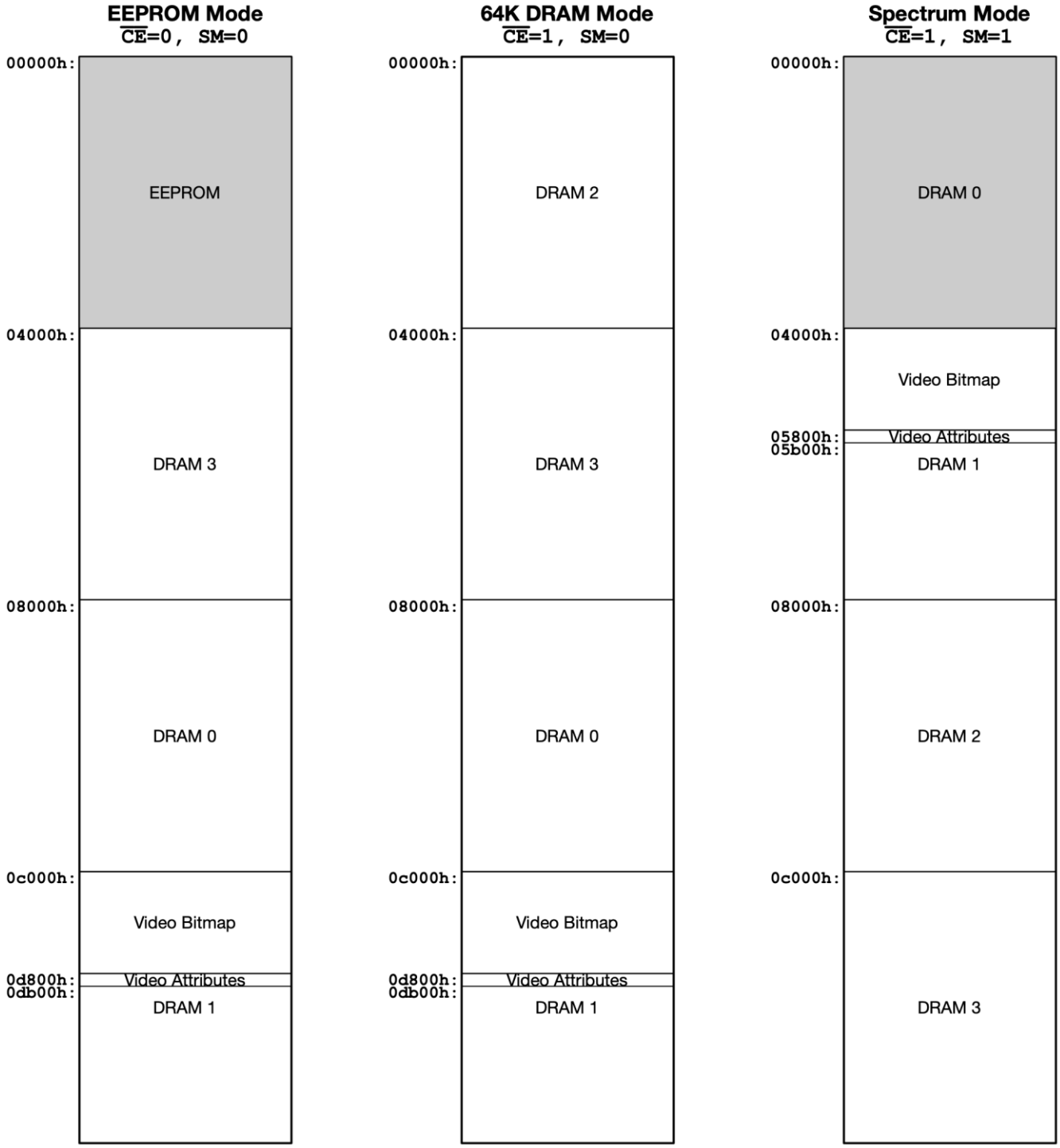


Figure 11: Memory Organization

2.8 DRAM Controller

The basic idea that led to the simple dual-port DRAM controller shown in Figure 12 is that CK2, the 3.5 MHz clock, is used both as the Z80 clock (Z.CLK) and as the DRAM RAS signal, driven by G1.6. RAS is enabled by DA (DRAM Access), which is set when CK3=0 (video access) OR DRQ=0 (DRAM Request), whenever Z80 wants DRAM access. This happens when MRQ (non-refresh Memory Request, MREQ=0 AND RFSH=1) AND DE (DRAM Enable) are high. DE=1 when access to memory beyond 04000h is desired (A15=1 OR A14=1) OR DO=0, which happens when CE=1 (disabled EEPROM, 64K DRAM) AND, this is a read (RD=0) OR a write in non-Spectrum mode (SM=1).

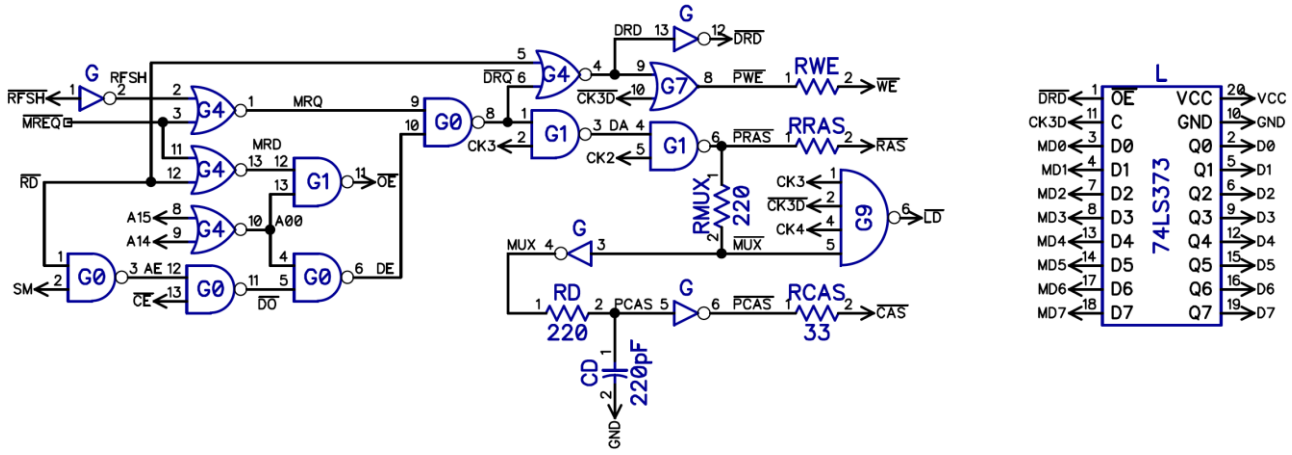


Figure 12: DRAM Controller

Figure 13 is the DRAM Timing Diagram that includes the CK, CK1, CK1, CK2, CK3, CK3D, CK3D, CK4 clocks, the H[5:0] character and V[8:0] video line counters, the DA, RAS, MUX, CAS, WE control signals, XA[7:0] multipleXed DRAM address, MD[7:0] DRAM read data, the LD strobe for the F4 bitmap shifter, S0-S4 serial bitmap bits, the GCK graphics character clock, the G[7:0] color attribute from F3 output, PS (Paper Select) that muxes Ink and Paper, G[RGBI] at the X5 output, HVB (Horizontal-Vertical Border) that muxes Screen and Border, X[RGBI] at the X6 output, and V[RGBI] the final pixel value out of F6.

Active clock edges have an arrow and dotted lines are possible levels when Z80 wants DRAM access. The Figure 13 DRAM Timing Diagram includes the typical signal delays caused by the intrinsic gate delays and by the extra delay introduced by resistors and capacitors.

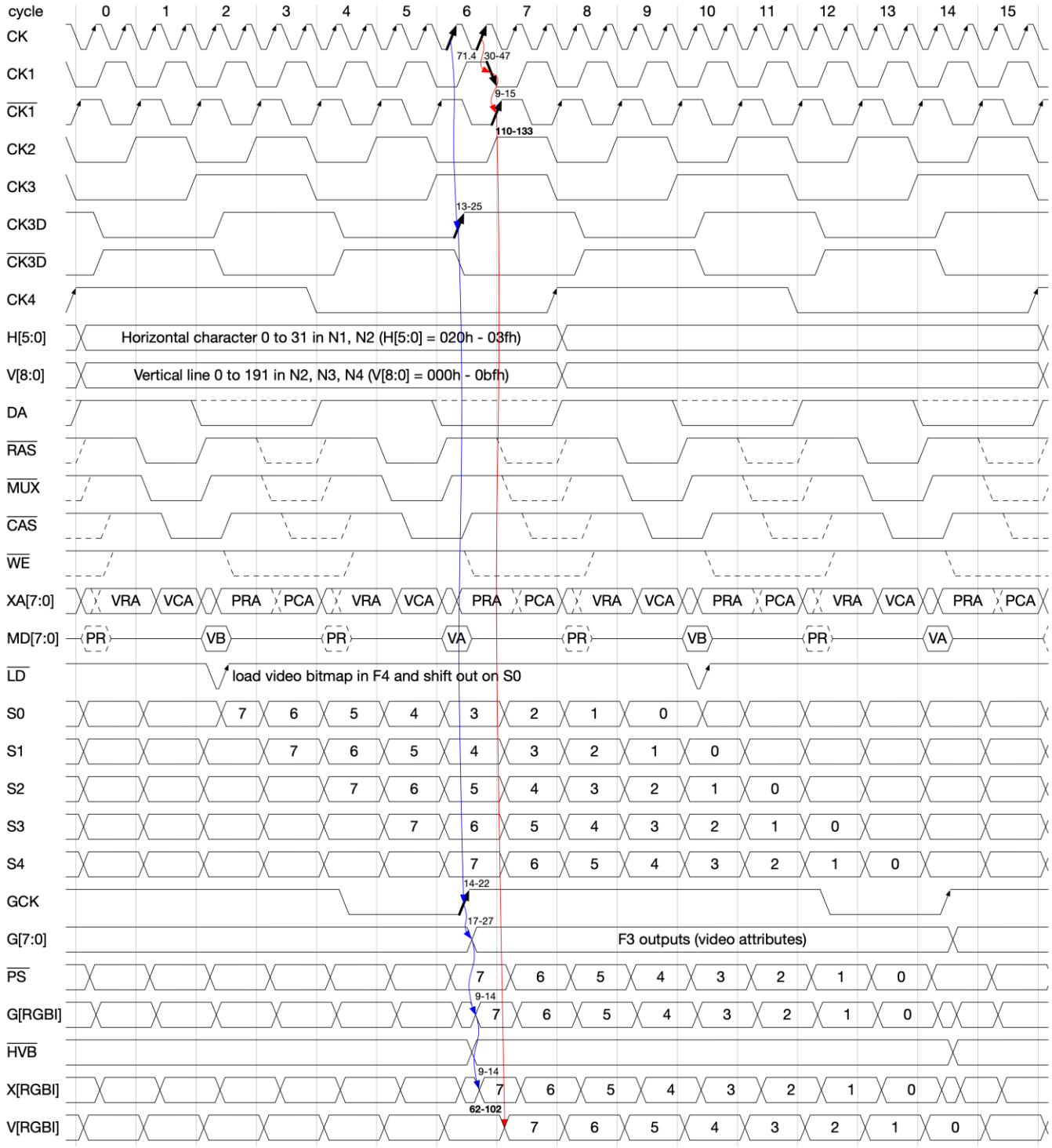
MUX is a delayed version of RAS, and is used as select for X1-X4, to multiplex the DRAM Row and Column Address. MUX is further delayed by the RD and CD circuit, with a time constant of 220Ω * 220pF = 48 ns to form CAS.

DRD=1 when DRQ=0 AND RD=0, i.e. whenever Z80 wants to read from DRAM. DRD, the inverted version, is used as Output Enable for the L latch that stores the MD[7:0] DRAM read data, which is latched on the falling edge of CK3D.

As Z80 drives WR late in the memory write cycle, DRD=0 is used to drive DRAM WE=0 when CK3D=0, i.e. during the Z80 DRAM access slot.

Figure 13 DRAM Timing Diagram also shows the critical timing path; relevant active clock edges are shown in bold. It starts from the first CK edge in cycle 6, shown in bold, and the timing dependencies are shown with blue arrow lines. The rising edge of CK3D, driven by F0.9, comes out with a typical delay of 13 ns and a maximum delay of 25 ns; the 13-25 interval is shown in the diagram above the rising edge of CK3D. The rising edge of GCK, driven by G5.6, comes out with a delay of 14-22 ns, then G[7:0] has a delay of 17-27 ns, G[RGBI] has 9-14 ns, X[RGBI] has 9-14 ns, for a total delay from the first CK edge in cycle 6 of **62-102** ns, shown in bold below the X[RGBI] waveform.

X[RGBI] is being latched into the final F6 register on the following rising edge of CK1. This is derived from the second CK edge in cycle 6, which comes with a delay of 71.4 ns after the first edge; timing dependencies of CK1 are shown with red arrow lines. The falling edge of CK1, driven by N0.3, comes out with a delay of 30-47 ns, and the following rising edge of CK1, driven by G.8, comes out with a delay of 9-15 ns, for a total delay from the first CK edge in cycle 6 of **110-133** ns. Inputs to F6 need min. 20 ns set-up time; this is typically met, but depending on the ICs used, the X[RGBI] set-up time can be anything between 8-71 ns. To avoid a possible timing violation, a 390pF CCK1 capacitor is used to delay CK1.



Legend for XA[7:0]:

VRA: Video Row Address
VCA: Video Column Address
PRA: Processor Row Address
PCA: Processor Column Address

Legend for MD[7:0]:

PR: Processor Read
VB: Video Bitmap
VA: Video Attributes

Figure 13: DRAM Timing Diagram

2.9 Z80 DRAM Access

Because Z80 can access DRAM only when CK3=1, i.e., every other Z80 clock cycle, there are two possible phases. Figure 14 through Figure 19 show all possible Z80 memory cycles for both phases. Relevant active clock edges have an arrow and are shown in bold. RAS and CAS cycles and MD[7:0] data used by Z80 are shown in bold, too. The dual-port DRAM controller does not impact the Z80 memory write cycles, but 50% of the Z80 memory read cycles have one WAIT state.

Figure 14 shows phase 0 of the Z80 M1 Cycle, which coincides with CK3=1, when no wait state is required. The rising edge of DRD is used to clear WAIT, but CK3D=0, the delayed and inverted CK3, prevents that by keeping the flop SET input LOW, so no wait state is inserted. The MD[7:0] data is latched into L on the falling edge of CK3D, and is read by Z80 on the following rising edge of CK2.

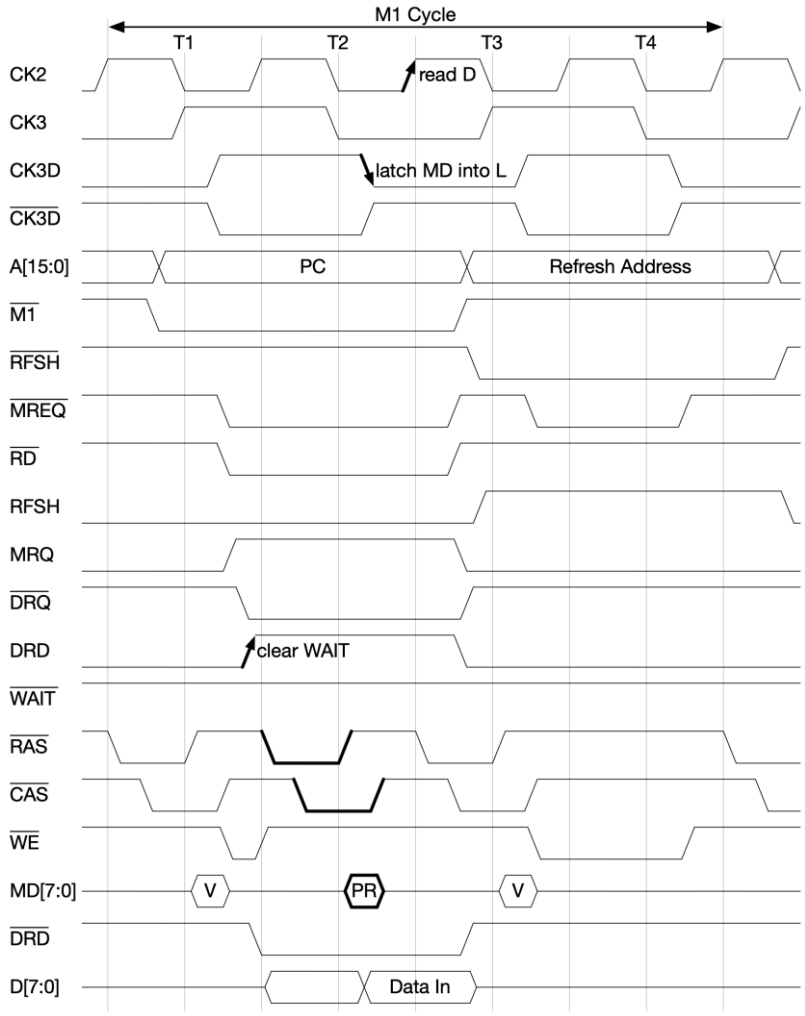


Figure 14: Z80 M1 Cycle, phase 0

Figure 15 shows phase 1 of the Z80 M1 Cycle, which coincides with CK3=0, when a wait state is required. The rising edge of DRD is used to clear WAIT; this time CK3D=1, so a wait state is inserted on the following falling edge of CK2. The MD[7:0] data is latched into L on the falling edge of CK3D, and is read by Z80 on the following rising edge of CK2.

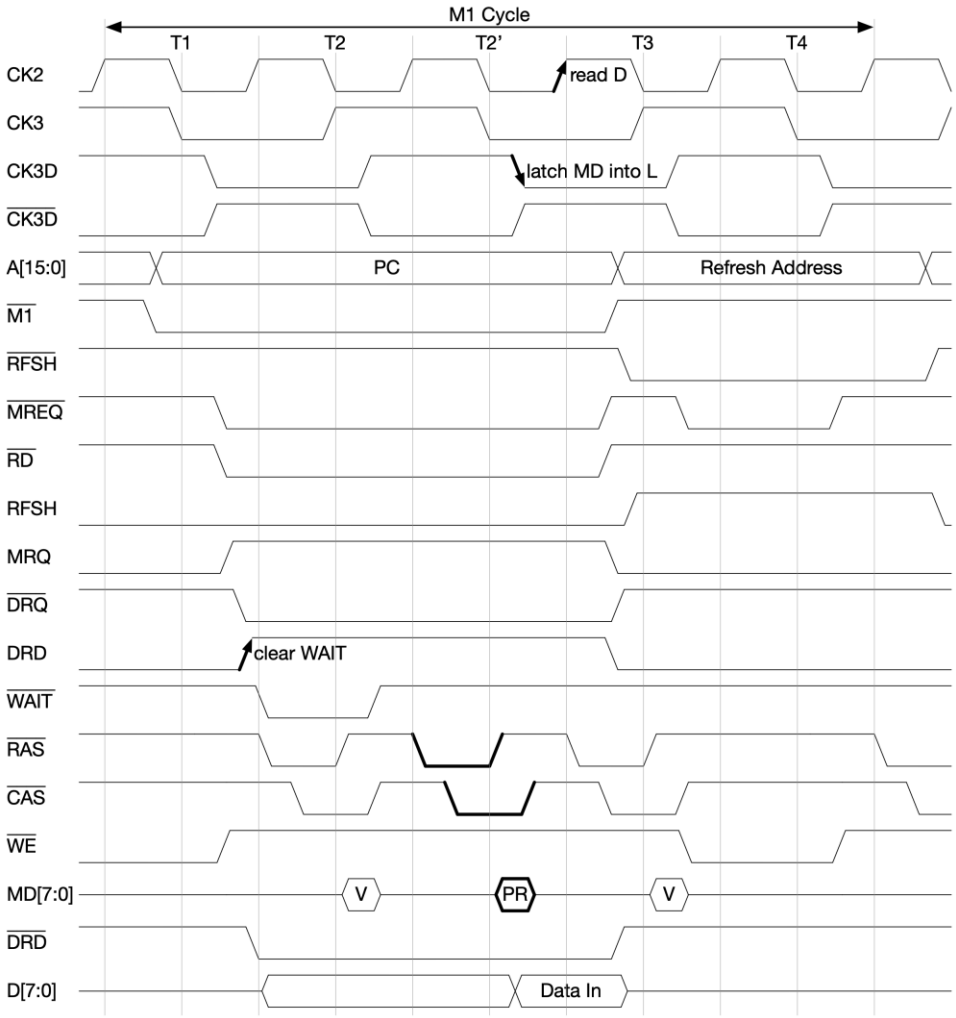


Figure 15: Z80 M1 Cycle, phase 1

Z80 Memory Read Cycles are half a CLK state longer than M1 Cycles, but that's not sufficient to avoid wait states. Figure 16 shows phase 0 of the Z80 Memory Read Cycle, which coincides with CK3=1, when no wait state is required. The rising edge of DRD is used to clear WAIT, but CK3D=0, so no wait state is inserted. The MD[7:0] data is latched into L on the falling edge of CK3D, and is read by Z80 on the following falling edge of CK2.

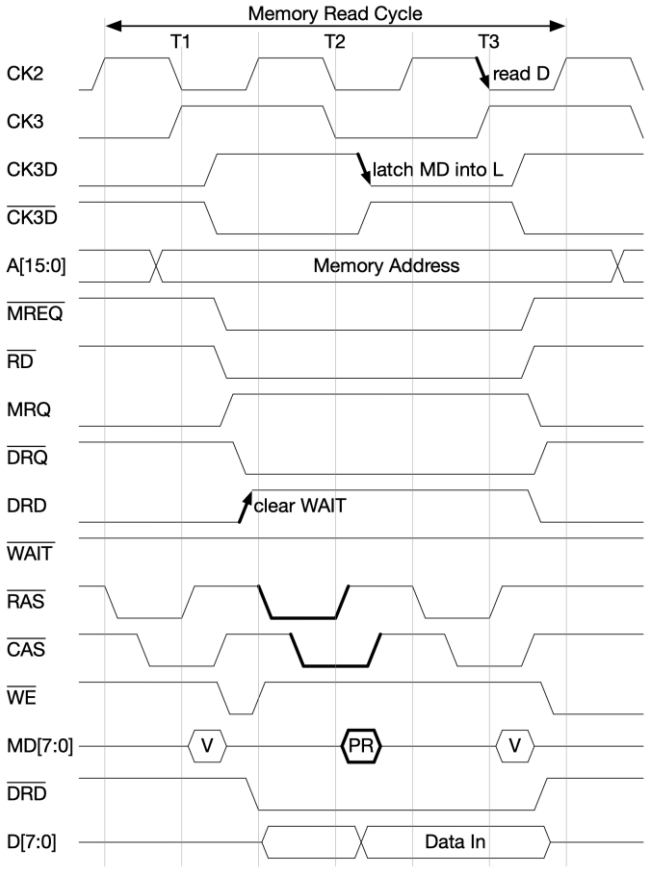


Figure 16: Z80 Memory Read Cycle, phase 0

Figure 17 shows phase 1 of the Z80 Memory Read Cycle, which coincides with CK3=0, when a wait state is required. The rising edge of DRD is used to clear WAIT; this time CK3D=1, so a wait state is inserted on the following falling edge of CK2. The MD[7:0] data is latched into L on the falling edge of CK3D, and is read by Z80 on the following falling edge of CK2.

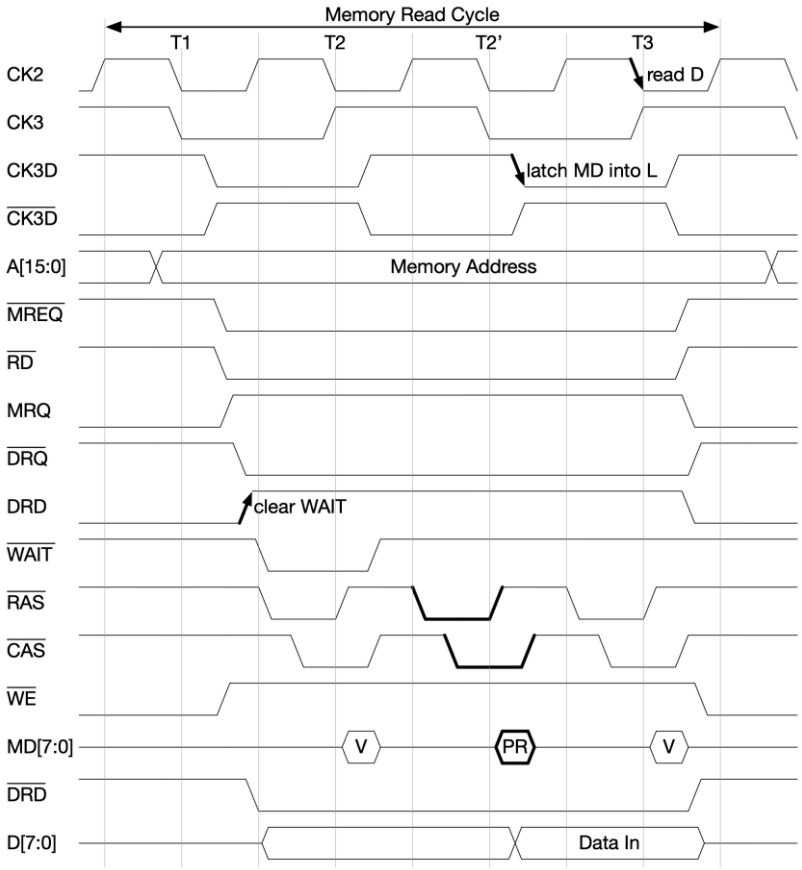


Figure 17: Z80 Memory Read Cycle, phase 1

Z80 Memory Write Cycles are longer than Read Cycles, so wait states can be avoided for both phases. Figure 18 shows phase 0 of the Z80 Memory Write Cycle, which coincides with `CK3=1`, when the Z80 `D[7:0]` data is written during the T2 cycle. Figure 19 shows phase 1 of the Z80 Memory Write Cycle, which coincides with `CK3=0`, when the Z80 `D[7:0]` data is written during the T3 cycle.

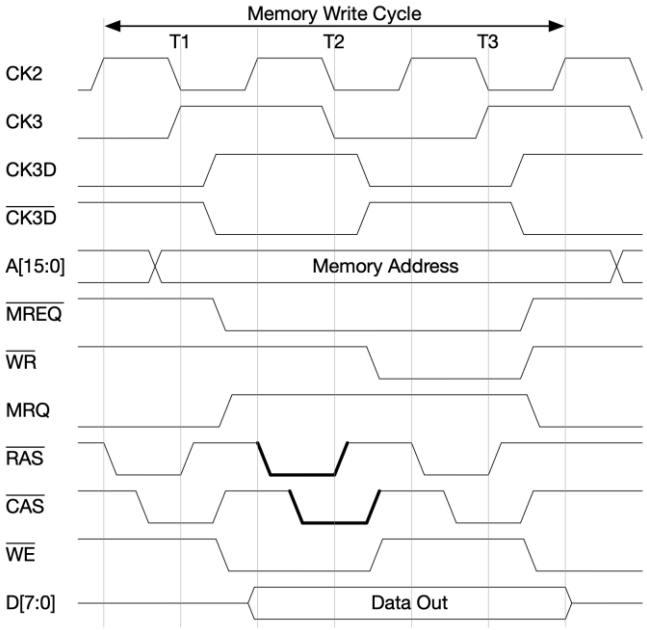


Figure 18: Z80 Memory Write Cycle, phase 0

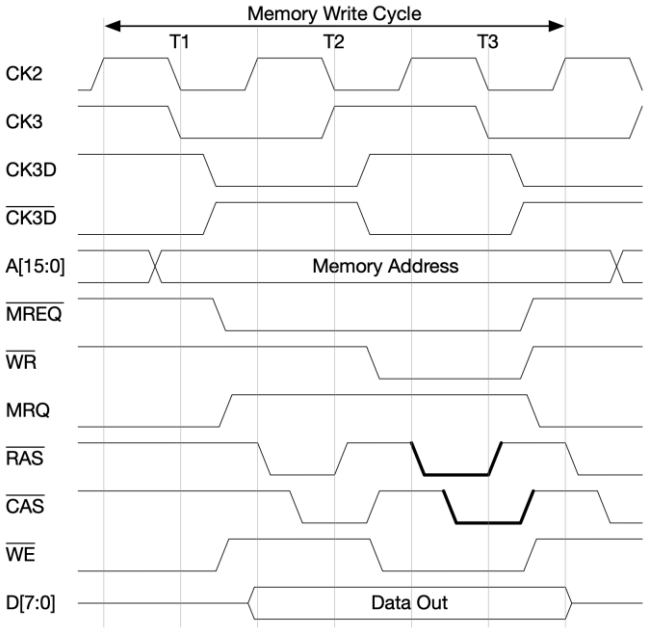


Figure 19: Z80 Memory Write Cycle, phase 1

2.10 Video Controller

The N1–N4 ripple counters implement the H[5:0] character and V[8:0] video line counters. CK4 is the input clock with a frequency of 7/8 MHz and a cycle time of 8/7 = 1.14µs. The two F1 flops are used to synchronize PHVS and PHVB (Pre-Horizontal & Vertical Sync and Border) with GCK, the graphics character clock; the outputs are HVS and HVB, the Horizontal & Vertical Sync and Border. The N5 counter is used to further subdivide the video frame frequency by 8 to obtain V12, the character flush frequency of 50/8 = 6.25 Hz. Figure 21 is the video line timing and Figure 22 is the video frame timing.

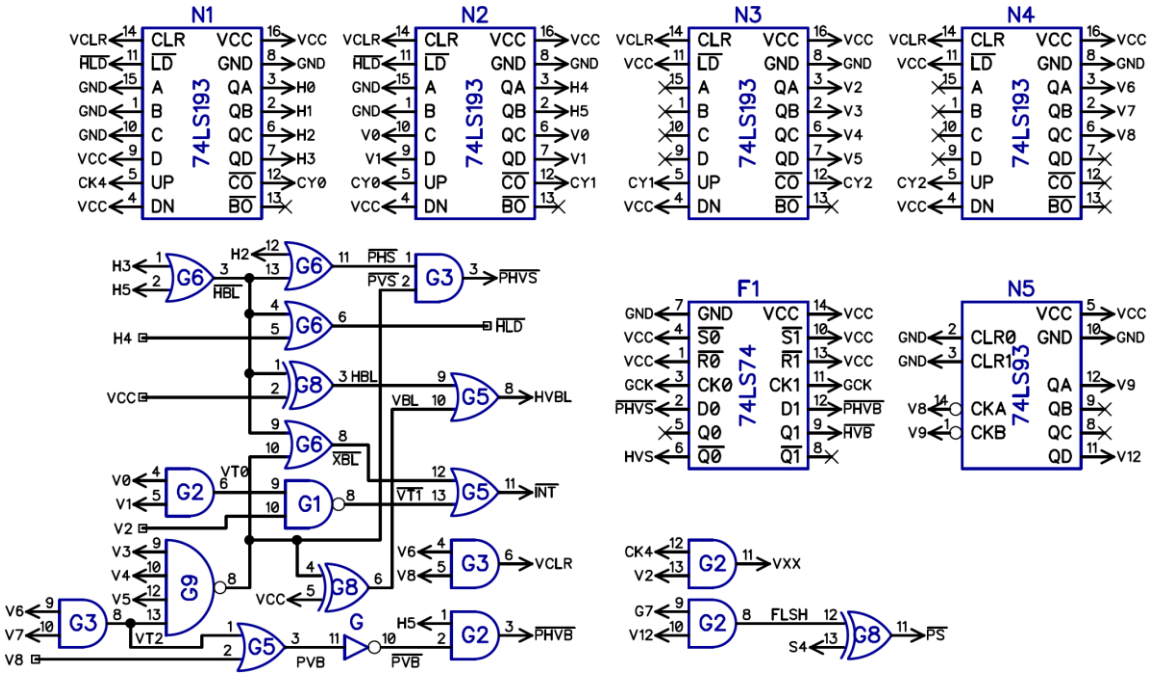


Figure 20: Video Controller

After H[5:0] wraps around, the HLD (Horizontal Load) pulse is generated which loads H[5:0] with 8, so H[5:0] counts from 8 to 63; there are 56 CK4 cycles, so the line time is 56 * 8/7 = 64 µs. Cycles 32 to 63 (H5=1) are the active line time, cycles 8 to 31 (H5=0) are the horizontal border, cycles 16 to 23 (HBL=0) are the Horizontal Blanking, and cycles 16 to 19 (PHS=0) are the Pre-Horizontal Sync.

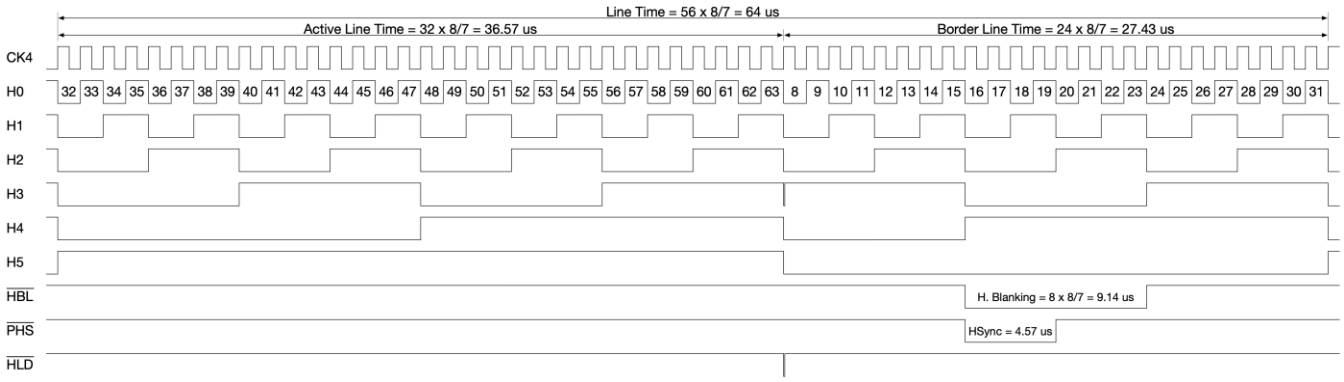


Figure 21: Video Line Timing

When V[8:0] reaches 320, VCLR is set which clears the N1–N4 counters, so V[8:0] counts from 0 to 319; there are 320 lines, so the frame time is 320 * 64µs = 20.48 ms. The first 192 lines (PVB=1) is the active frame time, lines 192 to 319 (PVB=0) is the Pre-Vertical Border, lines 248 to 255 (VBL=1) is the Pre-Vertical Blanking and (PVS=0) Pre-Vertical Sync. The INT Interrupt is asserted during the last Vertical Sync line 191, while HBL=0 (Horizontal Blanking).

VXX is used to generate one of the DRAM address bits, and PS (Paper Select) is the select signal that muxes Ink and Paper. PS is the XOR of S4, the final pixel value, with FLSH, the Flush character attribute (G7) AND V12, the character flush frequency of 50/8 = 6.25 Hz.

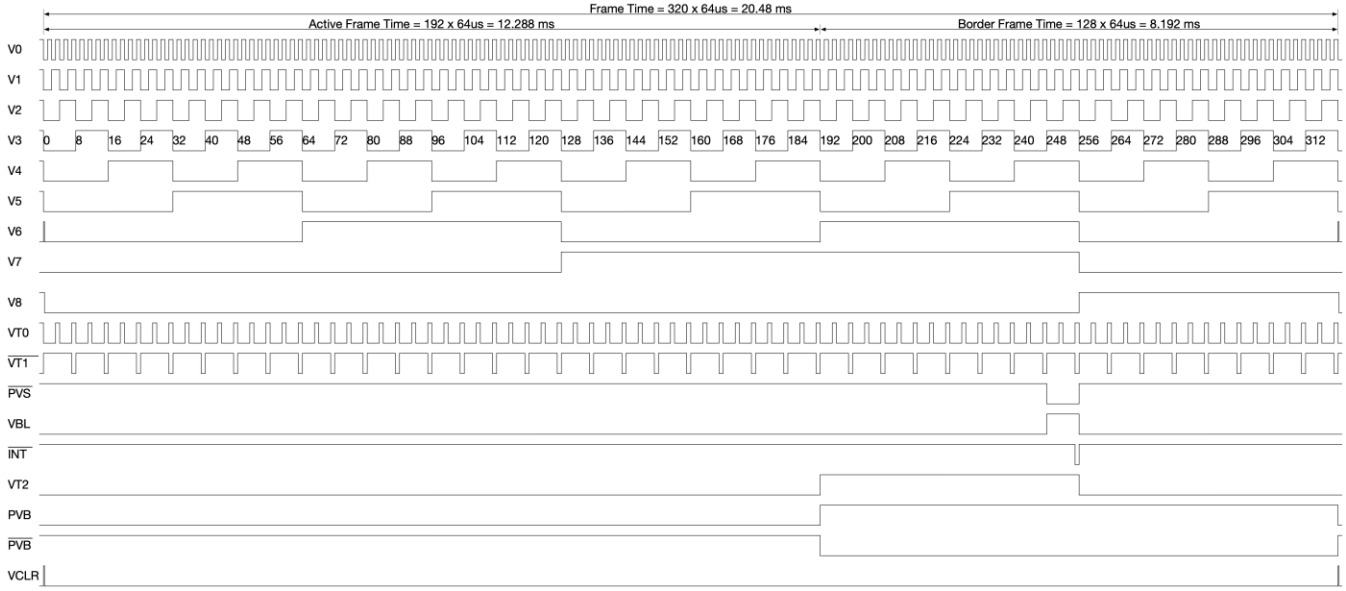


Figure 22: Video Frame Timing

2.11 DRAM Addressing

The DRAM address is driven by the X1−X4 4:1 multiplexers. CK3 selects either the video (CK3=0) or Z80 (CK3=1) address, and MUX, which is an inverted and delayed CK2, selects either the Row (MUX=1) or Column (MUX=0) address. For video address, the X0 2:1 multiplexer further selects either the bitmap (CK4=1) or attribute (CK4=0) video address.

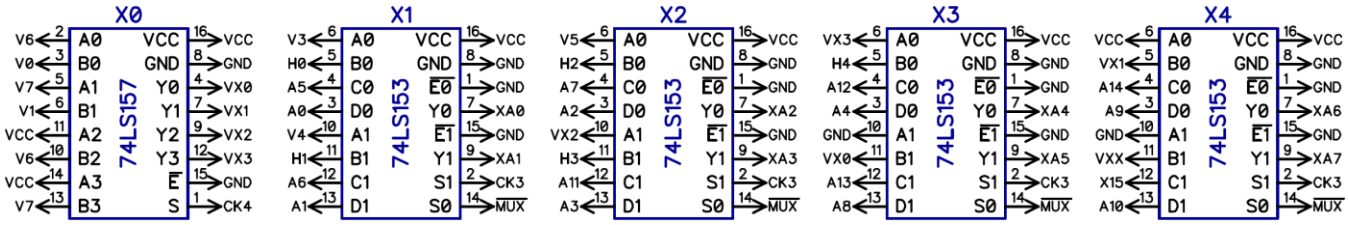


Figure 23: DRAM Address Multiplexers

Table 1 shows how the address bits are multiplexed. Low address bits are used as Row address (DRAM page), so every 8 lines (8 * 64µs = 0.512 ms) all DRAM pages are automatically refreshed, and no additional refresh cycles are needed.

CK4	CK3	CK2	{XA7,XA6,XA5,XA4,XA3,XA2,XA1,XA0}	Description
x	1	0	{A10, A9, A8, A4, A3, A2, A1, A0}	Z80 Row Address
x	1	1	{X15,A14,A13,A12,A11, A7, A6, A5}	Z80 Column Address
1	0	0	{ V2, V1, V0, H4, H3, H2, H1, H0}	Video Bitmap Row Address
1	0	1	{ 0, 1, 0, V7, V6, V5, V4, V3}	Video Bitmap Column Address
0	0	0	{ 0, V7, V6, H4, H3, H2, H1, H0}	Video Attribute Row Address
0	0	1	{ 0, 1, 0, 1, 1, V5, V4, V3}	Video Attribute Column Address

Table 1: DRAM Address

2.12 Input/Output

The F5 74LS377 register is used as output port. Whenever an out instruction is executed at any even address (A0=0), for example 0feh, the output data byte is stored to F5. Bits Q[2:0] set the border color, Q3 is used as output for tape recording, similar to CoBra, Q4 is the audio output, Q5 enables/disables the EEPROM, Q6 sets the Spectrum mode, and Q7 is not used. Any in instruction loads the Q5 and Q6 bits to the CE and SM flops (see section 2.7 Operating Modes on page 10).

The B 74LS244 buffer is used as input port. Whenever an in instruction is executed from any even address (A0=0), for example 0feh, Z80 reads the K[7:0] bus. Bits K[5:0] are driven by the keyboard, same as for CoBra, K6 is the tape input, and K7 is not used.

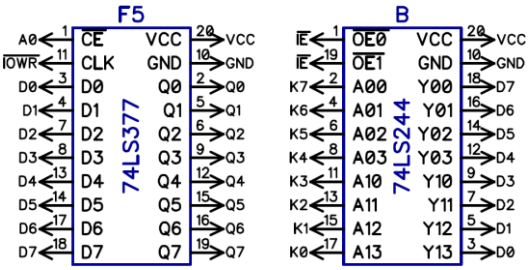


Figure 24: Input/Output

2.12.1 Keyboard and Audio Output

Turbo Spectrum uses the same 2x10 pin keyboard connector as CoBra, with eight 1N4148 diodes and six 4.7KΩ pull-up resistors. The Q4 audio output is connected to a 10µF electrolytic capacitor, with the other end (SPKR) connected to a speaker, to a pin of the keyboard connector, and to an audio jack through a 10KΩ resistor. The JAO Audio Output Jack has 2 extra pins, A and B, connected to GND; these are pads that coincide with the audio jack protruding pins.

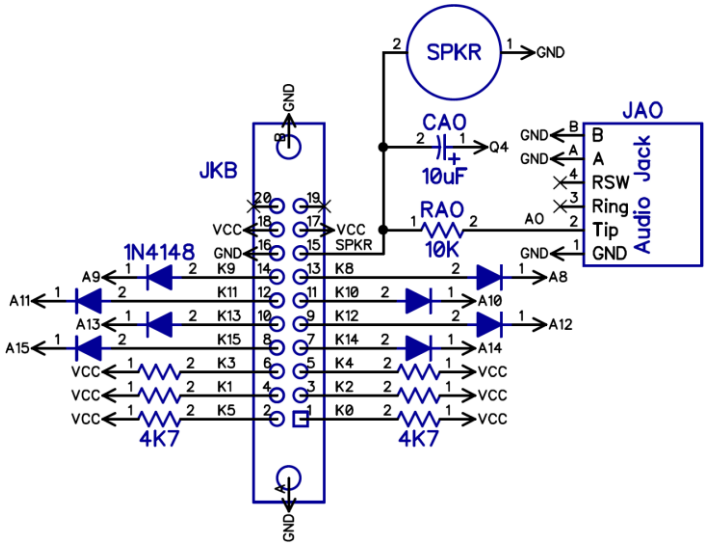


Figure 25: Keyboard and Audio Output

2.12.2 Tape Input and Output

The Tape Output is similar to CoBra, Q3 is connected to an audio jack through two 10KΩ resistors and two 1N4148 diodes for voltage surge protection.

The Tape Input no longer comes from a cassette player, but from CASduino which drives a digital signal. Therefore, the LM741 Op Amp and the -5V voltage are no longer needed and the input circuit is greatly simplified. As CASduino does not provide the sound we got used to from the 80-ties, I added another small speaker to reproduce that audio experience.

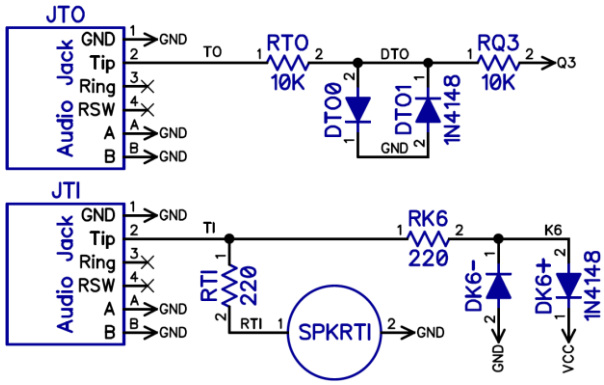


Figure 26: Tape Input and Output

2.13 Video Pipeline

The video bitmap byte is latched in the `F4` register on the rising edge of `LD` and is shifted out on the `S0` pin and is further delayed by the `F6` register, being shifted on the rising edge of `CK1` as `S1-S4`, until the attribute byte is loaded in the `F3` register on the rising edge of `GCK`.

The Ink `{G6,G2,G1,G0}` and Paper `{G6,G2,G1,G0}` bits are muxed by `X5` to `{GI,GG,GR,GB}` using `PS` as select, then the Screen pixel `{GG,GR,GB,GO}` and Border `{Q2,Q1,Q0, 0}` are muxed by `X6` to `{XG,XR,XB,XI}` using `HVB` as select and `HVBL` to blank out the output (set to 0), which is loaded on the rising edge of `CK1` in the final `F6` register that drives `{VG,VR,VB,VI}`.

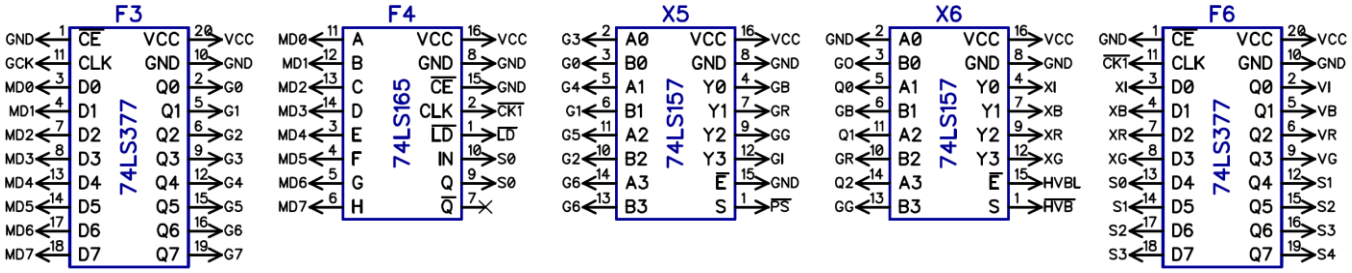


Figure 27: Video Output

In between the `X5` and `X6` multiplexers, `GI` is transformed to `GO` to convert bright black `{0,0,0,1}` to black `{0,0,0,0}`. The logical function for this is `GO = (R | G | B) & GI`, which can't be implemented with a single 74LSxx circuit. In Turbo Spectrum, this was implemented with 5 diodes and 2 resistors as shown in Figure 28.

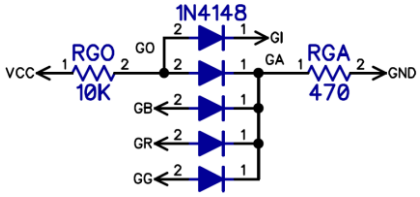


Figure 28: Bright Black to Black Conversion

2.13.1 RGBI and VGA Outputs

The original Turbo Spectrum had a DE-9 connector for the RGBI digital output and composite video signal for black-and-white monitors, which was preserved in the rebuilt version. In addition, a VGA connector and a simple RGBI to VGA conversion circuit was added, which consists of 3 diodes and 4 resistors, plus 4 capacitors to filter out some of the high frequency switching noise introduced by the final `F6` register. `HVS` is used as both HSync and VSync on pins 13 and 14 of the VGA connector; this proved to work well for the Lenovo Q24i-10 LED 24″ monitor.

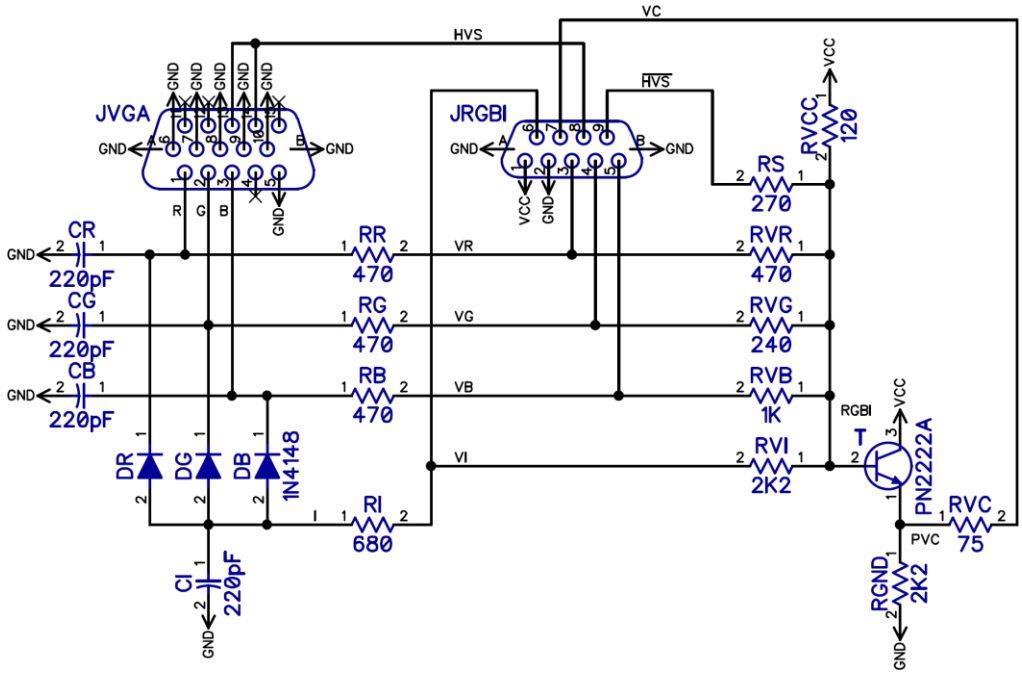


Figure 29: RGBI to VGA Conversion

2.14 Bill of Materials

RefDes	Name	Value
B	74LS244	
CAO	CAP080R	10uF
CB	CAP100	220pF
CCK1	CAP100	390pF
CD	CAP100	220pF
CF6	CAP100	0.1uF
CG	CAP100	220pF
CHR	CAP100	0.1uF
CI	CAP100	220pF
CM0	CAP100	0.1uF
CM1	CAP100	0.1uF
CM2	CAP100	0.1uF
CM3	CAP100	0.1uF
CM4	CAP100	0.1uF
CM5	CAP100	0.1uF
CM6	CAP100	0.1uF
CM7	CAP100	0.1uF
CPR	CAP080R	10uF
CPWR	CAP100R	100uF
CR	CAP100	220pF
CT	CAP100	0.1uF
CZ	CAP100	0.1uF
DA8	1N4148	
DA9	1N4148	
DA10	1N4148	
DA11	1N4148	
DA12	1N4148	
DA13	1N4148	
DA14	1N4148	
DA15	1N4148	
DB	1N4148	
DG	1N4148	
DGB	1N4148	
DGG	1N4148	
DGI	1N4148	
DGO	1N4148	

RefDes	Name	Value
DGR	1N4148	
DK6+	1N4148	
DK6-	1N4148	
DR	1N4148	
DTO0	1N4148	
DTO1	1N4148	
F0	74LS74	
F1	74LS74	
F2	74LS74	
F3	74LS377	
F4	74LS165	
F5	74LS377	
F6	74LS377	
G	74LS04	
G0	74LS00	
G1	74LS00	
G2	74LS08	
G3	74LS08	
G4	74LS02	
G5	74LS32	
G6	74LS32	
G7	74LS32	
G8	74LS86	
G9	74LS20	
JAO	Audio Jack	
JEXT	CON3x32	
JKB	CON2x10	
JPWR	Power Jack	
JRGBI	DB9	
JTI	Audio Jack	
JTO	Audio Jack	
JVGA	D-Sub15	
L	74LS373	
M	27C128	
M0	64Kx1	
M1	64Kx1	

RefDes	Name	Value
M2	64Kx1	
M3	64Kx1	
M4	64Kx1	
M5	64Kx1	
M6	64Kx1	
M7	64Kx1	
N0	74LS193	
N1	74LS193	
N2	74LS193	
N3	74LS193	
N4	74LS193	
N5	74LS93	
RAO	RES300	10K
RB	RES300	470
RBR	RES300	10K
RCAS	RES300	33
RCLK	RES300	330
RD	RES300	220
RD0	RES300	4K7
RD1	RES300	4K7
RD2	RES300	4K7
RD3	RES300	4K7
RD4	RES300	4K7
RD5	RES300	4K7
RD6	RES300	4K7
RD7	RES300	4K7
RG	RES300	470
RGA	RES300	470
RGND	RES300	2K2
RGO	RES300	10K
RHR	RES300	10K
RI	RES300	680
RK0	RES300	4K7
RK1	RES300	4K7
RK2	RES300	4K7
RK3	RES300	4K7

RefDes	Name	Value
RK4	RES300	4K7
RK5	RES300	4K7
RK6	RES300	220
RMUX	RES300	220
RNMI	RES300	10K
RPR	RES300	10K
RQ3	RES300	10K
RR	RES300	470
RRAS	RES300	33
RS	RES300	270
RTI	RES300	220
RTO	RES300	10K
RVB	RES300	1K
RVC	RES300	75
RVCC	RES300	120
RVG	RES300	240
RVI	RES300	2K2
RVR	RES300	470
RWE	RES300	33
RX0	RES300	680
RX1	RES300	680
SPKR	SPKR30	
SPKRTI	SPKR16	
SW	SWITCH	
T	PN2222A	
TS	74LS14	
X0	74LS157	
X1	74LS153	
X2	74LS153	
X3	74LS153	
X4	74LS153	
X5	74LS157	
X6	74LS157	
XTAL	XTAL	14MHz
Z	Z80A	

Table 2: Bill of Materials

2.15 Netlist

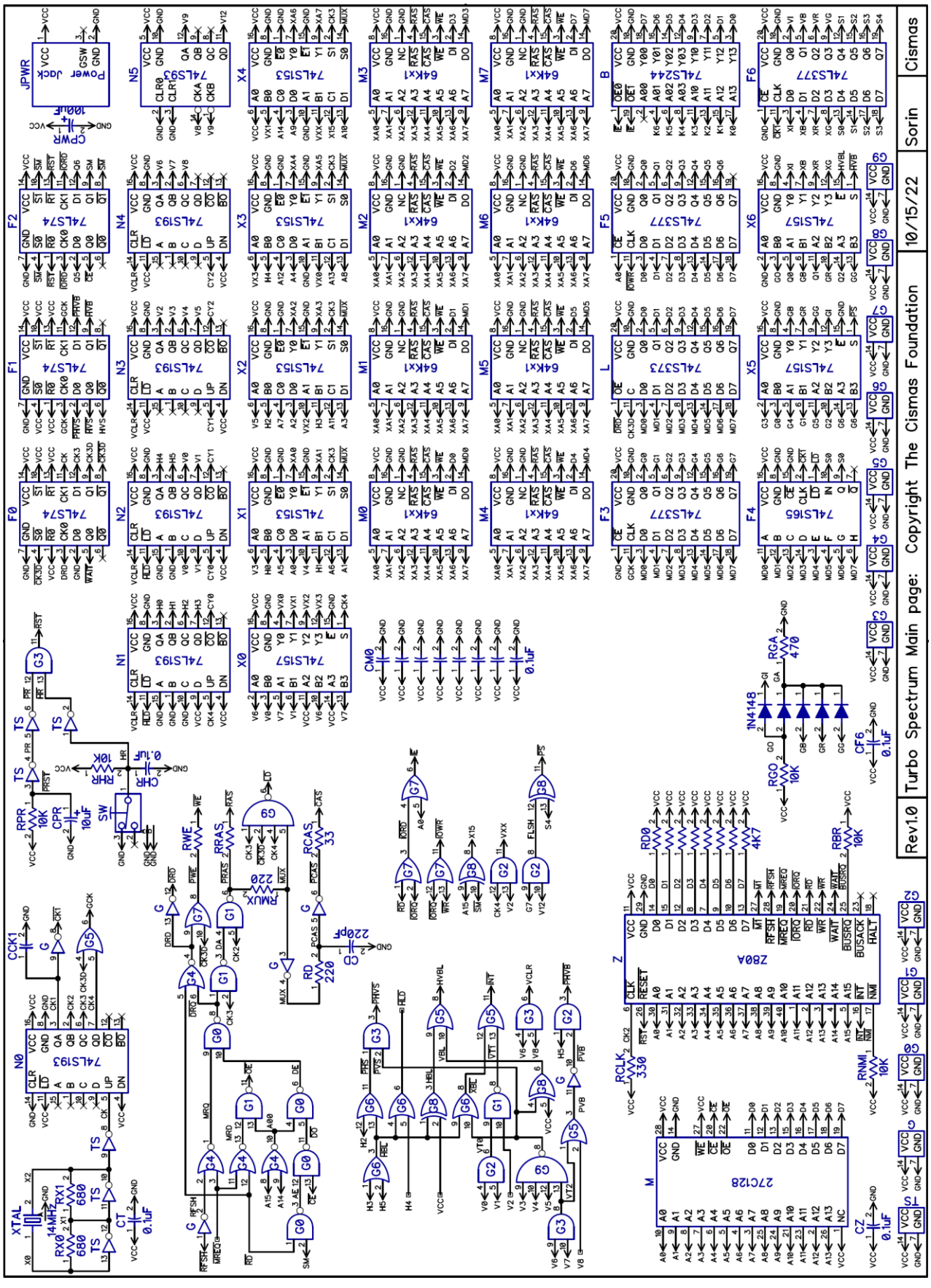
```
A0      F5.1 G7.5 JEXT.51 JEXT.50 M.10 X1.3 Z.30
A00     G0.4 G1.13 G4.10
A1      JEXT.54 JEXT.53 M.9 X1.13 Z.31
A2      JEXT.57 JEXT.56 M.8 X2.3 Z.32
A3      JEXT.60 JEXT.59 M.7 X2.13 Z.33
A4      JEXT.63 JEXT.62 M.6 X3.3 Z.34
A5      JEXT.66 JEXT.65 M.5 X1.4 Z.35
A6      JEXT.69 JEXT.68 M.4 X1.12 Z.36
A7      JEXT.72 JEXT.71 M.3 X2.4 Z.37
A8      DA8.1 JEXT.75 JEXT.74 M.25 X3.13 Z.38
A9      DA9.1 JEXT.78 JEXT.77 M.24 X4.3 Z.39
A10     DA10.1 JEXT.81 JEXT.80 M.21 X4.13 Z.40
A11     DA11.1 JEXT.84 JEXT.83 M.23 X2.12 Z.1
A12     DA12.1 JEXT.87 JEXT.86 M.2 X3.4 Z.2
A13     DA13.1 JEXT.90 JEXT.89 M.26 X3.12 Z.3
A14     DA14.1 G4.9 JEXT.93 JEXT.92 X4.4 Z.4
A15     DA15.1 G4.8 G8.9 JEXT.96 JEXT.95 Z.5
AE      G0.3 G0.12
AO      JAO.2 RAO.2
B       CB.1 DB.1 JVGA.3 RB.1
CK      F0.11 JEXT.10 JEXT.7 N0.5 TS.8
CK1     CCK1.1 G.9 JEXT.16 JEXT.13 N0.3
CK2     G1.5 JEXT.9 JEXT.8 N0.2 RCLK.2 Z.6
CK3     F0.12 G1.2 G9.1 JEXT.22 JEXT.19 N0.6 X1.2 X2.2 X3.2 X4.2
CK4     G2.12 G5.5 G9.4 JEXT.28 JEXT.25 N0.7 N1.5 X0.1
CK3D    F0.9 G5.4 JEXT.34 JEXT.31 L.11
CY0     N1.12 N2.5
CY1     N2.12 N3.5
CY2     N3.12 N4.5
D0      B.3 F5.3 JEXT.46 JEXT.43 L.2 M.11 M0.2 RD0.1 Z.14
D1      B.5 F5.4 JEXT.52 JEXT.49 L.5 M.12 M1.2 RD1.1 Z.15
D2      B.7 F5.7 JEXT.58 JEXT.55 L.6 M.13 M2.2 RD2.1 Z.12
D3      B.9 F5.8 JEXT.64 JEXT.61 L.9 M.15 M3.2 RD3.1 Z.8
D4      B.12 F5.13 JEXT.70 JEXT.67 L.12 M.16 M4.2 RD4.1 Z.7
D5      B.14 F5.14 JEXT.76 JEXT.73 L.15 M.17 M5.2 RD5.1 Z.9
D6      B.16 F5.17 JEXT.82 JEXT.79 L.16 M.18 M6.2 RD6.1 Z.10
D7      B.18 F5.18 JEXT.88 JEXT.85 L.19 M.19 M7.2 RD7.1 Z.13
DA      G1.3 G1.4
DE      G0.6 G0.10
DRD     F0.3 G.13 G4.4 G7.9
DTO     DTO0.2 DTO1.1 RQ3.1 RTO.2
FLSH    G2.8 G8.12
G       CG.1 DG.1 JVGA.2 RG.1
G0      F3.2 X5.3
G1      F3.5 X5.6
G2      F3.6 X5.10
G3      F3.9 X5.2
G4      F3.12 X5.5
G5      F3.15 X5.11
G6      F3.16 X5.13 X5.14
G7      F3.19 G2.9
GA      DGB.1 DGG.1 DGO.1 DGR.1 RGA.1
GB      DGB.2 X5.4 X6.6
GCK     F1.3 F1.11 F3.11 G5.6
GG      DGG.2 X5.9 X6.13
GI      DGI.1 X5.12
GND     B.10 CB.2 CCK1.2 CD.2 CF6.2 CG.2 CHR.2 CI.2 CM0.2 CM1.2
GND     CM2.2 CM3.2 CM4.2 CM5.2 CM6.2 CM7.2 CPR.2 CPWR.2 CR.2 CT.2
GND     CZ.2 DK6-.2 DTO0.1 DTO1.2 F0.2 F0.7 F1.7 F2.7 F3.1 F3.10
GND     F4.15 F4.8 F5.10 F6.1 F6.10 G.7 G0.7 G1.7 G2.7 G3.7
GND     G4.7 G5.7 G6.7 G7.7 G8.7 G9.7 JAO.1 JAO.A JAO.B JEXT.6
GND     JEXT.5 JEXT.4 JEXT.A JEXT.B JKB.16 JKB.A JKB.B JPWR.2 JRGBI.2 JRGBI.A
```

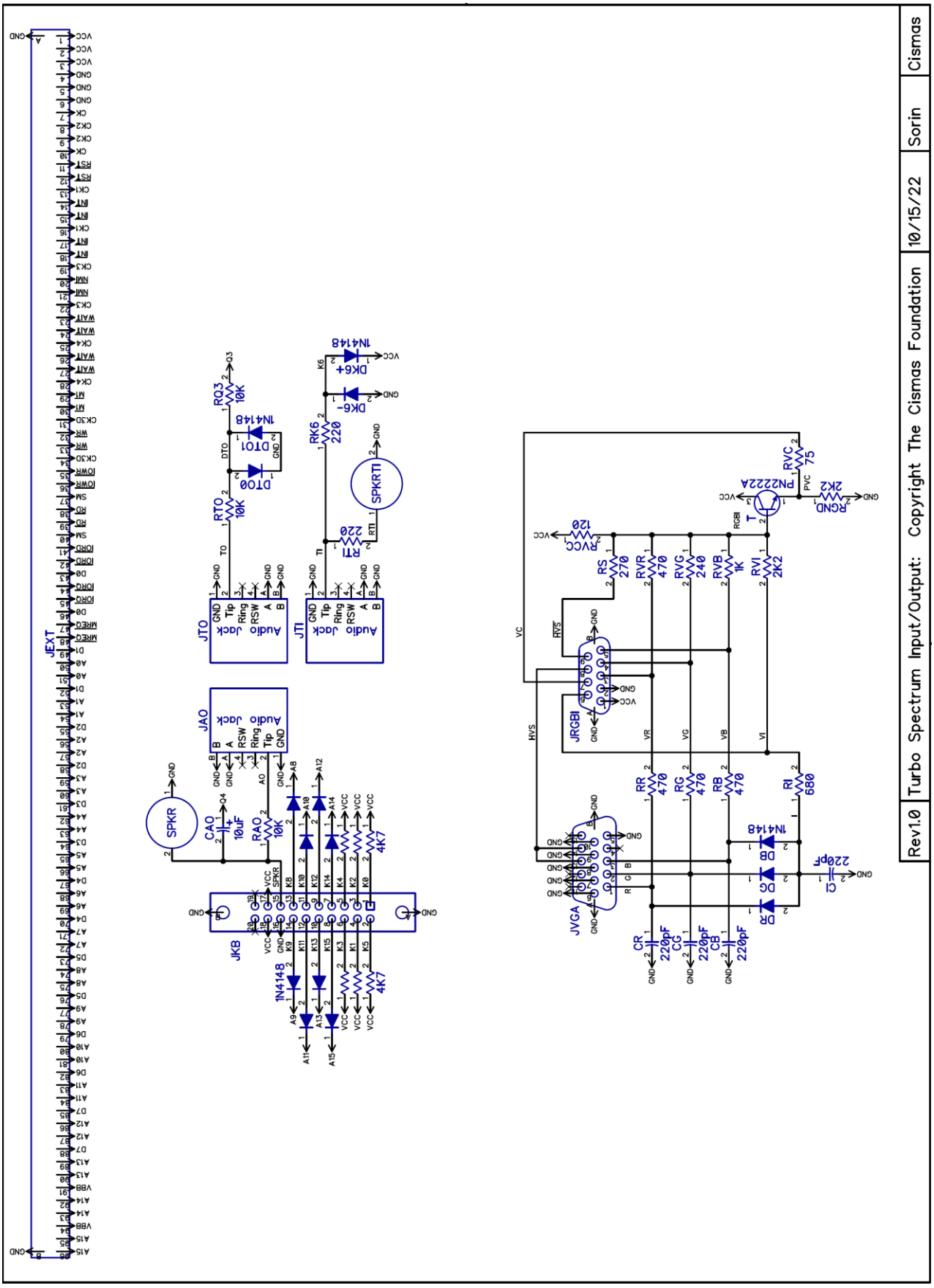
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GND     JRGBI.B JTI.1 JTI.A JTI.B JTO.1 JTO.A JTO.B JVGA.5 JVGA.6 JVGA.7
GND     JVGA.8 JVGA.9 JVGA.10 JVGA.A JVGA.B L.10 M.14 M0.16 M0.1 M1.16
GND     M1.1 M2.16 M2.1 M3.16 M3.1 M4.16 M4.1 M5.16 M5.1 M6.16
GND     M6.1 M7.16 M7.1 N0.14 N0.8 N1.1 N1.10 N1.15 N1.8 N2.1
GND     N2.15 N2.8 N3.8 N4.8 N5.2 N5.3 N5.10 RGA.2 RGND.2 SPKR.1
GND     SPKRTI.2 SW.3 SW.A SW.B TS.7 X0.15 X0.8 X1.1 X1.15 X1.8
GND     X2.1 X2.15 X2.8 X3.1 X3.10 X3.15 X3.8 X4.1 X4.10 X4.15
GND     X4.8 X5.15 X5.8 X6.2 X6.8 XTAL.A Z.29
GO      DGI.2 DGO.2 RGO.2 X6.3
GR      DGR.2 X5.7 X6.10
H0      N1.3 X1.5
H1      N1.2 X1.11
H2      G6.12 N1.6 X2.5
H3      G6.1 N1.7 X2.11
H4      G6.5 N2.3 X3.5
H5      G2.1 G6.2 N2.2
HBL     G5.9 G8.3
HR      CHR.1 RHR.2 SW.1 TS.1
HVBL    G5.8 X6.15
HVS     F1.6 JRGBI.8 JVGA.13 JVGA.14
I       CI.1 DB.2 DG.2 DR.2 RI.1
K0      B.17 JKB.1 RK0.2
K1      B.15 JKB.4 RK1.2
K2      B.13 JKB.3 RK2.2
K3      B.11 JKB.6 RK3.2
K4      B.8 JKB.5 RK4.2
K5      B.6 JKB.2 RK5.2
K6      B.4 DK6+.2 DK6-.1 RK6.2
K8      DA8.2 JKB.13
K9      DA9.2 JKB.14
K10     DA10.2 JKB.11
K11     DA11.2 JKB.12
K12     DA12.2 JKB.9
K13     DA13.2 JKB.10
K14     DA14.2 JKB.7
K15     DA15.2 JKB.8
MD0     F3.3 F4.11 L.3 M0.14
MD1     F3.4 F4.12 L.4 M1.14
MD2     F3.7 F4.13 L.7 M2.14
MD3     F3.8 F4.14 L.8 M3.14
MD4     F3.13 F4.3 L.13 M4.14
MD5     F3.14 F4.4 L.14 M5.14
MD6     F3.17 F4.5 L.17 M6.14
MD7     F3.18 F4.6 L.18 M7.14
MRD     G1.12 G4.13
MRQ     G0.9 G4.1
MUX     G.4 RD.1
VB      F6.5 JRGBI.5 RB.2 RVB.2
VG      F6.9 JRGBI.4 RG.2 RVG.2
VR      F6.6 JRGBI.3 RR.2 RVR.2
VC      JRGBI.7 RVC.2
VI      F6.2 JRGBI.6 RI.2 RVI.2
RGBI    RS.1 RVB.1 RVCC.2 RVG.1 RVI.1 RVR.1 T.2
PCAS    CD.1 G.5 RD.2
PR      TS.4 TS.5
PVB     G.11 G5.3
PVC     RGND.1 RVC.1 T.1
Q0      F5.2 X6.5
Q1      F5.5 X6.11
Q2      F5.6 X6.14
Q3      F5.9 RQ3.2
Q4      CAO.1 F5.12
Q5      F2.2 F5.15
Q6      F2.12 F5.16
R       CR.1 DR.1 JVGA.1 RR.1
RFSH    G.2 G4.2
```

```
RTI     RTI.2 SPKRTI.1
S0      F4.10 F4.9 F6.13
S1      F6.12 F6.14
S2      F6.15 F6.17
S3      F6.16 F6.18
S4      F6.19 G8.13
SM      F2.9 G0.2 JEXT.40 JEXT.37
SPKR    CAO.2 JKB.15 RAO.1 SPKR.2
TI      JTI.2 RK6.1 RTI.1
TO      JTO.2 RTO.1
V0      G2.4 N2.6 N2.10 X0.3
V1      G2.5 N2.7 N2.9 X0.6
V2      G1.10 G2.13 N3.3
V3      G9.9 N3.2 X1.6
V4      G9.10 N3.6 X1.10
V5      G9.12 N3.7 X2.6
V6      G3.4 G3.9 N4.3 X0.10 X0.2
V7      G3.10 N4.2 X0.13 X0.5
V8      G3.5 G5.2 N4.6 N5.14
V9      N5.1 N5.12
V12     G2.10 N5.11
VBB     JEXT.94 JEXT.91
VBL     G5.10 G8.6
VCC     B.20 CF6.1 CM0.1 CM1.1 CM2.1 CM3.1 CM4.1 CM5.1 CM6.1 CM7.1
VCC     CPWR.1 CT.1 CZ.1 DK6+.1 F0.1 F0.10 F0.13 F0.14 F1.1 F1.4
VCC     F1.10 F1.13 F1.14 F2.14 F3.20 F4.16 F5.20 F6.20 G.14 G0.14
VCC     G1.14 G2.14 G3.14 G4.14 G5.14 G6.14 G7.14 G8.2 G8.5 G8.14
VCC     G9.14 JEXT.3 JEXT.2 JEXT.1 JKB.17 JKB.18 JPWR.1 JRGBI.1 L.20 M.27
VCC     M.28 M.1 M0.8 M1.8 M2.8 M3.8 M4.8 M5.8 M6.8 M7.8
VCC     N0.4 N0.11 N0.16 N1.4 N1.9 N1.16 N2.4 N2.16 N3.4 N3.11
VCC     N3.16 N4.4 N4.11 N4.16 N5.5 RBR.1 RCLK.1 RD0.2 RD1.2 RD2.2
VCC     RD3.2 RD4.2 RD5.2 RD6.2 RD7.2 RGO.1 RHR.1 RK0.1 RK1.1 RK2.1
VCC     RK3.1 RK4.1 RK5.1 RNMI.2 RPR.2 RVCC.1 T.3 TS.14 X0.11 X0.14
VCC     X0.16 X1.16 X2.16 X3.16 X4.6 X4.16 X5.16 X6.16 Z.11
VCLR    G3.6 N1.14 N2.14 N3.14 N4.14
VT0     G1.9 G2.6
VT2     G3.8 G5.1 G9.13
VX0     X0.4 X3.11
VX1     X0.7 X4.5
VX2     X0.9 X2.10
VX3     X0.12 X3.6
VXX     G2.11 X4.11
X0      RX0.1 TS.13 XTAL.1
X1      RX0.2 RX1.1 TS.11 TS.12
X2      RX1.2 TS.9 TS.10 XTAL.2
X15     G8.8 X4.12
XA0     M0.5 M1.5 M2.5 M3.5 M4.5 M5.5 M6.5 M7.5 X1.7
XA1     M0.7 M1.7 M2.7 M3.7 M4.7 M5.7 M6.7 M7.7 X1.9
XA2     M0.6 M1.6 M2.6 M3.6 M4.6 M5.6 M6.6 M7.6 X2.7
XA3     M0.12 M1.12 M2.12 M3.12 M4.12 M5.12 M6.12 M7.12 X2.9
XA4     M0.11 M1.11 M2.11 M3.11 M4.11 M5.11 M6.11 M7.11 X3.7
XA5     M0.10 M1.10 M2.10 M3.10 M4.10 M5.10 M6.10 M7.10 X3.9
XA6     M0.13 M1.13 M2.13 M3.13 M4.13 M5.13 M6.13 M7.13 X4.7
XA7     M0.9 M1.9 M2.9 M3.9 M4.9 M5.9 M6.9 M7.9 X4.9
XB      F6.4 X6.7
XG      F6.8 X6.12
XI      F6.3 X6.4
XR      F6.7 X6.9
~BUSRQ  RBR.2 Z.25
~CAS    M0.15 M1.15 M2.15 M3.15 M4.15 M5.15 M6.15 M7.15 RCAS.1
~CE     F2.5 G0.13 M.20
~CK1    F4.2 F6.11 G.8
~CK3D   F0.4 F0.8 G7.10 G9.2
~DO     G0.5 G0.11
~DRD    G.12 L.1
~DRQ    G0.8 G1.1 G4.6
```

```
~HBL    G6.3 G6.4 G6.9 G6.13 G8.1
~HLD    G6.6 N1.11 N2.11
~HR     G3.13 TS.2
~HVB    F1.9 X6.1
~HVS    F1.5 JRGBI.9 RS.2
~IE     B.1 B.19 G7.6
~INT    G5.11 JEXT.18 JEXT.17 JEXT.15 JEXT.14 Z.16
~IORD   F2.3 F2.11 G7.3 G7.4 JEXT.42 JEXT.41
~IORQ   G7.2 G7.12 JEXT.45 JEXT.44 Z.20
~IOWR   F5.11 G7.11 JEXT.36 JEXT.35
~LD     F4.1 G9.6
~M1     JEXT.30 JEXT.29 Z.27
~MREQ   G4.3 G4.11 JEXT.48 JEXT.47 Z.19
~MUX    G.3 G9.5 RMUX.1 X1.14 X2.14 X3.14 X4.14
~NMI    JEXT.21 JEXT.20 RNMI.1 Z.17
~OE     G1.11 M.22
~PCAS   G.6 RCAS.2
~PHS    G3.1 G6.11
~PHVB   F1.12 G2.3
~PHVS   F1.2 G3.3
~PR     G3.12 TS.6
~PRAS   G1.6 RMUX.2 RRAS.2
~PRST   CPR.1 RPR.1 TS.3
~PS     G8.11 X5.1
~PVB    G.10 G2.2
~PVS    G3.2 G6.10 G8.4 G9.8
~PWE    G7.8 RWE.2
~RAS    M0.4 M1.4 M2.4 M3.4 M4.4 M5.4 M6.4 M7.4 RRAS.1
~RD     G0.1 G4.5 G4.12 G7.1 JEXT.39 JEXT.38 Z.21
~RFSH   G.1 Z.28
~RST    F2.1 F2.13 G3.11 JEXT.12 JEXT.11 Z.26
~SM     F2.4 F2.8 F2.10 G8.10
~VT1    G1.8 G5.13
~WAIT   F0.5 JEXT.27 JEXT.26 JEXT.24 JEXT.23 Z.24
~WE     M0.3 M1.3 M2.3 M3.3 M4.3 M5.3 M6.3 M7.3 RWE.1
~WR     G7.13 JEXT.33 JEXT.32 Z.22
~XBL    G5.12 G6.8
```

2.16 Schematics


Rev1.0
Turbo Spectrum Main page: Copyright The Cismas Foundation
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1N4148
2N2222A
Audio Jack
Tip
Ring
RSW
RTO
RK6
RT1
RQ3
CAO
RAO
RS
RVR
RVG
RVB
RVI
RVC
RGND
RR
RG
RB
RI
CR
CG
CB
CI
220pF
10uF
10K
4K7
270
470
240
1K
2K2
75
680
120
220
VCC
GND
HVS
VC
VR
VG
VB
VI
PVC
RGBI
RTS

2.17 Layout

Figure 30 through Figure 33 in this section are the rebuilt Turbo Spectrum PCB at 1:1 scale. Same design rules have been used as in the original 2-layer PCB, with 0.8 mm IC pins holes and 1.5 mm pads, with a single trace allowed in between IC pins, and minimum trace width and clearance of 0.3 mm. The layout is dense, and it required mostly manual routing, same as in the old days. DipTrace Schematic and PCB Design Software (https://diptrace.com) was used, which provided a discounted price for this non-profit project.

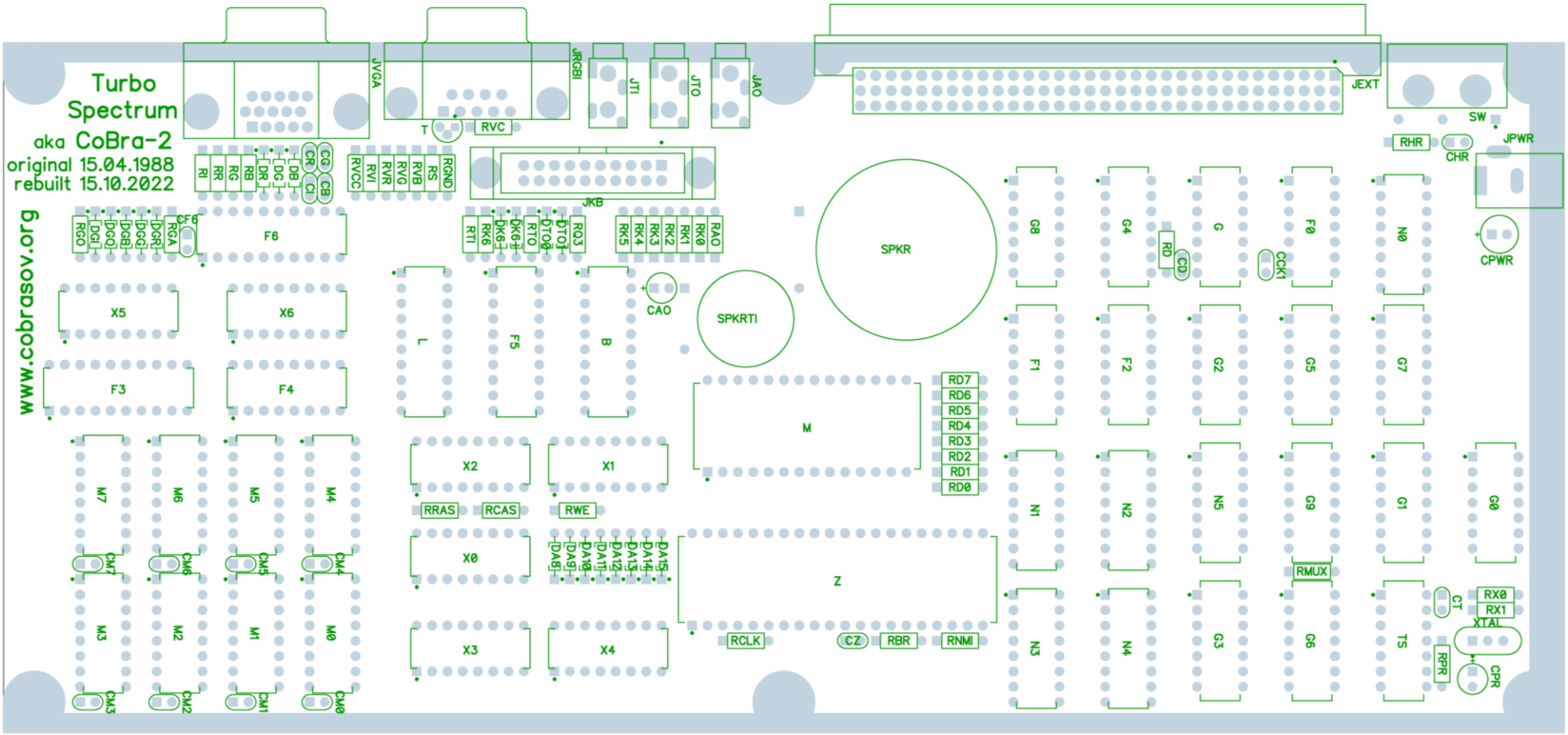


Figure 30: PCB Components

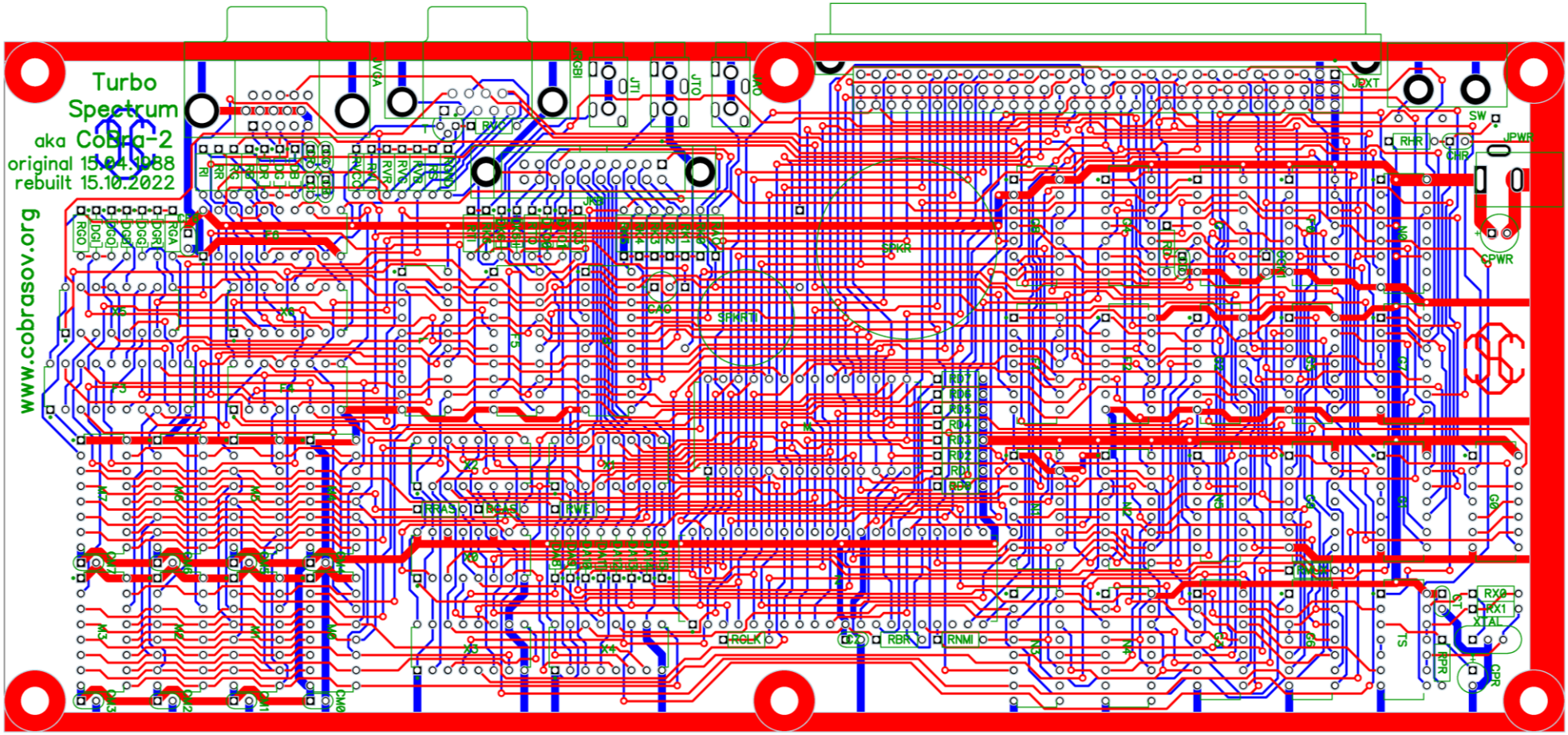


Figure 31: PCB All Layer

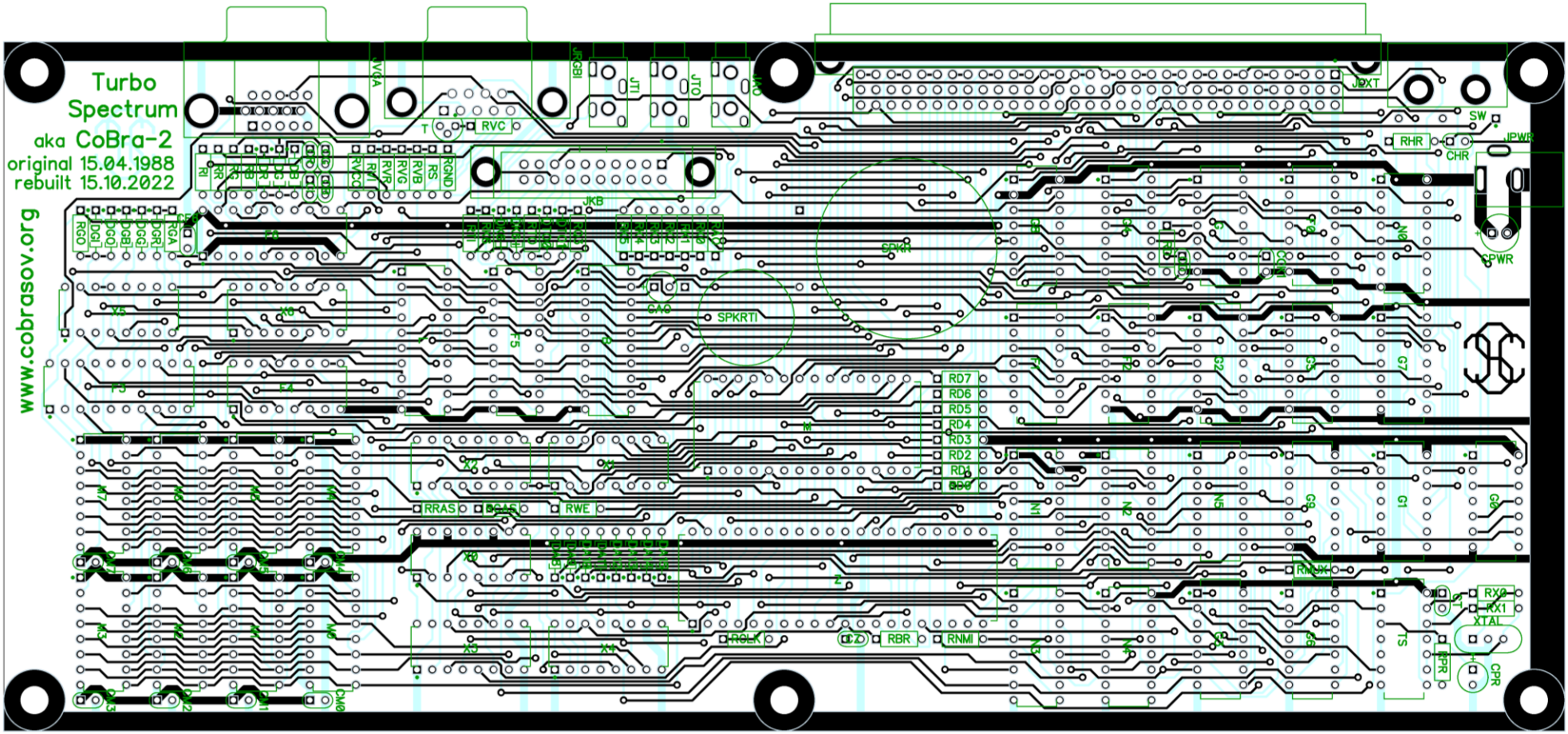


Figure 32: PCB Top Layer

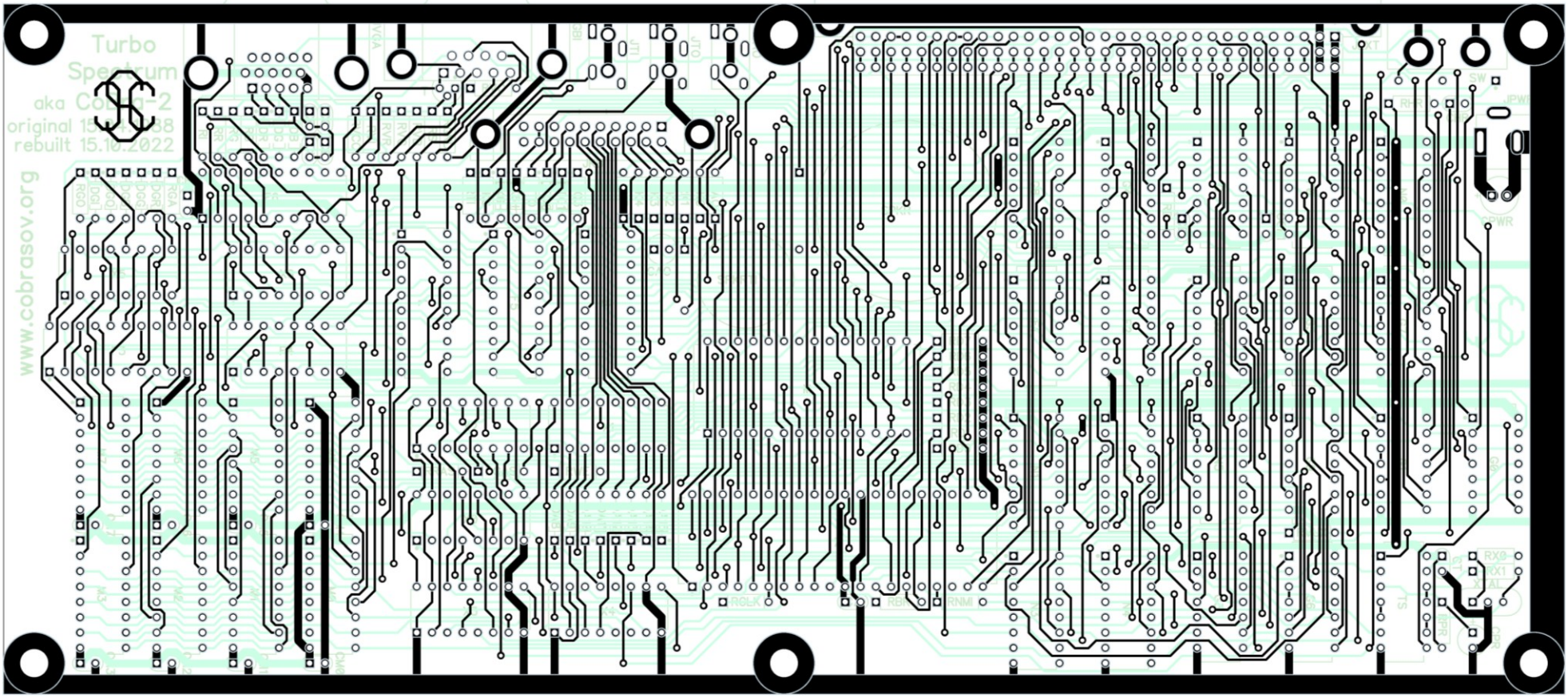


Figure 33: PCB Bottom Layer

3 Boot Code

After reset, Turbo Spectrum boot code starts executing in EEPROM Mode. The 16Kx8 EEPROM contains the Turbo Specrum logo generation code and a compacted Sinclair ZX Spectrum ROM. Figure 34 shows the TS boot screen, where every pixel is shown. The thin lines are Spectrum 8x8 character demarcation lines; they do not appear on the screen. The bright black at the left of the color test pattern does not appear on the screen because of the logic that forces bright black to be displayed as black.

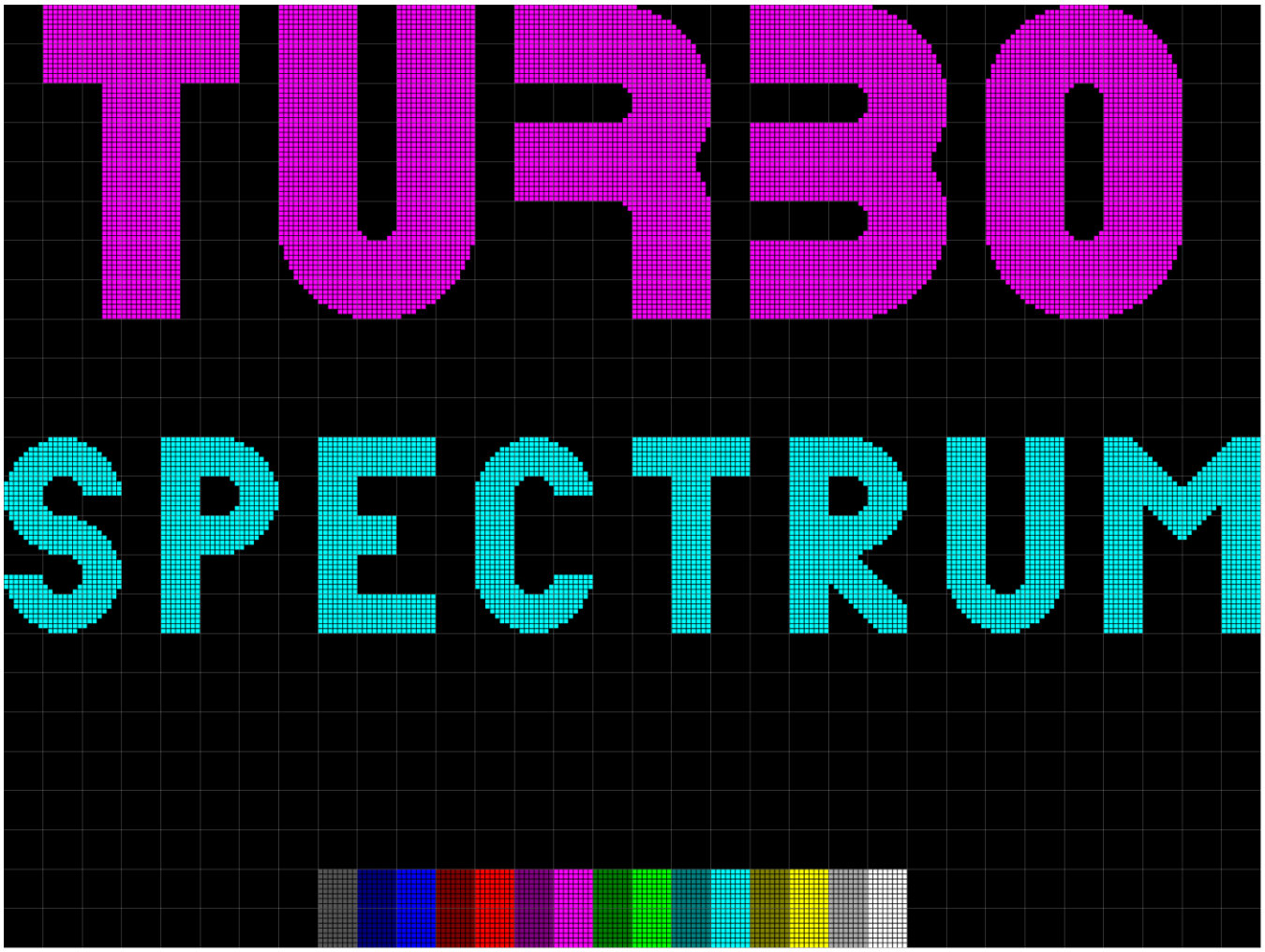


Figure 34: Turbo Spectrum Logo

Code 1 on page 60 is the TS boot source code with comments describing how it works. After some initializations are done, including setting Stack Pointer to `0e800h`, clearing the screen, and setting the color bar at the bottom of the screen, `7b4h` bytes are copied from EEPROM addresses `0004ch` to `007ffh` to DRAM addresses `0dfffh` to `0e7b2h`, then a jump is executed to `0e004h`, where a transition is made to the 64K DRAM Mode.

The Turbo Spectrum logo is stored compressed to reduce the code and data size. The total size of code and data for logo generation and rotation is 1231 bytes. The original Sinclair ZX Spectrum ROM had 1170 unused bytes, from `0386eh` to `03cffh`, which are not sufficient for the logo code and data. To make room in the 16KB EEPROM, the characters in the Sinclair ZX Spectrum ROM are stored using a light compression.

The TS logo is generated from the 9 special characters stored at `C_Base` (`0e2edh`) and their rotations by a multiple of 90 degrees (see Figure 35), using the data structure that starts at `C_struct` (`0e335h`), which consists of 9 sections, one for each special character. There are 61 unused bytes that start at `Spare`, from `0e491h` to `0e4cdh`, that are set to `0xff`.

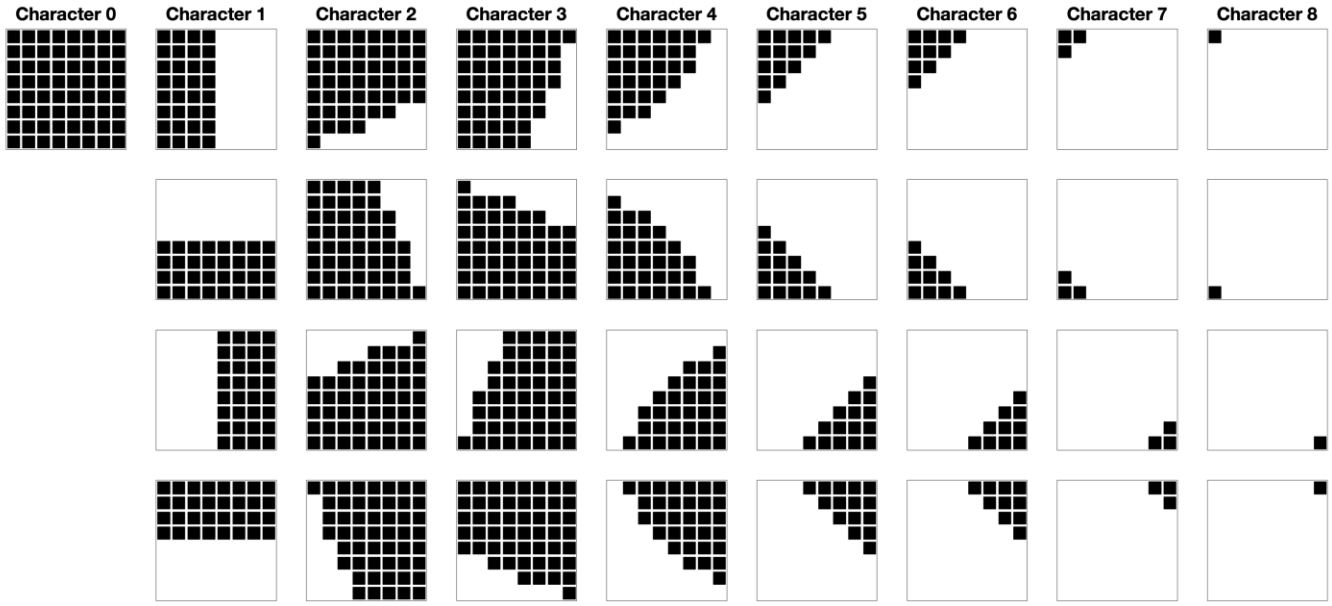


Figure 35: Turbo Spectrum Logo Special Characters

The TURBO part of the logo has 8 Spectrum characters rows, 8*8 = 64 screen lines, and is generated in `T_buffer`, addresses `09900h` to `0b0ffh`; the SPECTRUM part has 5 Spectrum characters rows, 5*8 = 40 screen lines, and is generated in `S_buffer`, addresses `0b100h` to `0bfffh`. Both the TURBO and SPECTRUM part use 96 bytes lines, 3 times longer than the 32 characters on the Spectrum screen. Same line interleaving is used as for the Spectrum bitmap, so going from the current character line address to the next, the high address is incremented by 3 instead of 1 for TURBO, and for SPECTRUM, the address is incremented by 5 * 96 = 480 = 1e0h. Figure 36 shows the memory organization for TS logo generation.

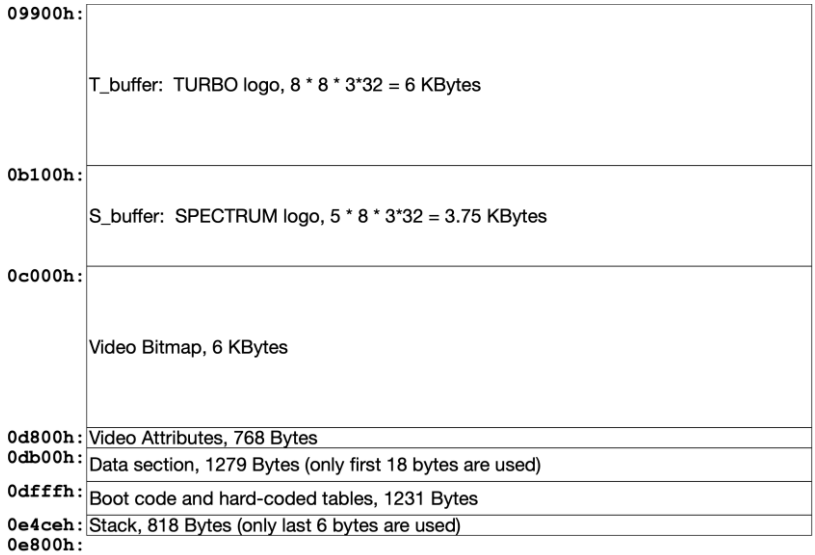


Figure 36: Bitmap for Turbo Spectrum Generation

The code from `0e006h` to `0e07eh` generates the TS logo in the left third of the 96 characters rows, the code from `0e07fh` to `0e092h` copies it to the right third, leaving the middle third black, and the code from `0e093h` to `0e0c2h` copies `T_buffer` to addresses `03000h` to `047ffh`, and generates 4 shifted versions of the TURBO and 8 of SPECTRUM; Figure 37 shows the memory map used. The shifted versions are generated by calling the `Rotate` subroutine at addresses `0e0d0h` to `0e0ech`, once for SPECTRUM and twice for TURBO.

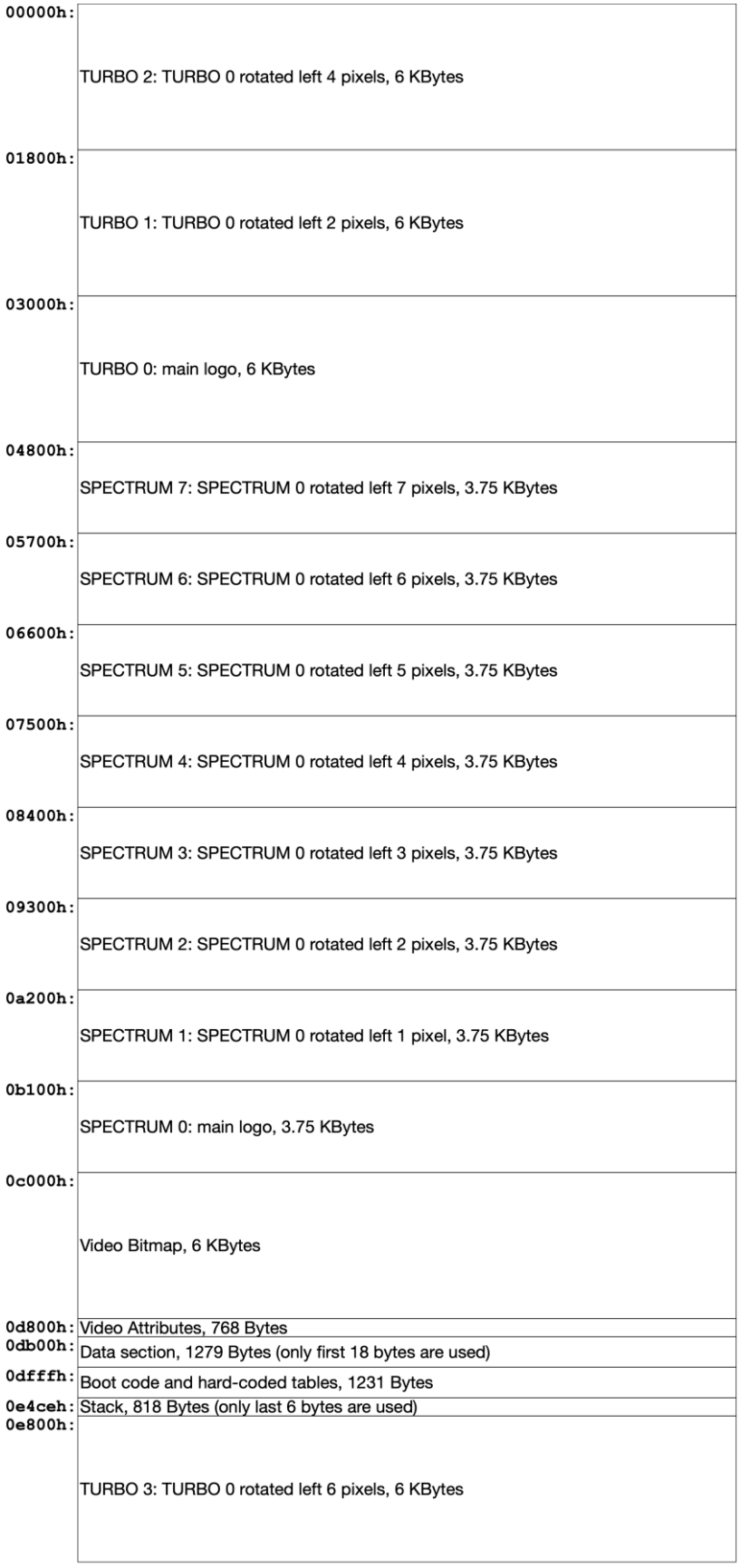


Figure 37: Memory Map for Logo Generation

Two nested loops are used to generate the TS logo. The outer loop goes through the 9 special characters and the inner loop through the 4 rotations, except for the first character. Each of the 9 `C_struct` data structure sections, has 4 subsections, except the first one, each subsection starting with a 16-bit base address of the first character instance, followed by the number of instances `n`, and a list of `n-1` increments; the increments are ordered in ascending order, first the TURBO increments, then the SPECTRUM increments.

The special character is first copied at the beginning of the data section, at address `C_buffer0` (`0db00h`). Each of the inner loops goes through 2 stages. In the first stage, two more nested loops are used, the outer loop sequencing through all increments, and the inner loop sequencing through the character lines. In the second stage, the character at `C_buffer0` is copied to `C_buffer1` (`0db08h`), then the character at `C_buffer1` is rotated by 90 degrees in trigonometric order and is written to `C_buffer0`.

In the inner loop of the second stage, each byte from `C_buffer0` is XOR-ed with the corresponding byte of the character in `T_buffer` or `S_buffer`, determined by the increment; this feature (XOR instead of copy) is used to generate more complex characters by superposition of multiple special characters. The line-to-line address increment is `300h` for TURBO and `1e0h` for SPECTRUM, but there is no explicit demarcation between lines belonging to the TURBO and SPECTRUM sections. To determine the address of the next line, the high address is incremented first by 1, and a comparison is made at address `0e02bh`; if the high address is lower than `0b2h`, this is a TURBO special character and the high address is incremented two more times; otherwise, this is a SPECTRUM special character and `0e0h` is added to the address.

3.1 Interrupt Handler

The rotation of the logo is done on the long 96 characters rows; TURBO is rotated 2 pixels per frame interrupt and SPECTRUM 1 pixel, so TURBO moves 2x faster. With the 3x longer lines, both TURBO and SPECTRUM completely disappear on the left edge before they reappear on the right edge.

The rotation is done on the frame interrupt to avoid tearing. TS uses Mode 2 Interrupt but does not include a CTC or other circuit that can drive the Z80 data bus during the interrupt. The Z80 documentation claims that in Mode 2, the interrupting device must supply the lower eight bits of the pointer where the address of the interrupt service routine is located, and the least-significant bit must be 0. That's not true, the least-significant bit does not need to be 0.

Although it is desirable for 16-bit addresses to be aligned, Z80 fetches the address of the interrupt service routine in 2 cycles, first the low byte at the address provided by the I register concatenated with the byte read during the interrupt, then it uses the internal 16-bit adder to compute the address of the high byte. TS uses this undocumented feature and does not drive anything during the interrupt and lets the pull-up resistors to provide the `0ffh` value.

The code from `0e0c3h` to `0e0cfh` initializes the rotation counters (bit counter B with 8 and character counter C with 64 modulo 256), loads `0dfh` to the I register and sets Z80 Interrupt Mode 2, then interrupts are enabled, and the `halt` instruction is executed, from where it can exit only on the interrupt. After the interrupt is serviced the boot code keeps looping until a key is pressed.

The interrupt handler is located at address `0e13ah`, which is stored at `0dfffh`, from where Z80 reads it on the interrupt, then starts executing it. The `0e13ah` to `0e158h` section implements the SPECTRUM rotation, the `0e159h` to `0e165h` section tests if a key was pressed (jumps at `Key`, `0e19ah`, in case a key was pressed), and the `0e166h` to `0e199h` section implements the TURBO rotation.

The rotation is done by copying from the SPECTRUM or TURBO area selected by the value in the B register a 32 byte wide section indexed by the value in the C register to the Video Bitmap area. The copying is implemented with the `Copy` subroutine at addresses `0e0edh` to `0e139h`, which copies one line at a time using 16 (`pop`, `push`) instruction pairs.

3.2 Motivation for the implementation choices

The reason the special `Copy` subroutine is used instead of the `ldir` or `lddr` instructions and most of the 64K DRAM is used for the 8 shifted versions of SPECTRUM and 4 of TURBO instead of doing the shift on the frame interrupt, is execution speed. The number of bytes that need to be shifted per frame interrupt are (8*8 + 5*8) * 32 = 104 * 32 = 3328 bytes. An `ldir` or `lddr` instruction takes 21 cycles per byte to execute and does 2 memory reads which adds in average 1 WAIT state for DRAM read access. Z80 runs at 3.5 MHz, so the time it takes for copy with the `ldir` or `lddr` instruction is 3328 * 22 / 3.5 = 20.919 ms, more than the 20.48 ms frame time.

The (pop, push) instruction pair takes 11 + 12 cycles per 2 bytes to execute and does 4 memory reads which adds in average 2 WAIT states for DRAM read access. The time it takes for copy with the (pop, push) instruction pair is 3328/2 * 25 / 3.5 = 11.886 ms, almost 2x faster. There are more instructions in the Copy subroutine to be executed in the interrupt handler, which does not leave sufficient time for the shift to be implemented on the frame interrupt.

3.3 Spectrum Mode

When a key is pressed, the interrupt handler jumps to Key. The 0e19ah to 0e1a9h section of the code checks if the T or S keys have been pressed; if a different key was pressed, the 0e1aah to 0e1f2h section of the code does the Spectrum code unpacking. It transitions Turbo Spectrum to EEPROM Mode, copies the Spectrum code from EEPROM addresses 00558h to 03dc5h to DRAM addresses 08000h to 0b86dh, then fills DRAM addresses 0b86eh to 0bcffh with 0ffh, then adds the blank character (fills 0bd00h to 0bd07h with 000h), and then unpacks the remaining 95 characters.

Most Spectrum characters have the first and last line 000h, so these 2 bytes are not stored to the TS boot EEPROM, saving 190 bytes. There are 12 exceptions, the first being at address 0bd27h, for which the Bytes table is provided with the 12 non-zero bytes and the relative address increment for the last 11.

After the unpacking is done, the code jumps to Done, 0e2dfh, that initializes the Stack Pointer to 00000h, does an out instruction that sets black border and the O5 and O6 bits to 1, then pushes 0fedbh, to the stack, which ends up as 0dbfeh at address 0fffeh, which is the in a,(0feh) instruction, then jumps at 0fffeh from where the in a,(0feh) instruction is executed that sets TS in Spectrum Mode, then continues execution from address 00000h.

3.4 T and S Modes

If the T key is pressed while the boot code is executing, the execution jumps to 0e20ah, and if the S key is pressed, the execution jumps to 0e20bh. It's not clear what the code from 0e20ah to 0e2deh does, it seems it wants to load something from the tape. I believe I wanted to add some extra features that I did not manage to implement before I left Romania. While in the T or S Modes, if the Space key is pressed, it returns where it left before the T or S key were pressed.

```
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; OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
; SOFTWARE.
;
; Turbo Spectrum boot code  v1.0  09/30/2022  sorin@cismas.org
; multi-line comments start with ";b", continue with ";|" and end with ";e"

        org     00000h

; After reset, Turbo Spectrum (TS) boot code starts executing in EEPROM Mode.
; The 16Kx8 EEPROM boot code contains the Turbo Spectrum logo generation and
; a compacted Sinclair ZX Spectrum ROM.  After some initializations are done,
; including setting Stack Pointer to 0e800h, clearing the screen, and setting
; the color bar at the bottom of the screen, 7b4h bytes are copied from EEPROM
; addresses 0004ch to 007ffh to DRAM addresses 0dfffh to 0e7b2h, then a jump
; is executed at 0e004h, where a transition is made to the 64K DRAM Mode.

; The Turbo Spectrum logo is stored compressed to reduce code and data size.
; The logo can be generated from 9 special characters and their rotations by
; a multiple of 90 degrees.  The total size of code and data for logo
; generation and rotation is 1231 bytes.  The original Sinclair ZX Spectrum
; ROM had 1170 unused bytes, 0386eh - 03cffh, which are not sufficient for
; the logo code and data. To make room in the 16KB EEPROM, the characters in
; the Sinclair ZX Spectrum ROM are stored using a light compression.

start:
        di              ;0000  f3         ;  disable interrupts
        ld sp,0e800h    ;0001  31 00 e8   ;  initialize SP=0e800h
        ld a,020h       ;0004  3e 20      ;b out(020feh)=020h
        out (0feh),a    ;0006  d3 fe      ;e set black border and O5 (input to !CE flop)
        xor a           ;0008  af         ;b clear (09900h)-(0daffh)
        ld hl,0daffh    ;0009  21 ff da   ;|
        ld (hl),a       ;000c  77         ;|
        ld de,0dafeh    ;000d  11 fe da   ;|
        ld bc,041ffh    ;0010  01 ff 41   ;|
        lddr            ;0013  ed b8      ;e
        ld hl,0dac7h    ;0015  21 c7 da   ;b add color bar in the center
        ld c,002h       ;0018  0e 02      ;e of the two bottom screen lines
l001ah:
        ld b,008h       ;001a  06 08      ;  8*2 characters <-- <--,
l001ch:                                   ;                       ^
        ld (hl),a       ;001c  77         ;  1st character INK <,  |
        inc hl          ;001d  23         ;                     ^  ^
        xor 040h        ;001e  ee 40      ;  set BRIGHT         |  |
        ld (hl),a       ;0020  77         ;  2nd character INK  ^  ^
        inc hl          ;0021  23         ;                    |  |
        xor 040h        ;0022  ee 40      ;  clear BRIGHT      ^  ^
        add a,008h      ;0024  c6 08      ;  next color        |  |
        djnz l001ch     ;0026  10 f4      ;  loop 8 times --> --'  ^
        ld l,0e7h       ;0028  2e e7      ;  character row below   |
        xor a           ;002a  af         ;  clear A               ^
        dec c           ;002b  0d         ;                        |
        jr nz,l001ah    ;002c  20 ec      ;  loop 2 times --> --> -'
        ld hl,0d800h    ;002e  21 00 d8   ;  color attributes base address
        ld a,043h       ;0031  3e 43      ;  magenta INK on black PAPER
        ld c,002h       ;0033  0e 02      ;  2 thirds of screen
l0035h:
        ld (hl),a       ;0035  77         ;  store color attribute <-,--,
        inc hl          ;0036  23         ;                          |  |
        djnz l0035h     ;0037  10 fc      ;  loop 1st 3rd of screen -'  ^
        ld a,045h       ;0039  3e 45      ;  cyan INK on black PAPER    |
        dec c           ;003b  0d         ;                             ^
        jr nz,l0035h    ;003c  20 f7      ;  loop 2nd 3rd of screen  -->'
        ld hl,0004ch    ;003e  21 4c 00   ;b copy from (0004ch)-(007ffh)
        ld de,0dfffh    ;0041  11 ff df   ;| to (0dfffh)-(0e7b2h)
        ld bc,007b3h+1  ;0044  01 b4 07   ;| 007b4h bytes
        ldir            ;0047  ed b0      ;e
        jp 0e004h       ;0049  c3 04 e0   ;  jump to (0e004h)
```

```
        org     09900h
T_buffer:

        org     0b100h
S_buffer:

        org     0c000h

        org     0db00h
C_buffer0:

        org     0db08h
C_buffer1:

        org     0db16h
HL_save:

        org     0dfffh

        defw    IntHandler      ;dfff   3a e1      ;

le001h:
        inc d           ;e001   14          ;b for TURBO only
        jr le035h       ;e002   18 31       ;e

; enter 64K DRAM Mode and generate TURBO SPECTRUM logo with 060h byte lines
        in a,(0feh)     ;e004   db fe       ;  !CE=1; A=in(045feh), rows KAf,KAd,KAc,KAb,KA9
        ld hl,C_struct  ;e006   21 35 e3    ;  Char structure Base address
        ld bc,00109h    ;e009   01 09 01    ;  1 rotation and 9 special characters
        exx             ;e00c   d9          ;
        ld hl,C_base    ;e00d   21 ed e2    ;  load HL with Special Character Base Address
le010h:
        ld de,C_buffer0 ;e010   11 00 db    ;b copy special character to (C_buffer0) <-- <,
        ld bc,00008h    ;e013   01 08 00    ;|                                           ^
        ldir            ;e016   ed b0       ;e                                           |
        exx             ;e018   d9          ;                                            ^
le019h:                                     ;                                            |
        push bc         ;e019   c5          ;  save BC <-- <-- <-- <-- <-- <-- <-- <-- <, ^
        ld e,(hl)       ;e01a   5e          ;b load DE with char start address          | |
        inc hl          ;e01b   23          ;|                                          ^ ^
        ld d,(hl)       ;e01c   56          ;|                                          | |
        inc hl          ;e01d   23          ;e                                          ^ ^
        ld c,(hl)       ;e01e   4e          ;  load C with number of char increments    | |
le01fh:                                     ;                                          ^ ^
        push hl         ;e01f   e5          ;  save HL <-- <-- <-- <-- <-- <-- <-- <--, | |
        ld hl,C_buffer0 ;e020   21 00 db    ;  load HL with C_buffer0                ^ ^ ^
        ld b,008h       ;e023   06 08       ;  8 bytes per character                 | | |
le025h:                                     ;                                        ^ ^ ^
        ld a,(de)       ;e025   1a          ;b xor char at (DE) with (C_buffer0) <--, | | |
        xor (hl)        ;e026   ae          ;|                                     ^ ^ ^ ^
        ld (de),a       ;e027   12          ;|                                     | | | |
        inc l           ;e028   2c          ;|                                     ^ ^ ^ ^
        inc d           ;e029   14          ;e                                     | | | |
        ld a,d          ;e02a   7a          ;b for TURBO, lines are 0300h apart    ^ ^ ^ ^
        cp 0b2h         ;e02b   fe b2       ;| and D is incremented 3 times        | | | |
        jr c,$-44       ;e02d   38 d2       ;e jump if TURBO                       ^ ^ ^ ^
        ld a,e          ;e02f   7b          ;b for SPECTRUM, lines are 01e0h apart | | | |
        add a,0e0h      ;e030   c6 e0       ;| and D is incremented once or twice  ^ ^ ^ ^
        ld e,a          ;e032   5f          ;|                                     | | | |
        jr nc,le036h    ;e033   30 01       ;e                                     ^ ^ ^ ^
le035h:                                     ;                                      | | | |
        inc d           ;e035   14          ;                                      ^ ^ ^ ^
le036h:                                     ;                                      | | | |
        djnz le025h     ;e036   10 ed       ;  close byte loop --> --> --> --> --> -' ^ ^ ^
        ld a,d          ;e038   7a          ;b TURBO char start address < 09c00h,     | | |
        sub 018h        ;e039   d6 18       ;| and it takes 8 * 0300h = 01800h bytes  ^ ^ ^
        cp 09ch         ;e03b   fe 9c       ;|                                     | | |
        jr c,le041h     ;e03d   38 02       ;e                                     ^ ^ ^
        add a,009h      ;e03f   c6 09       ;  SPECTRUM takes 8*01e0h = 00f00h bytes | | |
le041h:                 ;                                    ^ ^ ^
        ld d,a          ;e041   57          ;  restore DE with char start address     | | |
        pop hl          ;e042   e1          ;  restore HL                             ^ ^ ^
        inc hl          ;e043   23          ;                                         | | |
        ld a,(hl)       ;e044   7e          ;  load A with next char increment        ^ ^ ^
        add a,e         ;e045   83          ;b next char start address                | | |
        ld e,a          ;e046   5f          ;e                                        ^ ^ ^
        jr nc,le052h    ;e047   30 09       ;  jump if E does not wrap around         | | |
        inc d           ;e049   14          ;  need to increment D                    ^ ^ ^
        ld a,d          ;e04a   7a          ;                                         | | |
        cp 09ch         ;e04b   fe 9c       ;                                         ^ ^ ^
        jr nz,le052h    ;e04d   20 03       ;  jump if TURBO                          | | |
        add a,015h      ;e04f   c6 15       ;b transition to SPECTRUM                 ^ ^ ^
        ld d,a          ;e051   57          ;e                                        | | |
le052h:                                     ;                                         ^ ^ ^
        dec c           ;e052   0d          ;                                         | | |
```

```
        jr nz,le01fh    ;e053   20 ca      ;  close char loop --> --> --> --> --> -->' ^ ^
        push hl         ;e055   e5         ;  save HL                                  | |
        ld hl,C_buffer0 ;e056   21 00 db   ;b copy (C_buffer0) to (C_buffer1)          ^ ^
        ld de,C_buffer1 ;e059   11 08 db   ;| then rotate left char (C_buffer0)       | |
        ld bc,00008h    ;e05c   01 08 00   ;|                                         ^ ^
        ldir            ;e05f   ed b0      ;e HL holds C_buffer1                     | |
        ld e,000h       ;e061   1e 00      ;  DE holds C_buffer0                     ^ ^
        ld c,008h       ;e063   0e 08      ;  8 bytes per character                  | |
le065h:                                    ;                                         ^ ^
        ld b,008h       ;e065   06 08      ;  8 bits per byte <-- <-- <-- <-,        | |
        ld l,b          ;e067   68         ;                               ^         ^ ^
le068h:                                    ;                               |         | |
        rr (hl)         ;e068   cb 1e      ;  shift bit 0 of (HL) to CY <,  ^         ^ ^
        ex de,hl        ;e06a   eb         ;                             |  |        | |
        rl (hl)         ;e06b   cb 16      ;  shift CY to bit 0 of (DE)  ^  ^        ^ ^
        ex de,hl        ;e06d   eb         ;                             |  |        | |
        inc l           ;e06e   2c         ;                             ^  ^        ^ ^
        djnz le068h     ;e06f   10 f7      ;  close bit loop  --> --> -->'  |        | |
        inc e           ;e071   1c         ;                               ^         ^ ^
        dec c           ;e072   0d         ;                               |         | |
        jr nz,le065h    ;e073   20 f0      ;  close byte loop --> --> --> --'        ^ ^
        pop hl          ;e075   e1         ;  restore HL                             | |
        pop bc          ;e076   c1         ;  restore BC                             ^ ^
        djnz le019h     ;e077   10 a0      ;  close special char rotation loop  --> -->' |
        ld b,004h       ;e079   06 04      ;  4 character rotations                    ^
        dec c           ;e07b   0d         ;                                          |
        exx             ;e07c   d9         ;                                          ^
        jr nz,le010h    ;e07d   20 91      ;  close special char loop --> --> --> --> -->'

; copy left 3rd to the right third
        ld de,T_buffer  ;e07f   11 00 99   ;  load DE with T_buffer
        ld b,068h       ;e082   06 68      ;  (8 TURBO + 5 SPECTRUM) * 8 lines/char = 104 lines
le084h:
        push bc         ;e084   c5         ;  save BC <-- <-- <-- <,
        ld h,d          ;e085   62         ;b copy first 32 bytes  |
        ld l,e          ;e086   6b         ;| in the 96 byte line  ^
        ld bc,00040h    ;e087   01 40 00   ;| to the last 32       |
        add hl,bc       ;e08a   09         ;|                     ^
        ex de,hl        ;e08b   eb         ;|                     |
        ld c,020h       ;e08c   0e 20      ;|                     ^
        ldir            ;e08e   ed b0      ;e                     |
        pop bc          ;e090   c1         ;  restore BC          ^
        djnz le084h     ;e091   10 f1      ;  close line loop --> -'

; generate 4 shifted versions of TURBO and 8 of SPECTRUM
        ld hl,T_buffer  ;e093   21 00 99   ;b copy TURBO from 09900h to 03000h
        ld de,03000h    ;e096   11 00 30   ;|
        ld bc,01800h    ;e099   01 00 18   ;|
        ldir            ;e09c   ed b0      ;e
        ld de,0b0ffh    ;e09e   11 ff b0   ;b rotate left SPECTRUM
        ld hl,00f00h    ;e0a1   21 00 0f   ;| by 1 bit, 7 times
        ld b,023h       ;e0a4   06 23      ;| 7 x 5 = 35 character rows
        call Rotate     ;e0a6   cd d0 e0   ;e
        ld de,02fffh    ;e0a9   11 ff 2f   ;
        ld c,003h       ;e0ac   0e 03      ;  3 times
le0aeh:
        ld hl,01800h    ;e0ae   21 00 18   ;b rotate left TURBO by 1 bit <,
        ld b,008h       ;e0b1   06 08      ;| 8 character rows            ^
        call Rotate     ;e0b3   cd d0 e0   ;e                            |
        add hl,de       ;e0b6   19         ;b rotate left TURBO by 1 bit  ^
        ex de,hl        ;e0b7   eb         ;|                            |
        ld hl,00000h    ;e0b8   21 00 00   ;| in place                    ^
        ld b,008h       ;e0bb   06 08      ;| 8 character rows            |
        call Rotate     ;e0bd   cd d0 e0   ;e                            ^
        dec c           ;e0c0   0d         ;                            |
        jr nz,le0aeh    ;e0c1   20 eb      ;  close rotate TURBO loop  -->'

; set interrupt mode 2 with interrupt handler address at 0dfffh, i.e. 0e13ah
        ld bc,00800h    ;e0c3   01 00 08   ;  8 bit rotate, starting from horizontal char 0
        ld a,0dfh       ;e0c6   3e df      ;
        ld i,a          ;e0c8   ed 47      ;
        im 2            ;e0ca   ed 5e      ;
le0cch:
        ei              ;e0cc   fb         ;  enable interrupts <-- <-- <-,
        halt            ;e0cd   76         ;  halt                       |
        jr le0cch       ;e0ce   18 fc      ;  loop until a key pressed -->'

Rotate: ; subroutine to Rotate left B 060h byte characters lines ending at DE+HL to DE
        push bc         ;e0d0   c5         ;  save BC  <-- <-- <-- <-- <--,
        ld c,008h       ;e0d1   0e 08      ;  8 lines per character       ^
le0d3h:                                    ;                              |
        push hl         ;e0d3   e5         ;  save HL <-- <-- <-- <-- <,  ^
        add hl,de       ;e0d4   19         ;  sdd DE to HL             |  |
        ld b,060h       ;e0d5   06 60      ;  060h bytes per line      ^  ^
le0d7h:                                    ;                           |  |
```

```
        ld a,(hl)       ;e0d7   7e          ;b rotate left 1 bit <,     ^  ^
        rla             ;e0d8   17          ;|                     |     |  |
        ld (de),a       ;e0d9   12          ;|                     ^     ^  ^
        dec hl          ;e0da   2b          ;|                     |     |  |
        dec de          ;e0db   1b          ;e                     ^     ^  ^
        djnz le0d7h     ;e0dc   10 f9       ;  close byte loop -->'     |  |
        jr nc,le0e5h    ;e0de   30 05       ;                           ^  ^
        ld hl,00060h    ;e0e0   21 60 00    ;b increment the last byte  |  |
        add hl,de       ;e0e3   19          ;| in case a 1 bit was      ^  ^
        inc (hl)        ;e0e4   34          ;e shifted out on the left  |  |
le0e5h:                                     ;                           ^  ^
        pop hl          ;e0e5   e1          ;  restore HL               |  |
        dec c           ;e0e6   0d          ;                           ^  ^
        jr nz,le0d3h    ;e0e7   20 ea       ;  close line loop --> --> -'  |
        pop bc          ;e0e9   c1          ;  restore BC                  ^
        djnz Rotate     ;e0ea   10 e4       ;  close character row loop -->'
        ret             ;e0ec   c9          ;  return

Copy:   ; subroutine to copy 1 line of L character rows starting from IX to IY-010h
        ld (HL_save),hl ;e0ed   22 16 db    ;  save HL <-- <-- <-- <-- <-- <-,
        ld sp,ix        ;e0f0   dd f9       ;b read 16 bytes from (IX)           |
        pop af          ;e0f2   f1          ;|                                   ^
        pop bc          ;e0f3   c1          ;|                                   |
        pop de          ;e0f4   d1          ;|                                   ^
        pop hl          ;e0f5   e1          ;|                                   |
        ex af,af'       ;e0f6   08          ;|                                   ^
        exx             ;e0f7   d9          ;|                                   |
        pop af          ;e0f8   f1          ;|                                   ^
        pop bc          ;e0f9   c1          ;|                                   |
        pop de          ;e0fa   d1          ;|                                   ^
        pop hl          ;e0fb   e1          ;e                                   |
        ld sp,iy        ;e0fc   fd f9       ;b write them to (IY-010h)           ^
        push hl         ;e0fe   e5          ;|                                   |
        push de         ;e0ff   d5          ;|                                   ^
        push bc         ;e100   c5          ;|                                   |
        push af         ;e101   f5          ;|                                   ^
        ex af,af'       ;e102   08          ;|                                   |
        exx             ;e103   d9          ;|                                   ^
        push hl         ;e104   e5          ;|                                   |
        push de         ;e105   d5          ;|                                   ^
        push bc         ;e106   c5          ;|                                   |
        push af         ;e107   f5          ;e                                   ^
        ld de,00010h    ;e108   11 10 00    ;b next 16 bytes                     |
        add ix,de       ;e10b   dd 19       ;|                                   ^
        add iy,de       ;e10d   fd 19       ;e                                   |
        ld sp,ix        ;e10f   dd f9       ;b read 16 bytes from (IX)           ^
        pop af          ;e111   f1          ;|                                   |
        pop bc          ;e112   c1          ;|                                   ^
        pop de          ;e113   d1          ;|                                   |
        pop hl          ;e114   e1          ;|                                   ^
        ex af,af'       ;e115   08          ;|                                   |
        exx             ;e116   d9          ;|                                   ^
        pop af          ;e117   f1          ;|                                   |
        pop bc          ;e118   c1          ;|                                   ^
        pop de          ;e119   d1          ;|                                   |
        pop hl          ;e11a   e1          ;e                                   ^
        ld sp,iy        ;e11b   fd f9       ;b write them to (IY-010h)           |
        push hl         ;e11d   e5          ;|                                   ^
        push de         ;e11e   d5          ;|                                   |
        push bc         ;e11f   c5          ;|                                   ^
        push af         ;e120   f5          ;|                                   |
        ex af,af'       ;e121   08          ;|                                   ^
        exx             ;e122   d9          ;|                                   |
        push hl         ;e123   e5          ;|                                   ^
        push de         ;e124   d5          ;|                                   |
        push bc         ;e125   c5          ;|                                   ^
        push af         ;e126   f5          ;e                                   |
        ld de,00050h    ;e127   11 50 00    ;b move IX to next source line       ^
        add ix,de       ;e12a   dd 19       ;e                                   |
        ld e,010h       ;e12c   1e 10       ;b move IY to next destination line  ^
        add iy,de       ;e12e   fd 19       ;e                                   |
        ld hl,(HL_save) ;e130   2a 16 db    ;  restore HL                        ^
        dec l           ;e133   2d          ;                                    |
        jr nz,Copy      ;e134   20 b7       ;  close line loop --> --> --> --> --'
        ld sp,0e7fah    ;e136   31 fa e7    ;  restore SP
        ret             ;e139   c9          ;  return

IntHandler:                                 ;  (07ffeh) holds the 0e0ceh Return Address
; There are 8 * 64 rotation phases, 8 bits per character and 64 characters,
; which take 512 frame interrupts, or 17 seconds
; The bit phase is stored to B and the character phase to C.
; TURBO rotates left 2 pixels per frame interrupt and SPECTRUM 1 pixel.
; TURBO completes a rotation cycle 2 times faster than SPECTRUM.
; Both TURBO and SPECTRUM completely exit the screen on the left before they reappear on the right.
        push bc         ;e13a   c5          ;  save BC to (0e7fch)
```

```
; copy SPECTRUM starting at 0c860h, the 4th character line of the 2nd third of the screen
        ld a,039h       ;e13b   3e 39      ;b determine high address of SPECTRUM source
le13dh:                                    ;| 039h + 00fh * B
        add a,00fh      ;e13d   c6 0f      ;| B=8 ,B=7 ,B=6 ,B=5 ,B=4 ,B=3 ,B=2 ,B=1
        djnz le13dh     ;e13f   10 fc      ;e 0b1h,0a2h,093h,084h,075h,066h,057h,048h
        ld ixh,a        ;e141   dd 67      ;b source address
        ld ixl,c        ;e143   dd 69      ;e
        ld iy,0c870h    ;e145   fd 21 70 c8;  destination address + 010h
        ld hl,00805h    ;e149   21 05 08   ;  8 lines of 5 character rows
le14ch:
        call Copy       ;e14c   cd ed e0   ;  (0e7fah) holds 0e14fh <-- <-- <--,
        ld de,00060h    ;e14f   11 60 00   ;b set IY to next destination line  ^
        add iy,de       ;e152   fd 19      ;e                                  |
        ld l,005h       ;e154   2e 05      ;  5 character rows                 ^
        dec h           ;e156   25         ;                                   |
        jr nz,le14ch    ;e157   20 f3      ;  close SPECTRUM loop --> --> --> -'

; keyboard test
        xor a           ;e159   af         ;
        ld b,064h       ;e15a   06 64      ;  100 loops
le15ch:                                    ;  delay = 4495/3500 = 1.28 ms  cycles
        in a,(0feh)     ;e15c   db fe      ;  A=in(000feh) <-- <-- <-- <-,   11
        and 01fh        ;e15e   e6 1f      ;                            |    7
        cp 01fh         ;e160   fe 1f      ;                            ^    7
        jr nz,Key       ;e162   20 36      ;  jump if a key was pressed  |    7
        djnz le15ch     ;e164   10 f6      ;  close keyboard test --> -->'   13,8: 45*100-5=4495

; copy TURBO to the 1st third of the screen
le166h:
        pop bc          ;e166   c1         ;  restore BC
        push bc         ;e167   c5         ;  save BC to (0e7fch)
        ld a,b          ;e168   78         ;b TURBO is rotated twice as fast (2 bits at a time)
        dec a           ;e169   3d         ;| C *= 2;
        and 003h        ;e16a   e6 03      ;| if (B>4) B-=4;
        inc a           ;e16c   3c         ;| else C++;
        cp b            ;e16d   b8         ;|
        ld b,a          ;e16e   47         ;|
        ld a,c          ;e16f   79         ;|
        ccf             ;e170   3f         ;|
        rla             ;e171   17         ;|
        and 03fh        ;e172   e6 3f      ;|
        ld c,a          ;e174   4f         ;e
        ld a,0d0h       ;e175   3e d0      ;b determine high address of TURBO source
le177h:                                    ;| 0d0h + 018h * B
        add a,018h      ;e177   c6 18      ;| B=4 ,B=3 ,B=2 ,B=1
        djnz le177h     ;e179   10 fc      ;e 030h,018h,000h,0e8h
        ld ixh,a        ;e17b   dd 67      ;b source address
        ld ixl,c        ;e17d   dd 69      ;e
        ld iy,0c010h    ;e17f   fd 21 10 c0;  destination address + 010h
        ld hl,00808h    ;e183   21 08 08   ;  8 lines of 8 character rows
le186h:
        call Copy       ;e186   cd ed e0   ;  (0e7fah) holds 0e189h <-,
        ld l,008h       ;e189   2e 08      ;  8 character rows        ^
        dec h           ;e18b   25         ;                          |
        jr nz,le186h    ;e18c   20 f8      ;  close TURBO loop --> -->'
        pop bc          ;e18e   c1         ;  restore BC
        djnz le198h     ;e18f   10 07      ;  jump if not done with bits
        ld b,008h       ;e191   06 08      ;  8 bits
        inc c           ;e193   0c         ;b increment C modulo 64
        ld a,c          ;e194   79         ;|
        and 03fh        ;e195   e6 3f      ;|
        ld c,a          ;e197   4f         ;e
le198h:
        reti            ;e198   ed 4d      ;  return from interrupt

; a key was pressed
Key:
        ld a,0fbh       ;e19a   3e fb      ;
        in a,(0feh)     ;e19c   db fe      ;  A=in(0fbfeh), row KAa
        and 010h        ;e19e   e6 10      ;
        jr z,T_key      ;e1a0   28 69      ;  jump if T was pressed
        ld a,0fdh       ;e1a2   3e fd      ;
        in a,(0feh)     ;e1a4   db fe      ;  A=in(0fdfeh), row KA9
        and 002h        ;e1a6   e6 02      ;
        jr z,S_key      ;e1a8   28 60      ;  jump if S was pressed

; go to Spectrum mode
        xor a           ;e1aa   af         ;b out(000feh)=000h
        out (0feh),a    ;e1ab   d3 fe      ;e set black border and clear O5 (input to !CE flop)
        in a,(0feh)     ;e1ad   db fe      ;  !CE=0; A=in(000feh), row KA9
        ld hl,00558h    ;e1af   21 58 05   ;b copy Spectrum code
        ld de,08000h    ;e1b2   11 00 80   ;| from (00558h)-(03dc5h)
        ld bc,0386eh    ;e1b5   01 6e 38   ;| to   (08000h)-(0b86dh)
        ldir            ;e1b8   ed b0      ;e
```

```
        ld a,0ffh       ;e1ba   3e ff      ;b fill (0b86eh)-(0bcffh)
        ld bc,09205h    ;e1bc   01 05 92   ;| with 0ffh
le1bfh:                                    ;|
        ld (de),a       ;e1bf   12         ;|
        inc de          ;e1c0   13         ;|
        djnz le1bfh     ;e1c1   10 fc      ;|
        dec c           ;e1c3   0d         ;|
        jr nz,le1bfh    ;e1c4   20 f9      ;e
        inc a           ;e1c6   3c         ;b fill (0bd00h)-(0bd07h)
        ld b,008h       ;e1c7   06 08      ;| with 000h
le1c9h:                                    ;| (blank character)
        ld (de),a       ;e1c9   12         ;|
        inc de          ;e1ca   13         ;|
        djnz le1c9h     ;e1cb   10 fc      ;e
        ld b,05fh       ;e1cd   06 5f      ;  unpack remaining 05fh characters
le1cfh:                                    ;
        push bc         ;e1cf   c5         ;  save BC <-- <-- <-- <,
        ld bc,00006h    ;e1d0   01 06 00   ;                       |
        ld (de),a       ;e1d3   12         ;  first byte is 000h   ^
        inc de          ;e1d4   13         ;                       |
        ldir            ;e1d5   ed b0      ;  copy middle 6 bytes  ^
        ld (de),a       ;e1d7   12         ;   last byte is 000h   |
        inc de          ;e1d8   13         ;                       ^
        pop bc          ;e1d9   c1         ;  restore BC           |
        djnz le1cfh     ;e1da   10 f3      ;  close character loop '
        ld bc,0000ch    ;e1dc   01 0c 00   ;  12 bytes
        ld hl,Bytes     ;e1df   21 f3 e1   ;  Bytes structure
        ld de,0bd27h    ;e1e2   11 27 bd   ;  start address
le1e5h:
        ldi             ;e1e5   ed a0      ;  copy (HL) to (DE) <,
        jp po,Done      ;e1e7   e2 df e2   ;  jump if done       |
        ld a,(hl)       ;e1ea   7e         ;b add increment      ^
        inc hl          ;e1eb   23         ;|                    |
        add a,e         ;e1ec   83         ;|                    ^
        ld e,a          ;e1ed   5f         ;e                    |
        jr nc,le1e5h    ;e1ee   30 f5      ;b close              ^
        inc d           ;e1f0   14         ;| byte               |
        jr le1e5h       ;e1f1   18 f2      ;e loop --> --> --> --'

Bytes:                                     ; Bytes structure
        defb         008h                  ;            byte c
        defb    03fh,010h                  ; increment, byte b
        defb    077h,020h                  ; increment, byte a
        defb    08fh,000h                  ; increment, byte 9
        defb    08fh,0ffh                  ; increment, byte 8
        defb    03fh,038h                  ; increment, byte 7
        defb    017h,018h                  ; increment, byte 6
        defb    02fh,040h                  ; increment, byte 5
        defb    007h,006h                  ; increment, byte 4
        defb    03fh,038h                  ; increment, byte 3
        defb    028h,03ch                  ; increment, byte 2
        defb    006h,03ch                  ; increment, byte 1

S_key:                                     ;  E=060h
        add a,e         ;e20a   83         ;  A=060h
T_key:                                     ;  A=000h
        push af         ;e20b   f5         ;  save AF
        ld ix,08000h    ;e20c   dd 21 00 80;
        ld a,0ffh       ;e210   3e ff      ;
        ld de,04000h    ;e212   11 00 40   ;
        jr nz,le21ah    ;e215   20 03      ;  jump if S was pressed
        ld ixh,000h     ;e217   dd 26 00   ;
le21ah:                                    ;  IX = T_was_pressed ? 00000h : 08000h
        and a           ;e21a   a7         ;
        scf             ;e21b   37         ;
        ex af,af'       ;e21c   08         ;
        ld a,02fh       ;e21d   3e 2f      ;b out(002feh)=02fh
        out (0feh),a    ;e21f   d3 fe      ;e set white border, TO, and O5 (input to !CE flop)
        ld hl,le2c8h    ;e221   21 c8 e2   ;
        push hl         ;e224   e5         ;  push 0e2c8h
        in a,(0feh)     ;e225   db fe      ;  !CE=1; A=in(002feh)
        rra             ;e227   1f         ;
        and 020h        ;e228   e6 20      ;
        or 002h         ;e22a   f6 02      ;
        ld c,a          ;e22c   4f         ;  C is 022h 0r 002h
        cp a            ;e22d   bf         ;  Z=0
le22eh:
        ret nz          ;e22e   c0         ;  jump if NZ (Space was pressed) to le2c8h
le22fh:
        call sub_e2aah  ;e22f   cd aa e2   ;  check is Space was pressed
        jr nc,le22eh    ;e232   30 fa      ;  jump if Space was pressed
        ld hl,00415h    ;e234   21 15 04   ;  1045 loops
le237h:                                    ;  delay = 3499700/3500 = 1000 ms
        djnz le237h     ;e237   10 fe      ;  13*256-5=3323 <-, cycles
        dec hl          ;e239   2b         ;                   |    6
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```
        ld a,h          ;e23a   7c         ;                  ^    4
        or l            ;e23b   b5         ;                  |    4
        jr nz,le237h    ;e23c   20 f9      ;  close loop --> -'   12,7: 3349*1045-5=3499700
        call sub_e2a6h  ;e23e   cd a6 e2   ;  check is Space was pressed
        jr nc,le22eh    ;e241   30 eb      ;  jump if Space was pressed
le243h:
        ld b,09ch       ;e243   06 9c      ;
        call sub_e2a6h  ;e245   cd a6 e2   ;
        jr nc,le22eh    ;e248   30 e4      ;
        ld a,0c6h       ;e24a   3e c6      ;
        cp b            ;e24c   b8         ;
        jr nc,le22fh    ;e24d   30 e0      ;
        inc h           ;e24f   24         ;
        jr nz,le243h    ;e250   20 f1      ;
le252h:
        ld b,0c9h       ;e252   06 c9      ;
        call sub_e2aah  ;e254   cd aa e2   ;
        jr nc,le22eh    ;e257   30 d5      ;
        ld a,b          ;e259   78         ;
        cp 0d4h         ;e25a   fe d4      ;
        jr nc,le252h    ;e25c   30 f4      ;
        call sub_e2aah  ;e25e   cd aa e2   ;
        ret nc          ;e261   d0         ;
        ld a,c          ;e262   79         ;
        xor 003h        ;e263   ee 03      ;
        ld c,a          ;e265   4f         ;
        ld h,000h       ;e266   26 00      ;
        ld b,0b0h       ;e268   06 b0      ;
        jr le28bh       ;e26a   18 1f      ;
le26ch:
        ex af,af'       ;e26c   08         ;
        jr nz,le276h    ;e26d   20 07      ;
        jr nc,le280h    ;e26f   30 0f      ;
        ld (ix+000h),l  ;e271   dd 75 00   ;
        jr le285h       ;e274   18 0f      ;
le276h:
        rl c            ;e276   cb 11      ;
        xor l           ;e278   ad         ;
        ret nz          ;e279   c0         ;
        ld a,c          ;e27a   79         ;
        rra             ;e27b   1f         ;
        ld c,a          ;e27c   4f         ;
        inc de          ;e27d   13         ;
        jr le287h       ;e27e   18 07      ;
le280h:
        ld a,(ix+000h)  ;e280   dd 7e 00   ;
        xor l           ;e283   ad         ;
        ret nz          ;e284   c0         ;
le285h:
        inc ix          ;e285   dd 23      ;
le287h:
        dec de          ;e287   1b         ;
        ex af,af'       ;e288   08         ;
        ld b,0b2h       ;e289   06 b2      ;
le28bh:
        ld l,001h       ;e28b   2e 01      ;
le28dh:
        call sub_e2a6h  ;e28d   cd a6 e2   ;
        ret nc          ;e290   d0         ;
        ld a,0cbh       ;e291   3e cb      ;
        cp b            ;e293   b8         ;
        rl l            ;e294   cb 15      ;
        ld b,0b0h       ;e296   06 b0      ;
        jp nc,le28dh    ;e298   d2 8d e2   ;
        ld a,h          ;e29b   7c         ;
        xor l           ;e29c   ad         ;
        ld h,a          ;e29d   67         ;
        ld a,d          ;e29e   7a         ;
        or e            ;e29f   b3         ;
        jr nz,le26ch    ;e2a0   20 ca      ;
        ld a,h          ;e2a2   7c         ;
        cp 001h         ;e2a3   fe 01      ;
        ret             ;e2a5   c9         ;
sub_e2a6h:
        call sub_e2aah  ;e2a6   cd aa e2   ;
        ret nc          ;e2a9   d0         ;

sub_e2aah:                                 ;  B is 064h to 001h
        ld a,016h       ;e2aa   3e 16      ;  22 loops
le2ach:                                    ;  delay = 347/3500 = 0.099 ms
        dec a           ;e2ac   3d         ;  decrement <--,   4    cycles
        jr nz,le2ach    ;e2ad   20 fd      ;  close loop ->'  12,7: 16*22-5=347
        and a           ;e2af   a7         ;  CY=0
le2b0h:
        inc b           ;e2b0   04         ;
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        ret z           ;e2b1   c8          ;  return if B=0 and redo the loop
        ld a,07fh       ;e2b2   3e 7f       ;
        in a,(0feh)     ;e2b4   db fe       ;  A=in(07ffeh), row KAf
        rra             ;e2b6   1f          ;
        ret nc          ;e2b7   d0          ;  return if Space was pressed
        xor c           ;e2b8   a9          ;
        and 020h        ;e2b9   e6 20       ;  A is 020h if TI changed state, 000h if not
        jr z,le2b0h     ;e2bb   28 f3       ;  wait for TI to change state
        ld a,c          ;e2bd   79          ;
        cpl             ;e2be   2f          ;
        ld c,a          ;e2bf   4f          ;  C=0dfh
        and 007h        ;e2c0   e6 07       ;
        or 028h         ;e2c2   f6 28       ;b out(002feh)=02fh
        out (0feh),a    ;e2c4   d3 fe       ;e set white border, TO, and O5 (input to !CE flop)
        scf             ;e2c6   37          ;
        ret             ;e2c7   c9          ;
le2c8h:
        jr c,le2dbh     ;e2c8   38 11       ;  jump if CY to le2dbh
        pop af          ;e2ca   f1          ;
        ld h,003h       ;e2cb   26 03       ;  768 to 1024 loops
le2cdh:                                     ;  delay = 2572027/3500 = 735 to 980 ms
        djnz le2cdh     ;e2cd   10 fe       ;  13*256-5=3323 <-, cycles
        dec hl          ;e2cf   2b          ;                   |    6
        ld a,h          ;e2d0   7c          ;                   ^    4
        or l            ;e2d1   b5          ;                   |    4
        jr nz,le2cdh    ;e2d2   20 f9       ;  close loop --> -'   12,7: 3349*768-5=2572027
        ld a,020h       ;e2d4   3e 20       ;b out(020feh)=020h
        out (0feh),a    ;e2d6   d3 fe       ;e set black border and O5 (input to !CE flop)
        jp le166h       ;e2d8   c3 66 e1    ;  continue TURBO rotation
le2dbh:
        pop af          ;e2db   f1          ;
        jp z,00000h     ;e2dc   ca 00 00    ;  jump if Z to 00000h

Done:
        ld sp,00000h    ;e2df   31 00 00    ;  initialize SP=00000h
        ld a,060h       ;e2e2   3e 60       ;b out(060feh)=060h
        out (0feh),a    ;e2e4   d3 fe       ;e set black border, O5 and O6 (inputs to !CE and SM flops)
        ld hl,0fedbh    ;e2e6   21 db fe    ;
        push hl         ;e2e9   e5          ;  write 0dbfeh at (0fffeh) - this is: in a,(0feh)
        jp 0fffeh       ;e2ea   c3 fe ff    ;  jump and execute it;  A=in(060feh)
; then start executing in Spectrum mode from address 00000h

C_base:                                                 ; Character Base Address
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh; char 0
        defb    0f0h,0f0h,0f0h,0f0h,0f0h,0f0h,0f0h,0f0h; char 1
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0fch,0f0h,080h; char 2
        defb    0ffh,0feh,0feh,0feh,0fch,0fch,0f8h,0f8h; char 3
        defb    0feh,0fch,0fch,0f8h,0f0h,0e0h,080h,000h; char 4
        defb    0f8h,0f0h,0e0h,0c0h,080h,000h,000h,000h; char 5
        defb    0f0h,0e0h,0c0h,080h,000h,000h,000h,000h; char 6
        defb    0c0h,080h,000h,000h,000h,000h,000h,000h; char 7
        defb    080h,000h,000h,000h,000h,000h,000h,000h; char 8

C_struct:                                               ; Char structure Base address
        defw    09901h                                  ; Character 0 start address
        defb    0b7h                                    ; number of char increments
        defb    001h,001h,001h,001h,002h,001h,002h,001h; TURBO char increments
        defb    002h,001h,001h,004h,001h,001h,006h,046h;
        defb    001h,001h,001h,001h,002h,001h,002h,001h;
        defb    002h,001h,001h,001h,003h,001h,001h,001h;
        defb    004h,001h,001h,047h,004h,001h,002h,001h;
        defb    005h,001h,005h,001h,002h,001h,002h,001h;
        defb    046h,004h,001h,002h,001h,002h,001h,001h;
        defb    001h,003h,001h,001h,001h,003h,001h,002h;
        defb    001h,046h,004h,001h,002h,001h,002h,001h;
        defb    001h,001h,003h,001h,001h,001h,003h,001h;
        defb    002h,001h,046h,004h,001h,002h,001h,005h;
        defb    001h,005h,001h,002h,001h,002h,001h,046h;
        defb    005h,001h,001h,006h,001h,002h,001h,001h;
        defb    001h,004h,001h,001h,047h,006h,007h,001h;
        defb    002h,001h,001h,006h;
        defb                        046h,003h,001h,003h; SPECTRUM char increments
        defb    001h,001h,003h,003h,001h,001h,002h,001h;
        defb    003h,002h,002h,003h,041h,004h,002h,002h;
        defb    004h,005h,003h,002h,002h,002h,002h,001h;
        defb    001h,001h,042h,003h,001h,003h,001h,003h;
        defb    005h,003h,001h,003h,002h,002h,003h,043h;
        defb    002h,004h,004h,005h,003h,001h,003h,002h;
        defb    002h,003h,042h,003h,004h,001h,001h,003h;
        defb    004h,003h,002h,003h,003h,003h

        defw    099c4h                                  ; Character 1, rotation 0 start address
        defb    006h                                    ; number of char increments
        defb    060h,060h,060h,060h,060h                ; increments
        defw    0b220h                                  ; Character 1, rotation 1 start address
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        defb    002h                                    ; number of char increments
        defb    00eh                                    ; increments
        defw    099c2h                                  ; Character 1, rotation 2 start address
        defb    006h                                    ; number of char increments
        defb    060h,060h,060h,060h,060h                ; increments
        defw    0b162h                                  ; Character 1, rotation 3 start address
        defb    002h                                    ; number of char increments
        defb    00ch                                    ; increments

        defw    09baah                                  ; Character 2, rotation 0 start address
        defb    003h                                    ; number of char increments
        defb    00ch,006h                               ; increments
        defw    09971h                                  ; Character 2, rotation 1 start address
        defb    007h                                    ; number of char increments
        defb    006h,006h,0c0h,000h,054h,006h           ; increments
        defw    0991ah                                  ; Character 2, rotation 2 start address
        defb    001h                                    ; number of char increments
        defw    09b47h                                  ; Character 2, rotation 3 start address
        defb    002h                                    ; number of char increments
        defb    012h                                    ; increments

        defw    09a31h                                  ; Character 3, rotation 0 start address
        defb    007h                                    ; number of char increments
        defb    006h,0c0h,000h,054h,00ch,006h           ; increments
        defw    09910h                                  ; Character 3, rotation 1 start address
        defb    003h                                    ; number of char increments
        defb    006h,006h                               ; increments
        defw    09979h                                  ; Character 3, rotation 2 start address
        defb    001h                                    ; number of char increments
        defw    09ba8h                                  ; Character 3, rotation 3 start address
        defb    002h                                    ; number of char increments
        defb    012h                                    ; increments

        defw    0b1c6h                                  ; Character 4, rotation 0 start address
        defb    005h                                    ; number of char increments
        defb    010h,0ach,00ch,00ch                     ; increments
        defw    0b102h                                  ; Character 4, rotation 1 start address
        defb    005h                                    ; number of char increments
        defb    004h,008h,008h,0ach                     ; increments
        defw    0b100h                                  ; Character 4, rotation 2 start address
        defb    002h                                    ; number of char increments
        defb    00ch                                    ; increments
        defw    0b1c0h                                  ; Character 4, rotation 3 start address
        defb    004h                                    ; number of char increments
        defb    0c0h,00ch,00ch                          ; increments

        defw    0b1deh                                  ; Character 5, rotation 0 start address
        defb    001h                                    ; number of char increments
        defw    0b11dh                                  ; Character 5, rotation 1 start address
        defb    004h                                    ; number of char increments
        defb    0c0h,000h,059h                          ; increments
        defw    0b11eh                                  ; Character 5, rotation 2 start address
        defb    001h                                    ; number of char increments
        defw    0b1ddh                                  ; Character 5, rotation 3 start address
        defb    002h                                    ; number of char increments
        defb    0b8h                                    ; increments

        defw    09babh                                  ; Character 6, rotation 0 start address
        defb    003h                                    ; number of char increments
        defb    00ch,006h                               ; increments
        defw    09911h                                  ; Character 6, rotation 1 start address
        defb    003h                                    ; number of char increments
        defb    006h,006h                               ; increments
        defw    09919h                                  ; Character 6, rotation 2 start address
        defb    001h                                    ; number of char increments
        defw    09ba7h                                  ; Character 6, rotation 3 start address
        defb    002h                                    ; number of char increments
        defb    012h                                    ; increments

        defw    099dbh                                  ; Character 7, rotation 0 start address
        defb    009h                                    ; number of char increments
        defb    0c0h,000h,0c0h,000h,0c4h,042h,00ch,011h; increments
        defw    09ae9h                                  ; Character 7, rotation 1 start address
        defb    00bh                                    ; number of char increments
        defb    012h,0c0h,000h,0a6h,01ch,0a4h,00ch,008h; increments
        defb    004h,05dh                               ;
        defw    099cfh                                  ; Character 7, rotation 2 start address
        defb    00fh                                    ; number of char increments
        defb    006h,0c0h,000h,054h,00ch,006h,0c0h,000h; increments
        defb    0aah,010h,009h,0a3h,00ch,00ch           ;
        defw    099cfh                                  ; Character 7, rotation 3 start address
        defb    012h                                    ; number of char increments
        defb    006h,006h,0c0h,000h,054h,006h,0c0h,000h; increments
        defb    067h,045h,004h,008h,008h,008h,0a4h,014h;
        defb    061h                                    ;
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        defw    0b101h                                  ; Character 8, rotation 0 start address
        defb    004h                                    ; number of char increments
        defb    00ch,053h,00ch                          ; increments
        defw    0b160h                                  ; Character 8, rotation 1 start address
        defb    008h                                    ; number of char increments
        defb    061h,05fh,00ch,00ch,049h,00ch,00ch      ; increments
        defw    0b166h                                  ; Character 8, rotation 2 start address
        defb    00ah                                    ; number of char increments
        defb    010h,04fh,010h,04dh,00ch,00ch,047h,00ch; increments
        defb    00ch                                    ;
        defw    0b101h                                  ; Character 8, rotation 3 start address
        defb    00ah                                    ; number of char increments
        defb    004h,008h,008h,04dh,004h,008h,008h,04bh; increments
        defb    061h                                    ;

Spare:                                                  ; spare bytes
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    0ffh,0ffh,0ffh,0ffh,0ffh                ;

Spectrum:
        defb                                              0f3h,0afh,011h,0ffh,0ffh,0c3h,0cbh,011h;
        defb    02ah,05dh,05ch,022h,05fh,05ch,018h,043h,0c3h,0f2h,015h,0ffh,0ffh,0ffh,0ffh,0ffh;
        defb    02ah,05dh,05ch,07eh,0cdh,07dh,000h,0d0h,0cdh,074h,000h,018h,0f7h,0ffh,0ffh,0ffh;
        defb    0c3h,05bh,033h,0ffh,0ffh,0ffh,0ffh,0ffh,0c5h,02ah,061h,05ch,0e5h,0c3h,09eh,016h;
        defb    0f5h,0e5h,02ah,078h,05ch,023h,022h,078h,05ch,07ch,0b5h,020h,003h,0fdh,034h,040h;
        defb    0c5h,0d5h,0cdh,0bfh,002h,0d1h,0c1h,0e1h,0f1h,0fbh,0c9h,0e1h,06eh,0fdh,075h,000h;
        defb    0edh,07bh,03dh,05ch,0c3h,0c5h,016h,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0ffh,0f5h,0e5h;
        defb    02ah,0b0h,05ch,07ch,0b5h,020h,001h,0e9h,0e1h,0f1h,0edh,045h,02ah,05dh,05ch,023h;
        defb    022h,05dh,05ch,07eh,0c9h,0feh,021h,0d0h,0feh,00dh,0c8h,0feh,010h,0d8h,0feh,018h;
        defb    03fh,0d8h,023h,0feh,016h,038h,001h,023h,037h,022h,05dh,05ch,0c9h,0bfh,052h,04eh;
        defb    0c4h,049h,04eh,04bh,045h,059h,0a4h,050h,0c9h,046h,0ceh,050h,04fh,049h,04eh,0d4h;
        defb    053h,043h,052h,045h,045h,04eh,0a4h,041h,054h,054h,0d2h,041h,0d4h,054h,041h,0c2h;
        defb    056h,041h,04ch,0a4h,043h,04fh,044h,0c5h,056h,041h,0cch,04ch,045h,0ceh,053h,049h;
        defb    0ceh,043h,04fh,0d3h,054h,041h,0ceh,041h,053h,0ceh,041h,043h,0d3h,041h,054h,0ceh;
        defb    04ch,0ceh,045h,058h,0d0h,049h,04eh,0d4h,053h,051h,0d2h,053h,047h,0ceh,041h,042h;
        defb    0d3h,050h,045h,045h,0cbh,049h,0ceh,055h,053h,0d2h,053h,054h,052h,0a4h,043h,048h;
        defb    052h,0a4h,04eh,04fh,0d4h,042h,049h,0ceh,04fh,0d2h,041h,04eh,0c4h,03ch,0bdh,03eh;
        defb    0bdh,03ch,0beh,04ch,049h,04eh,0c5h,054h,048h,045h,0ceh,054h,0cfh,053h,054h,045h;
        defb    0d0h,044h,045h,046h,020h,046h,0ceh,043h,041h,0d4h,046h,04fh,052h,04dh,041h,0d4h;
        defb    04dh,04fh,056h,0c5h,045h,052h,041h,053h,0c5h,04fh,050h,045h,04eh,020h,0a3h,043h;
        defb    04ch,04fh,053h,045h,020h,0a3h,04dh,045h,052h,047h,0c5h,056h,045h,052h,049h,046h;
        defb    0d9h,042h,045h,045h,0d0h,043h,049h,052h,043h,04ch,0c5h,049h,04eh,0cbh,050h,041h;
        defb    050h,045h,0d2h,046h,04ch,041h,053h,0c8h,042h,052h,049h,047h,048h,0d4h,049h,04eh;
        defb    056h,045h,052h,053h,0c5h,04fh,056h,045h,0d2h,04fh,055h,0d4h,04ch,050h,052h,049h;
        defb    04eh,0d4h,04ch,04ch,049h,053h,0d4h,053h,054h,04fh,0d0h,052h,045h,041h,0c4h,044h;
        defb    041h,054h,0c1h,052h,045h,053h,054h,04fh,052h,0c5h,04eh,045h,0d7h,042h,04fh,052h;
        defb    044h,045h,0d2h,043h,04fh,04eh,054h,049h,04eh,055h,0c5h,044h,049h,0cdh,052h,045h;
        defb    0cdh,046h,04fh,0d2h,047h,04fh,020h,054h,0cfh,047h,04fh,020h,053h,055h,0c2h,049h;
        defb    04eh,050h,055h,0d4h,04ch,04fh,041h,0c4h,04ch,049h,053h,0d4h,04ch,045h,0d4h,050h;
        defb    041h,055h,053h,0c5h,04eh,045h,058h,0d4h,050h,04fh,04bh,0c5h,050h,052h,049h,04eh;
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        defb    04fh,04dh,049h,05ah,0c5h,049h,0c6h,043h,04ch,0d3h,044h,052h,041h,0d7h,043h,04ch;
        defb    045h,041h,0d2h,052h,045h,054h,055h,052h,0ceh,043h,04fh,050h,0d9h,042h,048h,059h;
        defb    036h,035h,054h,047h,056h,04eh,04ah,055h,037h,034h,052h,046h,043h,04dh,04bh,049h;
        defb    038h,033h,045h,044h,058h,00eh,04ch,04fh,039h,032h,057h,053h,05ah,020h,00dh,050h;
        defb    030h,031h,051h,041h,0e3h,0c4h,0e0h,0e4h,0b4h,0bch,0bdh,0bbh,0afh,0b0h,0b1h,0c0h;
        defb    0a7h,0a6h,0beh,0adh,0b2h,0bah,0e5h,0a5h,0c2h,0e1h,0b3h,0b9h,0c1h,0b8h,07eh,0dch;
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        defb    009h,00fh,0e2h,02ah,03fh,0cdh,0c8h,0cch,0cbh,05eh,0ach,02dh,02bh,03dh,02eh,02ch;
        defb    03bh,022h,0c7h,03ch,0c3h,03eh,0c5h,02fh,0c9h,060h,0c6h,03ah,0d0h,0ceh,0a8h,0cah;
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        defb    003h,0cbh,078h,0c0h,021h,005h,002h,019h,07eh,037h,0c9h,07bh,0feh,03ah,038h,02fh;
        defb    00dh,0fah,04fh,003h,028h,003h,0c6h,04fh,0c9h,021h,0ebh,001h,004h,028h,003h,021h;
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        defb    030h,0d8h,00dh,0fah,09dh,003h,020h,019h,021h,054h,002h,0cbh,068h,028h,0d3h,0feh;
        defb    038h,030h,007h,0d6h,020h,004h,0c8h,0c6h,008h,0c9h,0d6h,036h,004h,0c8h,0c6h,0feh;
        defb    0c9h,021h,030h,002h,0feh,039h,028h,0bah,0feh,030h,028h,0b6h,0e6h,007h,0c6h,080h;
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        defb    03dh,0cbh,03dh,02fh,0e6h,003h,04fh,006h,000h,0ddh,021h,0d1h,003h,0ddh,009h,03ah;
        defb    048h,05ch,0e6h,038h,00fh,00fh,00fh,0f6h,008h,000h,000h,000h,004h,00ch,00dh,020h;
        defb    0fdh,00eh,03fh,005h,0c2h,0d6h,003h,0eeh,010h,0d3h,0feh,044h,04fh,0cbh,067h,020h;
        defb    009h,07ah,0b3h,028h,009h,079h,04dh,01bh,0ddh,0e9h,04dh,00ch,0ddh,0e9h,0fbh,0c9h;
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        defb    092h,05ch,07eh,0a7h,020h,05eh,023h,04eh,023h,046h,078h,017h,09fh,0b9h,020h,054h;
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        defb    03eh,001h,037h,0c3h,025h,005h,06ch,018h,0f4h,079h,0cbh,078h,010h,0feh,030h,004h;
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        defb    013h,02bh,046h,02bh,04eh,02bh,003h,003h,003h,0ddh,022h,05fh,05ch,0cdh,0e8h,019h;
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        defb    0e1h,018h,0ech,0cdh,02ch,009h,018h,0e2h,07eh,04fh,0feh,080h,0c8h,0e5h,02ah,04bh;
        defb    05ch,07eh,0feh,080h,028h,025h,0b9h,028h,008h,0c5h,0cdh,0b8h,019h,0c1h,0ebh,018h;
        defb    0f0h,0e6h,0e0h,0feh,0a0h,020h,012h,0d1h,0d5h,0e5h,023h,013h,01ah,0beh,020h,006h;
        defb    017h,030h,0f7h,0e1h,018h,003h,0e1h,018h,0e0h,03eh,0ffh,0d1h,0ebh,03ch,037h,0cdh;
        defb    02ch,009h,018h,0c4h,020h,010h,008h,022h,05fh,05ch,0ebh,0cdh,0b8h,019h,0cdh,0e8h;
        defb    019h,0ebh,02ah,05fh,05ch,008h,008h,0d5h,0cdh,0b8h,019h,022h,05fh,05ch,02ah,053h;
        defb    05ch,0e3h,0c5h,008h,038h,007h,02bh,0cdh,055h,016h,023h,018h,003h,0cdh,055h,016h;
        defb    023h,0c1h,0d1h,0edh,053h,053h,05ch,0edh,05bh,05fh,05ch,0c5h,0d5h,0ebh,0edh,0b0h;
        defb    0e1h,0c1h,0d5h,0cdh,0e8h,019h,0d1h,0c9h,0e5h,03eh,0fdh,0cdh,001h,016h,0afh,011h;
        defb    0a1h,009h,0cdh,00ah,00ch,0fdh,0cbh,002h,0eeh,0cdh,0d4h,015h,0ddh,0e5h,011h,011h;
        defb    000h,0afh,0cdh,0c2h,004h,0ddh,0e1h,006h,032h,076h,010h,0fdh,0ddh,05eh,00bh,0ddh;
        defb    056h,00ch,03eh,0ffh,0ddh,0e1h,0c3h,0c2h,004h,080h,053h,074h,061h,072h,074h,020h;
        defb    074h,061h,070h,065h,02ch,020h,074h,068h,065h,06eh,020h,070h,072h,065h,073h,073h;
        defb    020h,061h,06eh,079h,020h,06bh,065h,079h,0aeh,00dh,050h,072h,06fh,067h,072h,061h;
        defb    06dh,03ah,0a0h,00dh,04eh,075h,06dh,062h,065h,072h,020h,061h,072h,072h,061h,079h;
        defb    03ah,0a0h,00dh,043h,068h,061h,072h,061h,063h,074h,065h,072h,020h,061h,072h,072h;
        defb    061h,079h,03ah,0a0h,00dh,042h,079h,074h,065h,073h,03ah,0a0h,0cdh,003h,00bh,0feh;
        defb    020h,0d2h,0d9h,00ah,0feh,006h,038h,069h,0feh,018h,030h,065h,021h,00bh,00ah,05fh;
        defb    016h,000h,019h,05eh,019h,0e5h,0c3h,003h,00bh,04eh,057h,010h,029h,054h,053h,052h;
        defb    037h,050h,04fh,05fh,05eh,05dh,05ch,05bh,05ah,054h,053h,00ch,03eh,022h,0b9h,020h;
        defb    011h,0fdh,0cbh,001h,04eh,020h,009h,004h,00eh,002h,03eh,018h,0b8h,020h,003h,005h;
        defb    00eh,021h,0c3h,0d9h,00dh,03ah,091h,05ch,0f5h,0fdh,036h,057h,001h,03eh,020h,0cdh;
        defb    065h,00bh,0f1h,032h,091h,05ch,0c9h,0fdh,0cbh,001h,04eh,0c2h,0cdh,00eh,00eh,021h;
        defb    0cdh,055h,00ch,005h,0c3h,0d9h,00dh,0cdh,003h,00bh,079h,03dh,03dh,0e6h,010h,018h;
        defb    05ah,03eh,03fh,018h,06ch,011h,087h,00ah,032h,00fh,05ch,018h,00bh,011h,06dh,00ah;
        defb    018h,003h,011h,087h,00ah,032h,00eh,05ch,02ah,051h,05ch,073h,023h,072h,0c9h,011h;
        defb    0f4h,009h,0cdh,080h,00ah,02ah,00eh,05ch,057h,07dh,0feh,016h,0dah,011h,022h,020h;
        defb    029h,044h,04ah,03eh,01fh,091h,038h,00ch,0c6h,002h,04fh,0fdh,0cbh,001h,04eh,020h;
        defb    016h,03eh,016h,090h,0dah,09fh,01eh,03ch,047h,004h,0fdh,0cbh,002h,046h,0c2h,055h;
        defb    00ch,0fdh,0beh,031h,0dah,086h,00ch,0c3h,0d9h,00dh,07ch,0cdh,003h,00bh,081h,03dh;
        defb    0e6h,01fh,0c8h,057h,0fdh,0cbh,001h,0c6h,03eh,020h,0cdh,03bh,00ch,015h,020h,0f8h;
        defb    0c9h,0cdh,024h,00bh,0fdh,0cbh,001h,04eh,020h,01ah,0fdh,0cbh,002h,046h,020h,008h;
        defb    0edh,043h,088h,05ch,022h,084h,05ch,0c9h,0edh,043h,08ah,05ch,0edh,043h,082h,05ch;
        defb    022h,086h,05ch,0c9h,0fdh,071h,045h,022h,080h,05ch,0c9h,0fdh,0cbh,001h,04eh,020h;
        defb    014h,0edh,04bh,088h,05ch,02ah,084h,05ch,0fdh,0cbh,002h,046h,0c8h,0edh,04bh,08ah;
        defb    05ch,02ah,086h,05ch,0c9h,0fdh,04eh,045h,02ah,080h,05ch,0c9h,0feh,080h,038h,03dh;
        defb    0feh,090h,030h,026h,047h,0cdh,038h,00bh,0cdh,003h,00bh,011h,092h,05ch,018h,047h;
        defb    021h,092h,05ch,0cdh,03eh,00bh,0cbh,018h,09fh,0e6h,00fh,04fh,0cbh,018h,09fh,0e6h;
        defb    0f0h,0b1h,00eh,004h,077h,023h,00dh,020h,0fbh,0c9h,0d6h,0a5h,030h,009h,0c6h,015h;
        defb    0c5h,0edh,04bh,07bh,05ch,018h,00bh,0cdh,010h,00ch,0c3h,003h,00bh,0c5h,0edh,04bh;
        defb    036h,05ch,0ebh,021h,03bh,05ch,0cbh,086h,0feh,020h,020h,002h,0cbh,0c6h,026h,000h;
        defb    06fh,029h,029h,029h,009h,0c1h,0ebh,079h,03dh,03eh,021h,020h,00eh,005h,04fh,0fdh;
        defb    0cbh,001h,04eh,028h,006h,0d5h,0cdh,0cdh,00eh,0d1h,079h,0b9h,0d5h,0cch,055h,00ch;
        defb    0d1h,0c5h,0e5h,03ah,091h,05ch,006h,0ffh,01fh,038h,001h,004h,01fh,01fh,09fh,04fh;
        defb    03eh,008h,0a7h,0fdh,0cbh,001h,04eh,028h,005h,0fdh,0cbh,030h,0ceh,037h,0ebh,008h;
        defb    01ah,0a0h,0aeh,0a9h,012h,008h,038h,013h,014h,023h,03dh,020h,0f2h,0ebh,025h,0fdh;
        defb    0cbh,001h,04eh,0cch,0dbh,00bh,0e1h,0c1h,00dh,023h,0c9h,008h,03eh,020h,083h,05fh;
        defb    008h,018h,0e6h,07ch,00fh,00fh,00fh,0e6h,003h,0f6h,058h,067h,0edh,05bh,08fh,05ch;
        defb    07eh,0abh,0a2h,0abh,0fdh,0cbh,057h,076h,028h,008h,0e6h,0c7h,0cbh,057h,020h,002h;
        defb    0eeh,038h,0fdh,0cbh,057h,066h,028h,008h,0e6h,0f8h,0cbh,06fh,020h,002h,0eeh,007h;
        defb    077h,0c9h,0e5h,026h,000h,0e3h,018h,004h,011h,095h,000h,0f5h,0cdh,041h,00ch,038h;
        defb    009h,03eh,020h,0fdh,0cbh,001h,046h,0cch,03bh,00ch,01ah,0e6h,07fh,0cdh,03bh,00ch;
        defb    01ah,013h,087h,030h,0f5h,0d1h,0feh,048h,028h,003h,0feh,082h,0d8h,07ah,0feh,003h;
        defb    0d8h,03eh,020h,0d5h,0d9h,0d7h,0d9h,0d1h,0c9h,0f5h,0ebh,03ch,0cbh,07eh,023h,028h;
        defb    0fbh,03dh,020h,0f8h,0ebh,0f1h,0feh,020h,0d8h,01ah,0d6h,041h,0c9h,0fdh,0cbh,001h;
        defb    04eh,0c0h,011h,0d9h,00dh,0d5h,078h,0fdh,0cbh,002h,046h,0c2h,002h,00dh,0fdh,0beh;
        defb    031h,038h,01bh,0c0h,0fdh,0cbh,002h,066h,028h,016h,0fdh,05eh,02dh,01dh,028h,05ah;
        defb    03eh,000h,0cdh,001h,016h,0edh,07bh,03fh,05ch,0fdh,0cbh,002h,0a6h,0c9h,0cfh,004h;
        defb    0fdh,035h,052h,020h,045h,03eh,018h,090h,032h,08ch,05ch,02ah,08fh,05ch,0e5h,03ah;
        defb    091h,05ch,0f5h,03eh,0fdh,0cdh,001h,016h,0afh,011h,0f8h,00ch,0cdh,00ah,00ch,0fdh;
        defb    0cbh,002h,0eeh,021h,03bh,05ch,0cbh,0deh,0cbh,0aeh,0d9h,0cdh,0d4h,015h,0d9h,0feh;
        defb    020h,028h,045h,0feh,0e2h,028h,041h,0f6h,020h,0feh,06eh,028h,03bh,03eh,0feh,0cdh;
        defb    001h,016h,0f1h,032h,091h,05ch,0e1h,022h,08fh,05ch,0cdh,0feh,00dh,0fdh,046h,031h;
        defb    004h,00eh,021h,0c5h,0cdh,09bh,00eh,07ch,00fh,00fh,00fh,0e6h,003h,0f6h,058h,067h;
        defb    011h,0e0h,05ah,01ah,04eh,006h,020h,0ebh,012h,071h,013h,023h,010h,0fah,0c1h,0c9h;
        defb    080h,073h,063h,072h,06fh,06ch,06ch,0bfh,0cfh,00ch,0feh,002h,038h,080h,0fdh,086h;
        defb    031h,0d6h,019h,0d0h,0edh,044h,0c5h,047h,02ah,08fh,05ch,0e5h,02ah,091h,05ch,0e5h;
        defb    0cdh,04dh,00dh,078h,0f5h,021h,06bh,05ch,046h,078h,03ch,077h,021h,089h,05ch,0beh;
        defb    038h,003h,034h,006h,018h,0cdh,000h,00eh,0f1h,03dh,020h,0e8h,0e1h,0fdh,075h,057h;
        defb    0e1h,022h,08fh,05ch,0edh,04bh,088h,05ch,0fdh,0cbh,002h,086h,0cdh,0d9h,00dh,0fdh;
        defb    0cbh,002h,0c6h,0c1h,0c9h,0afh,02ah,08dh,05ch,0fdh,0cbh,002h,046h,028h,004h,067h;
        defb    0fdh,06eh,00eh,022h,08fh,05ch,021h,091h,05ch,020h,002h,07eh,00fh,0aeh,0e6h,055h;
        defb    0aeh,077h,0c9h,0cdh,0afh,00dh,021h,03ch,05ch,0cbh,0aeh,0cbh,0c6h,0cdh,04dh,00dh;
        defb    0fdh,046h,031h,0cdh,044h,00eh,021h,0c0h,05ah,03ah,08dh,05ch,005h,018h,007h,00eh;
        defb    020h,02bh,077h,00dh,020h,0fbh,010h,0f7h,0fdh,036h,031h,002h,03eh,0fdh,0cdh,001h;
        defb    016h,02ah,051h,05ch,011h,0f4h,009h,0a7h,073h,023h,072h,023h,011h,0a8h,010h,03fh;
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        defb    086h,0cdh,094h,00dh,03eh,0feh,0cdh,001h,016h,0cdh,04dh,00dh,006h,018h,0cdh,044h;
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        defb    018h,021h,000h,05bh,0fdh,0cbh,001h,04eh,020h,012h,078h,0fdh,0cbh,002h,046h,028h;
        defb    005h,0fdh,086h,031h,0d6h,018h,0c5h,047h,0cdh,09bh,00eh,0c1h,03eh,021h,091h,05fh;
        defb    016h,000h,019h,0c3h,0dch,00ah,006h,017h,0cdh,09bh,00eh,00eh,008h,0c5h,0e5h,078h;
        defb    0e6h,007h,078h,020h,00ch,0ebh,021h,0e0h,0f8h,019h,0ebh,001h,020h,000h,03dh,0edh;
        defb    0b0h,0ebh,021h,0e0h,0ffh,019h,0ebh,047h,0e6h,007h,00fh,00fh,00fh,04fh,078h,006h;
        defb    000h,0edh,0b0h,006h,007h,009h,0e6h,0f8h,020h,0dbh,0e1h,024h,0c1h,00dh,020h,0cdh;
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        defb    0cdh,088h,00eh,021h,0e0h,0ffh,019h,0ebh,0edh,0b0h,006h,001h,0c5h,0cdh,09bh,00eh;
        defb    00eh,008h,0c5h,0e5h,078h,0e6h,007h,00fh,00fh,00fh,04fh,078h,006h,000h,00dh,054h;
        defb    05dh,036h,000h,013h,0edh,0b0h,011h,001h,007h,019h,03dh,0e6h,0f8h,047h,020h,0e5h;
        defb    0e1h,024h,0c1h,00dh,020h,0dch,0cdh,088h,00eh,062h,06bh,013h,03ah,08dh,05ch,0fdh;
        defb    0cbh,002h,046h,028h,003h,03ah,048h,05ch,077h,00bh,0edh,0b0h,0c1h,00eh,021h,0c9h;
        defb    07ch,00fh,00fh,00fh,03dh,0f6h,050h,067h,0ebh,061h,068h,029h,029h,029h,029h,029h;
        defb    044h,04dh,0c9h,03eh,018h,090h,057h,00fh,00fh,00fh,0e6h,0e0h,06fh,07ah,0e6h,018h;
        defb    0f6h,040h,067h,0c9h,0f3h,006h,0b0h,021h,000h,040h,0e5h,0c5h,0cdh,0f4h,00eh,0c1h;
        defb    0e1h,024h,07ch,0e6h,007h,020h,00ah,07dh,0c6h,020h,06fh,03fh,09fh,0e6h,0f8h,084h;
        defb    067h,010h,0e7h,018h,00dh,0f3h,021h,000h,05bh,006h,008h,0c5h,0cdh,0f4h,00eh,0c1h;
        defb    010h,0f9h,03eh,004h,0d3h,0fbh,0fbh,021h,000h,05bh,0fdh,075h,046h,0afh,047h,077h;
        defb    023h,010h,0fch,0fdh,0cbh,030h,08eh,00eh,021h,0c3h,0d9h,00dh,078h,0feh,003h,09fh;
        defb    0e6h,002h,0d3h,0fbh,057h,0cdh,054h,01fh,038h,00ah,03eh,004h,0d3h,0fbh,0fbh,0cdh;
        defb    0dfh,00eh,0cfh,00ch,0dbh,0fbh,087h,0f8h,030h,0ebh,00eh,020h,05eh,023h,006h,008h;
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        defb    0cdh,0d4h,015h,0f5h,016h,000h,0fdh,05eh,0ffh,021h,0c8h,000h,0cdh,0b5h,003h,0f1h;
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        defb    016h,07ah,0b3h,0cah,097h,010h,0e5h,023h,04eh,023h,046h,021h,00ah,000h,009h,044h;
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        defb    037h,07eh,028h,0a8h,0c3h,081h,00fh,0fdh,0cbh,030h,066h,028h,0a1h,0fdh,036h,000h;
        defb    0ffh,016h,000h,0fdh,05eh,0feh,021h,090h,01ah,0cdh,0b5h,003h,0c3h,030h,00fh,0e5h;
        defb    0cdh,090h,011h,02bh,0cdh,0e5h,019h,022h,05bh,05ch,0fdh,036h,007h,000h,0e1h,0c9h;
        defb    0fdh,0cbh,002h,05eh,0c4h,01dh,011h,0a7h,0fdh,0cbh,001h,06eh,0c8h,03ah,008h,05ch;
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        defb    0d8h,0d6h,00dh,021h,041h,05ch,0beh,077h,020h,002h,036h,000h,0fdh,0cbh,002h,0deh;
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        defb    02ah,08ah,05ch,0e5h,02ah,03dh,05ch,0e5h,021h,067h,011h,0e5h,0edh,073h,03dh,05ch;
        defb    02ah,082h,05ch,0e5h,037h,0cdh,095h,011h,0ebh,0cdh,07dh,018h,0ebh,0cdh,0e1h,018h;
        defb    02ah,08ah,05ch,0e3h,0ebh,0cdh,04dh,00dh,03ah,08bh,05ch,092h,038h,026h,020h,006h;
        defb    07bh,0fdh,096h,050h,030h,01eh,03eh,020h,0d5h,0cdh,0f4h,009h,0d1h,018h,0e9h,016h;
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        defb    022h,082h,05ch,0fdh,036h,026h,000h,0c9h,02ah,061h,05ch,02bh,0a7h,0edh,05bh,059h;
        defb    05ch,0fdh,0cbh,037h,06eh,0c8h,0edh,05bh,061h,05ch,0d8h,02ah,063h,05ch,0c9h,07eh;
        defb    0feh,00eh,001h,006h,000h,0cch,0e8h,019h,07eh,023h,0feh,00dh,020h,0f1h,0c9h,0f3h;
        defb    03eh,0ffh,0edh,05bh,0b2h,05ch,0d9h,0edh,04bh,0b4h,05ch,0edh,05bh,038h,05ch,02ah;
        defb    07bh,05ch,0d9h,047h,03eh,007h,0d3h,0feh,03eh,03fh,0edh,047h,000h,000h,000h,000h;
        defb    000h,000h,062h,06bh,036h,002h,02bh,0bch,020h,0fah,0a7h,0edh,052h,019h,023h,030h;
        defb    006h,035h,028h,003h,035h,028h,0f3h,02bh,0d9h,0edh,043h,0b4h,05ch,0edh,053h,038h;
        defb    05ch,022h,07bh,05ch,0d9h,004h,028h,019h,022h,0b4h,05ch,011h,0afh,03eh,001h,0a8h;
        defb    000h,0ebh,0edh,0b8h,0ebh,023h,022h,07bh,05ch,02bh,001h,040h,000h,0edh,043h,038h;
        defb    05ch,022h,0b2h,05ch,021h,000h,03ch,022h,036h,05ch,02ah,0b2h,05ch,036h,03eh,02bh;
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        defb    0cbh,000h,07eh,020h,012h,0fdh,0cbh,030h,066h,028h,040h,02ah,059h,05ch,0cdh,0a7h;
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        defb    005h,000h,030h,002h,00eh,012h,017h,023h,07eh,030h,0fbh,018h,006h,023h,023h,04eh;
        defb    023h,046h,023h,009h,0d1h,0a7h,0edh,052h,044h,04dh,019h,0ebh,0c9h,0cdh,0ddh,019h;
        defb    0c5h,078h,02fh,047h,079h,02fh,04fh,003h,0cdh,064h,016h,0ebh,0e1h,019h,0d5h,0edh;
        defb    0b0h,0e1h,0c9h,02ah,059h,05ch,02bh,022h,05dh,05ch,0e7h,021h,092h,05ch,022h,065h;
        defb    05ch,0cdh,03bh,02dh,0cdh,0a2h,02dh,038h,004h,021h,0f0h,0d8h,009h,0dah,08ah,01ch;
        defb    0c3h,0c5h,016h,0d5h,0e5h,0afh,0cbh,078h,020h,020h,060h,069h,01eh,0ffh,018h,008h;
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        defb    037h,077h,044h,00fh,059h,02bh,043h,02dh,051h,03ah,06dh,042h,00dh,049h,05ch,044h;
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        defb    0b2h,02ah,0cdh,0ffh,02ah,018h,003h,0cdh,0fch,01fh,0cdh,04eh,020h,0cah,0c1h,020h;
        defb    0c9h,02ah,061h,05ch,022h,05dh,05ch,0dfh,0feh,0e2h,028h,00ch,03ah,071h,05ch,0cdh;
        defb    059h,01ch,0dfh,0feh,00dh,0c8h,0cfh,00bh,0cdh,030h,025h,0c8h,0cfh,010h,02ah,051h;
        defb    05ch,023h,023h,023h,023h,07eh,0feh,04bh,0c9h,0e7h,0cdh,0f2h,021h,0d8h,0dfh,0feh;
        defb    02ch,028h,0f6h,0feh,03bh,028h,0f2h,0c3h,08ah,01ch,0feh,0d9h,0d8h,0feh,0dfh,03fh;
        defb    0d8h,0f5h,0e7h,0f1h,0d6h,0c9h,0f5h,0cdh,082h,01ch,0f1h,0a7h,0cdh,0c3h,01fh,0f5h;
        defb    0cdh,094h,01eh,057h,0f1h,0d7h,07ah,0d7h,0c9h,0d6h,011h,0ceh,000h,028h,01dh,0d6h;
        defb    002h,0ceh,000h,028h,056h,0feh,001h,07ah,006h,001h,020h,004h,007h,007h,006h,004h;
        defb    04fh,07ah,0feh,002h,030h,016h,079h,021h,091h,05ch,018h,038h,07ah,006h,007h,038h;
        defb    005h,007h,007h,007h,006h,038h,04fh,07ah,0feh,00ah,038h,002h,0cfh,013h,021h,08fh;
        defb    05ch,0feh,008h,038h,00bh,07eh,028h,007h,0b0h,02fh,0e6h,024h,028h,001h,078h,04fh;
        defb    079h,0cdh,06ch,022h,03eh,007h,0bah,09fh,0cdh,06ch,022h,007h,007h,0e6h,050h,047h;
        defb    03eh,008h,0bah,09fh,0aeh,0a0h,0aeh,077h,023h,078h,0c9h,09fh,07ah,00fh,006h,080h;
        defb    020h,003h,00fh,006h,040h,04fh,07ah,0feh,008h,028h,004h,0feh,002h,030h,0bdh,079h;
        defb    021h,08fh,05ch,0cdh,06ch,022h,079h,00fh,00fh,00fh,018h,0d8h,0cdh,094h,01eh,0feh;
        defb    008h,030h,0a9h,0d3h,0feh,007h,007h,007h,0cbh,06fh,020h,002h,0eeh,007h,032h,048h;
        defb    05ch,0c9h,03eh,0afh,090h,0dah,0f9h,024h,047h,0a7h,01fh,037h,01fh,0a7h,01fh,0a8h;
        defb    0e6h,0f8h,0a8h,067h,079h,007h,007h,007h,0a8h,0e6h,0c7h,0a8h,007h,007h,06fh,079h;
        defb    0e6h,007h,0c9h,0cdh,007h,023h,0cdh,0aah,022h,047h,004h,07eh,007h,010h,0fdh,0e6h;
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        defb    05ch,0cdh,0aah,022h,047h,004h,03eh,0feh,00fh,010h,0fdh,047h,07eh,0fdh,04eh,057h;
        defb    0cbh,041h,020h,001h,0a0h,0cbh,051h,020h,002h,0a8h,02fh,077h,0c3h,0dbh,00bh,0cdh;
        defb    014h,023h,047h,0c5h,0cdh,014h,023h,059h,0c1h,051h,04fh,0c9h,0cdh,0d5h,02dh,0dah;
        defb    0f9h,024h,00eh,001h,0c8h,00eh,0ffh,0c9h,0dfh,0feh,02ch,0c2h,08ah,01ch,0e7h,0cdh;
        defb    082h,01ch,0cdh,0eeh,01bh,0efh,02ah,03dh,038h,07eh,0feh,081h,030h,005h,0efh,002h;
        defb    038h,018h,0a1h,0efh,0a3h,038h,036h,083h,0efh,0c5h,002h,038h,0cdh,07dh,024h,0c5h;
        defb    0efh,031h,0e1h,004h,038h,07eh,0feh,080h,030h,008h,0efh,002h,002h,038h,0c1h,0c3h;
        defb    0dch,022h,0efh,0c2h,001h,0c0h,002h,003h,001h,0e0h,00fh,0c0h,001h,031h,0e0h,001h;
        defb    031h,0e0h,0a0h,0c1h,002h,038h,0fdh,034h,062h,0cdh,094h,01eh,06fh,0e5h,0cdh,094h;
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        defb    01eh,0e1h,067h,022h,07dh,05ch,0c1h,0c3h,020h,024h,0dfh,0feh,02ch,028h,006h,0cdh;
        defb    0eeh,01bh,0c3h,077h,024h,0e7h,0cdh,082h,01ch,0cdh,0eeh,01bh,0efh,0c5h,0a2h,004h;
        defb    01fh,031h,030h,030h,000h,006h,002h,038h,0c3h,077h,024h,0c0h,002h,0c1h,002h,031h;
        defb    02ah,0e1h,001h,0e1h,02ah,00fh,0e0h,005h,02ah,0e0h,001h,03dh,038h,07eh,0feh,081h;
        defb    030h,007h,0efh,002h,002h,038h,0c3h,077h,024h,0cdh,07dh,024h,0c5h,0efh,002h,0e1h;
        defb    001h,005h,0c1h,002h,001h,031h,0e1h,004h,0c2h,002h,001h,031h,0e1h,004h,0e2h,0e5h;
        defb    0e0h,003h,0a2h,004h,031h,01fh,0c5h,002h,020h,0c0h,002h,0c2h,002h,0c1h,0e5h,004h;
        defb    0e0h,0e2h,004h,00fh,0e1h,001h,0c1h,002h,0e0h,004h,0e2h,0e5h,004h,003h,0c2h,02ah;
        defb    0e1h,02ah,00fh,002h,038h,01ah,0feh,081h,0c1h,0dah,077h,024h,0c5h,0efh,001h,038h;
        defb    03ah,07dh,05ch,0cdh,028h,02dh,0efh,0c0h,00fh,001h,038h,03ah,07eh,05ch,0cdh,028h;
        defb    02dh,0efh,0c5h,00fh,0e0h,0e5h,038h,0c1h,005h,028h,03ch,018h,014h,0efh,0e1h,031h;
        defb    0e3h,004h,0e2h,0e4h,004h,003h,0c1h,002h,0e4h,004h,0e2h,0e3h,004h,00fh,0c2h,002h;
        defb    038h,0c5h,0efh,0c0h,002h,0e1h,00fh,031h,038h,03ah,07dh,05ch,0cdh,028h,02dh,0efh;
        defb    003h,0e0h,0e2h,00fh,0c0h,001h,0e0h,038h,03ah,07eh,05ch,0cdh,028h,02dh,0efh,003h;
        defb    038h,0cdh,0b7h,024h,0c1h,010h,0c6h,0efh,002h,002h,001h,038h,03ah,07dh,05ch,0cdh;
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        defb    0e6h,018h,0eeh,040h,057h,006h,060h,0c5h,0d5h,0e5h,01ah,0aeh,028h,004h,03ch,020h;
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        defb    0c1h,03eh,080h,090h,001h,001h,000h,0f7h,012h,018h,00ah,0e1h,011h,008h,000h,019h;
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        defb    0aah,0bfh,0abh,0c7h,0a9h,0ceh,000h,0e7h,0c3h,0ffh,024h,0dfh,023h,0e5h,001h,000h;
        defb    000h,0cdh,00fh,025h,020h,01bh,0cdh,00fh,025h,028h,0fbh,0cdh,030h,025h,028h,011h;
        defb    0f7h,0e1h,0d5h,07eh,023h,012h,013h,0feh,022h,020h,0f8h,07eh,023h,0feh,022h,028h;
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        defb    0a2h,02dh,0edh,043h,076h,05ch,07eh,0a7h,028h,003h,0d6h,010h,077h,018h,009h,0cdh;
        defb    030h,025h,028h,004h,0efh,0a3h,038h,034h,0e7h,0c3h,0c3h,026h,001h,05ah,010h,0e7h;
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        defb    002h,00eh,000h,020h,013h,0cdh,01eh,003h,030h,00eh,015h,05fh,0cdh,033h,003h,0f5h;
        defb    001h,001h,000h,0f7h,0f1h,012h,00eh,001h,006h,000h,0cdh,0b2h,02ah,0c3h,012h,027h;
        defb    0cdh,022h,025h,0c4h,035h,025h,0e7h,0c3h,0dbh,025h,0cdh,022h,025h,0c4h,080h,025h;
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        defb    056h,0feh,041h,030h,03ch,0cdh,030h,025h,020h,023h,0cdh,09bh,02ch,0dfh,001h,006h;
        defb    000h,0cdh,055h,016h,023h,036h,00eh,023h,0ebh,02ah,065h,05ch,00eh,005h,0a7h,0edh;
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        defb    014h,0cdh,0b2h,028h,0dah,02eh,01ch,0cch,096h,029h,03ah,03bh,05ch,0feh,0c0h,038h;
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        defb    010h,0feh,0aeh,028h,020h,0d6h,0afh,0dah,08ah,01ch,001h,0f0h,004h,0feh,014h,028h;
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        defb    0eeh,038h,002h,0cbh,0b9h,0c5h,0e7h,0c3h,0ffh,024h,0dfh,0feh,028h,020h,00ch,0fdh;
        defb    0cbh,001h,076h,020h,017h,0cdh,052h,02ah,0e7h,018h,0f0h,006h,000h,04fh,021h,095h;
        defb    027h,0cdh,0dch,016h,030h,006h,04eh,021h,0edh,026h,009h,046h,0d1h,07ah,0b8h,038h;
        defb    03ah,0a7h,0cah,018h,000h,0c5h,021h,03bh,05ch,07bh,0feh,0edh,020h,006h,0cbh,076h;
        defb    020h,002h,01eh,099h,0d5h,0cdh,030h,025h,028h,009h,07bh,0e6h,03fh,047h,0efh,03bh;
        defb    038h,018h,009h,07bh,0fdh,0aeh,001h,0e6h,040h,0c2h,08ah,01ch,0d1h,021h,03bh,05ch;
        defb    0cbh,0f6h,0cbh,07bh,020h,002h,0cbh,0b6h,0c1h,018h,0c1h,0d5h,079h,0fdh,0cbh,001h;
        defb    076h,020h,015h,0e6h,03fh,0c6h,008h,04fh,0feh,010h,020h,004h,0cbh,0f1h,018h,008h;
        defb    038h,0d7h,0feh,017h,028h,002h,0cbh,0f9h,0c5h,0e7h,0c3h,0ffh,024h,02bh,0cfh,02dh;
        defb    0c3h,02ah,0c4h,02fh,0c5h,05eh,0c6h,03dh,0ceh,03eh,0cch,03ch,0cdh,0c7h,0c9h,0c8h;
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        defb    01ch,0e7h,0feh,024h,0f5h,020h,001h,0e7h,0feh,028h,020h,012h,0e7h,0feh,029h,028h;
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        defb    040h,028h,007h,02bh,0cdh,0abh,028h,023h,016h,000h,023h,0e5h,0d5h,0cdh,0fbh,024h;
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        defb    05ch,0d5h,0e7h,0e7h,0cdh,0fbh,024h,0e1h,022h,05dh,05ch,0e1h,022h,00bh,05ch,0e7h;
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        defb    08dh,02ch,0d2h,08ah,01ch,0e5h,0e6h,01fh,04fh,0e7h,0e5h,0feh,028h,028h,028h,0cbh;
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        defb    017h,087h,0f2h,03fh,029h,038h,030h,0d1h,0d5h,0e5h,023h,01ah,013h,0feh,020h,028h;
        defb    0fah,0f6h,020h,0beh,028h,0f4h,0f6h,080h,0beh,020h,006h,01ah,0cdh,088h,02ch,030h;
        defb    015h,0e1h,0c5h,0cdh,0b8h,019h,0ebh,0c1h,018h,0ceh,0cbh,0f8h,0d1h,0dfh,0feh,028h;
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        defb    0e7h,018h,0f8h,0e1h,0cbh,010h,0cbh,070h,0c9h,02ah,00bh,05ch,07eh,0feh,029h,0cah;
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        defb    029h,0cah,0efh,028h,0cdh,0abh,028h,018h,0d9h,0cbh,069h,020h,00ch,023h,0edh,05bh;
        defb    065h,05ch,0cdh,0c0h,033h,0ebh,022h,065h,05ch,0d1h,0d1h,0afh,03ch,0c9h,0afh,047h;
        defb    0cbh,079h,020h,04bh,0cbh,07eh,020h,00eh,03ch,023h,04eh,023h,046h,023h,0ebh,0cdh;
        defb    0b2h,02ah,0dfh,0c3h,049h,02ah,023h,023h,023h,046h,0cbh,071h,028h,00ah,005h,028h;
        defb    0e8h,0ebh,0dfh,0feh,028h,020h,061h,0ebh,0ebh,018h,024h,0e5h,0dfh,0e1h,0feh,02ch;
        defb    028h,020h,0cbh,079h,028h,052h,0cbh,071h,020h,006h,0feh,029h,020h,03ch,0e7h,0c9h;
        defb    0feh,029h,028h,06ch,0feh,0cch,020h,032h,0dfh,02bh,022h,05dh,05ch,018h,05eh,021h;
        defb    000h,000h,0e5h,0e7h,0e1h,079h,0feh,0c0h,020h,009h,0dfh,0feh,029h,028h,051h,0feh;
        defb    0cch,028h,0e5h,0c5h,0e5h,0cdh,0eeh,02ah,0e3h,0ebh,0cdh,0cch,02ah,038h,019h,00bh;
        defb    0cdh,0f4h,02ah,009h,0d1h,0c1h,010h,0b3h,0cbh,079h,020h,066h,0e5h,0cbh,071h,020h;
        defb    013h,042h,04bh,0dfh,0feh,029h,028h,002h,0cfh,002h,0e7h,0e1h,011h,005h,000h,0cdh;
        defb    0f4h,02ah,009h,0c9h,0cdh,0eeh,02ah,0e3h,0cdh,0f4h,02ah,0c1h,009h,023h,042h,04bh;
        defb    0ebh,0cdh,0b1h,02ah,0dfh,0feh,029h,028h,007h,0feh,02ch,020h,0dbh,0cdh,052h,02ah;
        defb    0e7h,0feh,028h,028h,0f8h,0fdh,0cbh,001h,0b6h,0c9h,0cdh,030h,025h,0c4h,0f1h,02bh;
        defb    0e7h,0feh,029h,028h,050h,0d5h,0afh,0f5h,0c5h,011h,001h,000h,0dfh,0e1h,0feh,0cch;
        defb    028h,017h,0f1h,0cdh,0cdh,02ah,0f5h,050h,059h,0e5h,0dfh,0e1h,0feh,0cch,028h,009h;
        defb    0feh,029h,0c2h,08ah,01ch,062h,06bh,018h,013h,0e5h,0e7h,0e1h,0feh,029h,028h,00ch;
        defb    0f1h,0cdh,0cdh,02ah,0f5h,0dfh,060h,069h,0feh,029h,020h,0e6h,0f1h,0e3h,019h,02bh;
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        defb    0a9h,033h,0c1h,02ah,065h,05ch,077h,023h,073h,023h,072h,023h,071h,023h,070h,023h;
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defb    0d1h,0d9h,0e1h,0d9h,0c9h,0cfh,005h,0cdh,093h,032h,0ebh,0afh,0cdh,0c0h,030h,038h;
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defb    060h,069h,0d9h,061h,068h,0afh,006h,0dfh,018h,010h,017h,0cbh,011h,0d9h,0cbh,011h;
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Characters:
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        defb    040h,040h,040h,040h,040h,07eh, 042h,066h,05ah,042h,042h,042h;
        defb    042h,062h,052h,04ah,046h,042h, 03ch,042h,042h,042h,042h,03ch;
        defb    07ch,042h,042h,07ch,040h,040h, 03ch,042h,042h,052h,04ah,03ch;
        defb    07ch,042h,042h,07ch,044h,042h, 03ch,040h,03ch,002h,042h,03ch;
        defb    0feh,010h,010h,010h,010h,010h, 042h,042h,042h,042h,042h,03ch;
        defb    042h,042h,042h,042h,024h,018h, 042h,042h,042h,042h,05ah,024h;
        defb    042h,024h,018h,018h,024h,042h, 082h,044h,028h,010h,010h,010h;
        defb    07eh,004h,008h,010h,020h,07eh, 00eh,008h,008h,008h,008h,00eh;
        defb    000h,040h,020h,010h,008h,004h, 070h,010h,010h,010h,010h,070h;
        defb    010h,038h,054h,010h,010h,010h, 000h,000h,000h,000h,000h,000h;
        defb    01ch,022h,078h,020h,020h,07eh, 000h,038h,004h,03ch,044h,03ch;
        defb    020h,020h,03ch,022h,022h,03ch, 000h,01ch,020h,020h,020h,01ch;
        defb    004h,004h,03ch,044h,044h,03ch, 000h,038h,044h,078h,040h,03ch;
        defb    00ch,010h,018h,010h,010h,010h, 000h,03ch,044h,044h,03ch,004h;
        defb    040h,040h,078h,044h,044h,044h, 010h,000h,030h,010h,010h,038h;
        defb    004h,000h,004h,004h,004h,024h, 020h,028h,030h,030h,028h,024h;
        defb    010h,010h,010h,010h,010h,00ch, 000h,068h,054h,054h,054h,054h;
        defb    000h,078h,044h,044h,044h,044h, 000h,038h,044h,044h,044h,038h;
        defb    000h,078h,044h,044h,078h,040h, 000h,03ch,044h,044h,03ch,004h;
        defb    000h,01ch,020h,020h,020h,020h, 000h,038h,040h,038h,004h,078h;
        defb    010h,038h,010h,010h,010h,00ch, 000h,044h,044h,044h,044h,038h;
        defb    000h,044h,044h,028h,028h,010h, 000h,044h,054h,054h,054h,028h;
        defb    000h,044h,028h,010h,028h,044h, 000h,044h,044h,044h,03ch,004h;
        defb    000h,07ch,008h,010h,020h,07ch, 00eh,008h,030h,008h,008h,00eh;
        defb    008h,008h,008h,008h,008h,008h, 070h,010h,00ch,010h,010h,070h;
        defb    014h,028h,000h,000h,000h,000h, 042h,099h,0a1h,0a1h,099h,042h;

        end
```

Code 1. Turbo Spectrum Boot Code

4 CoBra Logo

The CoBra logo appearing on posters and documentation was designed by architect Alexandru Antal. The CoBra logo in the boot code is generated in the center third of the screen; see Figure 38, where every pixel is shown. The thin lines are Spectrum 8x8 character demarcation lines; they do not appear on the screen.

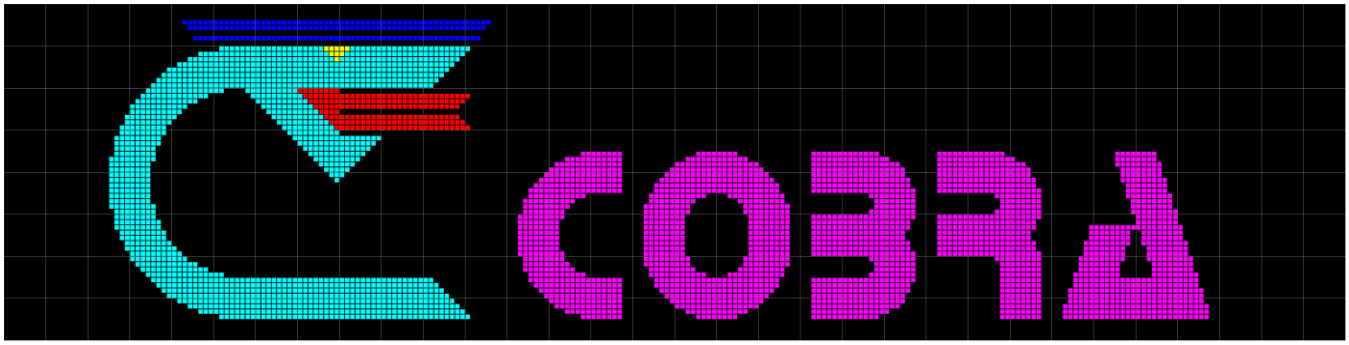


Figure 38: CoBra Logo

To reduce code and data size, the CoBra logo is compressed. The total size of code and data for logo generation and rotation is 820 bytes. The width of the letters and snake in the vertical sections is 8 pixels but is wider in the curved areas. There are many instances of 8 to 16 consecutive INK pixels in two consecutive bytes. The 44 possible patterns are shown in Figure 39. Code 2 is the CoBra boot code which includes the CoBra logo source code with comments describing how it works.

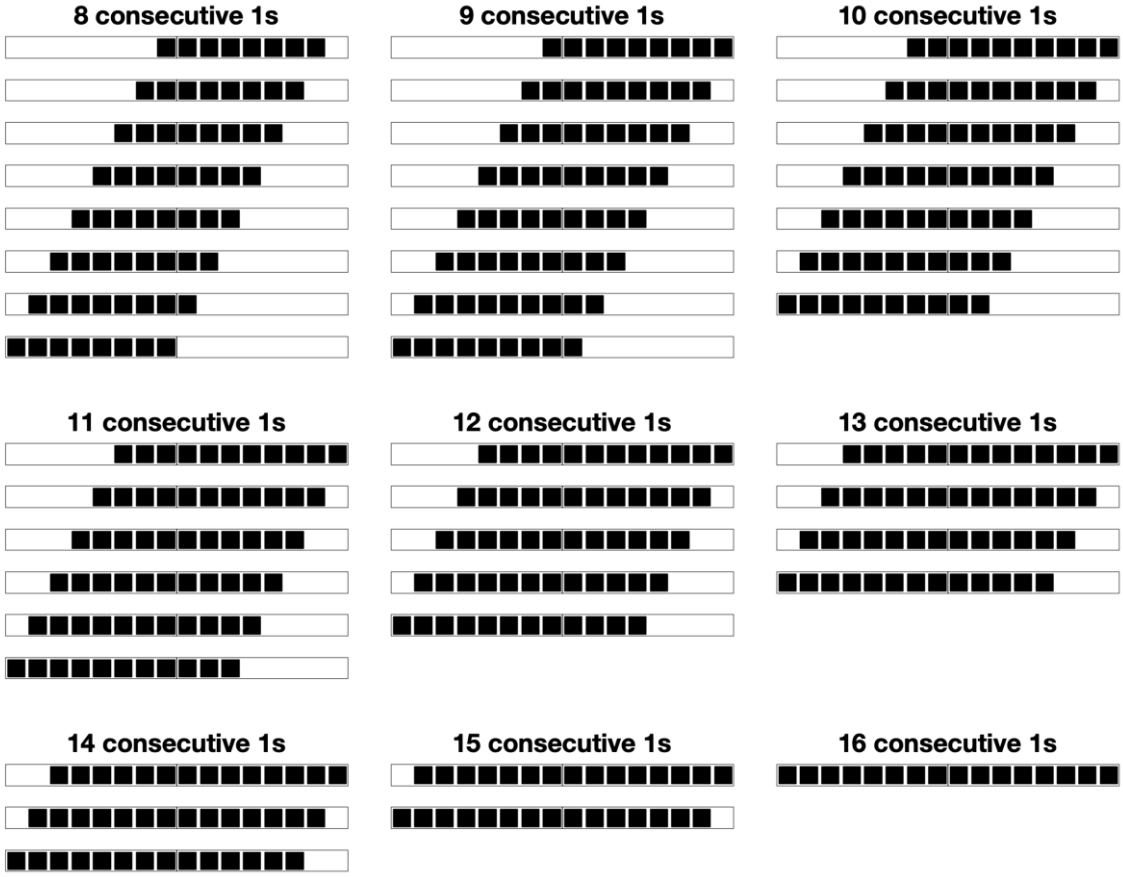


Figure 39: CoBra Logo Patterns

```
; CoBra logo code disassembled and commented on 5 February 2022 by sorin@cismas.org
; about 36 years after it was written (the original source code was lost).
;
; The CoBra logo appearing on posters and documentation was designed by architect
; Alexandru Antal.  We wanted to have that logo on the CoBra start-up screen.
; The original Sinclair ZX Spectrum ROM had 1170 unused bytes, 0386eh - 03cffh.
; The CoBra logo generation and rotation takes 820 bytes, consisting of:
;   608 bytes table                         (address 00122h - 00381h)
;   129 bytes code for logo generation      (address 00026h - 000afh)
;    34 bytes code for full screen rotation (address 000cdh - 000efh)
;    49 bytes code for 4 pixel rotation     (address 000f1h - 00122h)
; The remaining 1170 - 820 = 350 bytes were sufficient for CoBra initialization.
;
; After support for CP/M was added, the boot code become larger and the 1170 bytes
; were no longer sufficient.  A decision was made to add a 2KB boot ROM.
; Initially, the boot ROM did not have the CoBra logo.  It was added at the last
; minute because it would fit in the 2KB boot ROM.
;
; To reduce the size, the CoBra logo is compressed.  The width of the letters
; and snake in the vertical sections is 8 pixels but is wider in the curved areas.
; There are many instances of 8 to 16 consecutive INK pixels in 2 consecutive bytes.
; Here are the 44 possible patterns:
;   01fe  01ff  03ff  07ff  0fff  1fff  3fff  7fff  ffff
;   03fc  03fe  07fe  0ffe  1ffe  3ffe  7ffe  fffe
;   07f8  07fc  0ffc  1ffc  3ffc  7ffc  fffc
;   0ff0  0ff8  1ff8  3ff8  7ff8  fff8
;   1fe0  1ff0  3ff0  7ff0  fff0
;   3fc0  3fe0  7fe0  ffe0
;   7f80  7fc0  ffc0
;   ff00  ff80
;
; CoBra_tbl has 3 sections.  The first (00122h - 0034dh) holds information
; about where these patterns are present, the second (0034e - 00379h) has
; final additions, and the third holds the 8 bytes for the color pattern.
;
; The structure of the first section is:
;     1 byte   Number of Word patterns (N)
;     2 bytes  Base Address (lowest) of the first instance
;   N-1 bytes  Adress Increments
; The Increments are only 8 bits.  In case the delta between two consecutive
; addresses is larger than 255, an intermediate write is done to a location that
; will be later overwritten.  The patterns are ordered as in the above table,
; vertical first, then horizontal, as the "Word_xxxx" table labels below show.
; There are 3 nested loops, the outer loop goes through the columns, the middle
; loop through the rows, and the inner loop through the pattern instances.
;
; The second section is a list of byte patterns that could not be generated with
; the 44 16-bit patterns.  There are 11 of them, the first 5 are in the 6KB
; screen area and remaining 6 are color attributes (blue crown, yellow eye,
; red tongue, leading edge of cyan C).  The structure of the second section is:
;     1 byte   Number of consecutive Byte patterns
;     2 bytes  Base Address (lowest) of the first instance
;     1 byte   Byte pattern
;
; CoBra logo generation goes through 4 stages:
;    1. Fill color attributes in the middle 3rd of the screen: first 12 colums
;       with magenta INK (snake) and the last 20 with cyan INK (COBRA).
;    2. Use section 1 of CoBra_tbl to unpack the 44 16-bit patterns.
;    3. Use section 2 of CoBra_tbl to unpack the finishing bytes.
;    4. Use section 3 of CoBra_tbl to write the color pattern.
;
; CoBra logo rotation is not synchronized with the frame interrupt, which causes
; tearing.  This was by design, not a mistake.  Rotation starts from the bottom
; line and propagates up.  Tearing will give the sensation of undulating motion.
; CoBra rotation is done in 3 steps and is repeated 32 times to bring it back in
; the same position:
;    1. Rotate left CoBra logo 4 pixels.
;    2. Rotate left color attributes.
;    3. Rotate left CoBra logo 4 pixels.

        org     00000h

start:
; initialize CoBra
        ld a,080h       ;0000  3e 80   ;| R[7]=1
        ld r,a          ;0002  ed 4f   ;|
        ld hl,0069ch    ;0004  21 9c 06;| (0f000h)=0069ch
        ld (0f000h),hl  ;0007  22 00 f0;|
        ld sp,0ffffh    ;000a  31 ff ff;  initialize SP=ffff
        ld h,0c0h       ;000d  26 c0   ;| 0c000h is video base address
        ld l,000h       ;000f  2e 00   ;|
        ld (hl),000h    ;0011  36 00   ;  (0c000h)=0
        ld d,h          ;0013  54      ;|  0c001h
        ld e,001h       ;0014  1e 01   ;|
        ld bc,01affh    ;0016  01 ff 1a;  01b00h is video memory size
```

```
        ldir            ;0019  ed b0   ;  clear video memory
        ld a,092h       ;001b  3e 92   ;| out(092dfh)=092h
        out (0dfh),a    ;001d  d3 df   ;|
        ld a,040h       ;001f  3e 40   ;| out(040feh)=040h
        out (0feh),a    ;0021  d3 fe   ;|

        ld bc,078feh    ;0023  01 fe 78;  NOT NEEDED
; The initial boot did not have the CoBra logo and the keyboard test
; that starts at 000adh was starting here. This is a leftover instruction
; after the CoBra logo was added to the CoBra boot code at the last minute

; generate CoBra logo
        ld h,0d9h       ;0026  26 d9   ;| 0d900h is start Address
        ld l,000h       ;0028  2e 00   ;| of middle 3rd of screen attributes
        ld b,008h       ;002a  06 08   ;  8 rows
l002ch:                             ;
        push bc         ;002c  c5      ;  save BC  <-- <-- <-- <-- <--,
        ld bc,00c05h    ;002d  01 05 0c;| fill first 12 bytes with 5  |
l0030h:                             ;| cyan INK on black PAPER     ^
        ld (hl),c       ;0030  71      ;|                             |
        inc hl          ;0031  23      ;|                             ^
        djnz l0030h     ;0032  10 fc   ;|                             |
        ld bc,01403h    ;0034  01 03 14;| fill next 20 bytes with 3   ^
l0037h:                             ;| magenta INK on black PAPER  |
        ld (hl),c       ;0037  71      ;|                             ^
        inc hl          ;0038  23      ;|                             |
        djnz l0037h     ;0039  10 fc   ;|                             ^
        pop bc          ;003b  c1      ;  restore BC                  |
        djnz l002ch     ;003c  10 ee   ;  loop 8 times --> --> --> -->'
        exx             ;003e  d9      ;
        ld hl,CoBra_tbl ;003f  21 22 01;  load CoBra_tbl base address to H'L'
        exx             ;0042  d9      ;
        ld h,0cbh       ;0043  26 cb   ;| load HL with first Base Address
        ld l,0b9h       ;0045  2e b9   ;|
        ld de,001feh    ;0047  11 fe 01;  load DE with Word 001feh
        ld a,004h       ;004a  3e 04   ;  load A with Number of Words
        ld b,009h       ;004c  06 09   ;  9 outer loops
l004eh:                             ;
        push bc         ;004e  c5      ;  save BC  <-- <-- <-- <-- <-- <-- <-,
        ld b,a          ;004f  47      ;  Number of Words                    |
l0050h:                             ;                                        ^
        ld (hl),d       ;0050  72      ;  write upper byte <-- <--,<-- <--,  |
        inc hl          ;0051  23      ;| write lower byte        ^       ^  ^
        ld (hl),e       ;0052  73      ;|                         |       |  |
        exx             ;0053  d9      ;                          ^       ^  ^
        ld a,(hl)       ;0054  7e      ;  load A with Increment   |       |  |
        inc hl          ;0055  23      ;  increment H'L' pointer  ^       ^  ^
        exx             ;0056  d9      ;                          |       |  |
        ld c,a          ;0057  4f      ;  save byte read to C     ^       ^  ^
        add a,l         ;0058  85      ;| add hl,a                |       |  |
        ld l,a          ;0059  6f      ;|                         ^       ^  ^
        jr nc,l005dh    ;005a  30 01   ;|                         |       |  |
        inc h           ;005c  24      ;|                         ^       ^  ^
l005dh:                             ;                            |       |  |
        djnz l0050h     ;005d  10 f1   ;  close inner loop --> -->'       ^  ^
        ld b,c          ;005f  41      ;  byte read from H'L' pointer     |  |
        exx             ;0060  d9      ;                                  ^  ^
        ld e,(hl)       ;0061  5e      ;| load DE with next Base Address  |  |
        inc hl          ;0062  23      ;|                                 ^  ^
        ld d,(hl)       ;0063  56      ;|                                 |  |
        inc hl          ;0064  23      ;|                                 ^  ^
        push de         ;0065  d5      ;  save DE                         |  |
        exx             ;0066  d9      ;                                  ^  ^
        pop hl          ;0067  e1      ;  HL contains next Base Address   |  |
        sla e           ;0068  cb 23   ;| shift left arithmetic DE        ^  ^
        rl d            ;006a  cb 12   ;|                                 |  |
        jr nc,l0050h    ;006c  30 e2   ;  close middle loop -> --> --> -->'  ^
        rr d            ;006e  cb 1a   ;| unshift DE                       |
        rr e            ;0070  cb 1b   ;|                                  ^
        ld d,0ffh       ;0072  16 ff   ;  NOT NEEDED (redundant)           |
l0074h:                             ;                                      ^
        sla d           ;0074  cb 22   ;| determine next Word              |
        rl e            ;0076  cb 13   ;| with one extra 1 bit             ^
        jr c,l0074h     ;0078  38 fa   ;| right aligned                    |
        ld a,d          ;007a  7a      ;|                                  ^
        cpl             ;007b  2f      ;|                                  |
        ld d,a          ;007c  57      ;|                                  ^
        ld e,0ffh       ;007d  1e ff   ;|                                  |
        ld a,c          ;007f  79      ;  Number of Words/Bytes            ^
        pop bc          ;0080  c1      ;  restore BC                       |
        djnz l004eh     ;0081  10 cb   ;  close outer loop --> --> --> --> --'
        ld e,l          ;0083  5d      ;| save Address to DE
        ld d,h          ;0084  54      ;|
        exx             ;0085  d9      ;| copy H'L' pointer to HL
        push hl         ;0086  e5      ;|
```

```
        exx             ;0087  d9      ;|
        pop hl          ;0088  e1      ;|
        ld b,00bh       ;0089  06 0b   ;  11 finishing loops (should have been 12)
l008bh:                                ;
        push bc         ;008b  c5      ;  save BC <-- <-- <-- <-- <-- <,
        ld b,a          ;008c  47      ;  Number of Bytes              ^
        ld a,(hl)       ;008d  7e      ;| load A with Byte              |
        inc hl          ;008e  23      ;|                               ^
l008fh:                                ;                               |
        ld (de),a       ;008f  12      ;| write Byte B times             ^
        inc de          ;0090  13      ;|                               |
        djnz l008fh     ;0091  10 fc   ;|                               ^
        ld a,(hl)       ;0093  7e      ;| load A with Number of Bytes   |
        inc hl          ;0094  23      ;|                               ^
        ld e,(hl)       ;0095  5e      ;| load DE with next Address     |
        inc hl          ;0096  23      ;|                               ^
        ld d,(hl)       ;0097  56      ;|                               |
        inc hl          ;0098  23      ;|                               ^
        pop bc          ;0099  c1      ;  restore BC                    |
        djnz l008bh     ;009a  10 ef   ;  close finishing loop -> --> -'
        ld bc,00008h    ;009c  01 08 00;| add color pattern
        ld de,0da8ch    ;009f  11 8c da;|
        ld hl,Colors    ;00a2  21 7a 03;|
        ldir            ;00a5  ed b0   ;|

; screen cycle
l00a7h:
        ld d,007h       ;00a7  16 07   ;  7 outer loops  <-- <-- <-- <-- <-- <-- <--,
        ld bc,00000h    ;00a9  01 00 00;  256 middle (C) and 256 inner (B) loops    ^
l00ach:                                ;  which introduces a delay of 19.4 seconds  | cycles
        push bc         ;00ac  c5      ;  save BC  <-- <-- <-- <-- <-- <-- <-,-,-,  ^   11
; keyboard test                         ;                                     | | |  |
        ld bc,078feh    ;00ad  01 fe 78;                                      ^ ^ ^  ^   10
        in a,(c)        ;00b0  ed 78   ;  A=in(078feh), rows KAf,KAa,KA9,KA8 | | |  |   12
        nop             ;00b2  00      ;| old instruction(s) nullified       ^ ^ ^  ^    4
        nop             ;00b3  00      ;| at the last minute                 | | |  |    4
        jp 00705h       ;00b4  c3 05 07;  jump to partial keyboard test      ^ ^ ^  ^   38
KB_return:                             ;  which returns here                | | |  |
        rra             ;00b7  1f      ;  rotate right all, CY=or(col 2)     ^ ^ ^  ^    4
        jp nc,003d8h    ;00b8  d2 d8 03;  jump if "M,"E","D","X" pressed     | | |  |   10
        rra             ;00bb  1f      ;  rotate right all, CY=or(col 3)     ^ ^ ^  ^    4
        jp nc,006dfh    ;00bc  d2 df 06;  jump if "N,"R","F","C" pressed     | | |  |   10
        rra             ;00bf  1f      ;  rotate right all, CY=or(col 4)     ^ ^ ^  ^    4
        jp nc,00719h    ;00c0  d2 19 07;  jump if "B,"T","G","V" pressed     | | |  |   10
        rra             ;00c3  1f      ;  rotate right all, CY=or(col 5)     ^ ^ ^  ^    4
        pop bc          ;00c4  c1      ;  restore BC                         | | |  |   10
        djnz l00ach     ;00c5  10 e5   ;  close inner loop --> --> --> --> --' ^ ^  ^   13,8: 148*256-5=37883
        dec c           ;00c7  0d      ;                                       | |  |    4
        jr nz,l00ach    ;00c8  20 e2   ;  close middle loop -> --> --> --> --> ' ^  ^   12,7: 37899*256-5=9702139
        dec d           ;00ca  15      ;                                         |  |    4
        jr nz,l00ach    ;00cb  20 df   ;  close outer loop --> --> --> --> --> --'  ^   12,7: 9702155*7-5=67915080
; rotate left CoBra logo, full screen  ;                                           ^
        ld b,020h       ;00cd  06 20   ;  32 character columns                     ^
l00cfh:                                ;                                         |
        push bc         ;00cf  c5      ;  save BC <-- <-- <-- <-- <-- <-- <-- <,    ^
        call Rotate     ;00d0  cd f1 00;  rotate left CoBra logo 4 pixels      |    |
; rotate left CoBra logo attributes    ;                                     ^    ^
        ld h,0d9h       ;00d3  26 d9   ;| 0d901h is start Address + 1        |    |
        ld l,001h       ;00d5  2e 01   ;| of middle 3rd of screen attributes  ^    ^
        ld b,008h       ;00d7  06 08   ;  rotate left 8 attribute rows        |    |
l00d9h:                                ;                                     ^    ^
        push bc         ;00d9  c5      ;  save BC <-- <-- <-- <-- <-- <,      |    |
        ld bc,0001fh    ;00da  01 1f 00;  32 bytes per attribute row   ^      ^    ^
        ld d,h          ;00dd  54      ;| load DE with HL Address - 1  |      |    |
        ld e,l          ;00de  5d      ;|                              ^      ^    ^
        dec e           ;00df  1d      ;|                              |      |    |
        ld a,(de)       ;00e0  1a      ;  save first byte to A         ^      ^    ^
        ldir            ;00e1  ed b0   ;  rotate left attribute row    |      |    |
        ld (de),a       ;00e3  12      ;  write last byte with A       ^      ^    ^
        inc l           ;00e4  2c      ;  next L                       |      |    |
        inc e           ;00e5  1c      ;  next E                       ^      ^    ^
        pop bc          ;00e6  c1      ;  restore BC                   |      |    |
        djnz l00d9h     ;00e7  10 f0   ;  close attribute row loop -> -'      ^    ^
        call Rotate     ;00e9  cd f1 00;  rotate left CoBra logo 4 pixels      |    |
        pop bc          ;00ec  c1      ;  restore BC                         ^    ^
        djnz l00cfh     ;00ed  10 e0   ;  close character rotate left loop -> -'    |
        jr l00a7h       ;00ef  18 b6   ;  close screen cycle --> --> --> --> --> -->'

Rotate:
        ld b,004h       ;00f1  06 04   ;  rotate left CoBra logo 4 pixels
l00f3h:                                ;
        push bc         ;00f3  c5      ;  save BC <-- <-- <-- <-- <-- <-- <-- <,
        ld h,0cfh       ;00f4  26 cf   ;| 0cfffh is end Address               ^
        ld l,0ffh       ;00f6  2e ff   ;| of middle 3rd of screen             |
        ld b,008h       ;00f8  06 08   ;  rotate left 8 character rows        ^
```

```
l00fah:                                 ;                                       |
        push bc         ;00fa  c5       ;  save BC <-- <-- <-- <-- <-- <-- <-,  ^
        ld b,008h       ;00fb  06 08    ;  8 screen lines per character row  |  |
l00fdh:                                 ;                                    ^  ^
        push bc         ;00fd  c5       ;  save BC <-- <-- <-- <-- <--,      |  |
        ld d,h          ;00fe  54       ;  save Address to DE         ^      ^  ^
        ld e,l          ;00ff  5d       ;                             |      |  |
        ld b,020h       ;0100  06 20    ;  32 bytes per screen line   ^      ^  ^
        scf             ;0102  37       ;  CY=1                       |      |  |
l0103h:                                 ;                             ^      ^  ^
        rl (hl)         ;0103  cb 16    ;| rotate left                 |      |  |
        dec hl          ;0105  2b       ;|                             ^      ^  ^
        djnz l0103h     ;0106  10 fb    ;|                             |      |  |
        jr c,l010dh     ;0108  38 03    ;  jump if 1 was rotated out  ^      ^  ^
        ld a,(de)       ;010a  1a       ;| otherwise                   |      |  |
        dec a           ;010b  3d       ;| clear bit on the right      ^      ^  ^
        ld (de),a       ;010c  12       ;|                             |      |  |
l010dh:                                 ;                             ^      ^  ^
        pop bc          ;010d  c1       ;  restore BC                 |      |  |
        scf             ;010e  37       ;| CY=0                       ^      ^  ^
        ccf             ;010f  3f       ;|                             |      |  |
        ld de,000e0h    ;0110  11 e0 00;| load HL with end Address    ^      ^  ^
        sbc hl,de       ;0113  ed 52    ;| of previous screen line     |      |  |
        djnz l00fdh     ;0115  10 e6    ;  close screen line loop  -->'      ^  ^
        ld bc,007e0h    ;0117  01 e0 07;| load HL with end Address           |  |
        add hl,bc       ;011a  09       ;| of previous character row          ^  ^
        pop bc          ;011b  c1       ;  restore BC                         |  |
        djnz l00fah     ;011c  10 dc    ;  close character row loop -> --> --'  ^
        pop bc          ;011e  c1       ;  restore BC                           |
        djnz l00f3h     ;011f  10 d2    ;  close rotate left loop  --> --> --> -'
        ret             ;0121  c9       ;

CoBra_tbl:
; outer loop 0 writes Words with 8 consecutive 1's (8 middle loops)
Word_01fe:
;       defb    004h   (initial value in A)             ; Number of Words
;       defw    0cbb9h (initial value in HL)            ; Base Address
        defb    0ffh,0e0h,0ffh
Word_03fc:
        defb    01fh                                    ; Number of Words
        defw    0c8b1h                                  ; Base Address
        defb    045h,0b9h,022h,002h,01fh,0a2h,016h,022h; Increments
        defb    002h,01fh,0a2h,016h,025h,015h,009h,075h
        defb    043h,025h,0bch,002h,019h,025h,0bch,002h
        defb    019h,007h,01dh,0d9h,007h,01dh
Word_07f8:
        defb    00ch                                    ; Number of Words
        defw    0c89ah                                  ; Base Address
        defb    007h,0a5h,090h,0ffh,088h,0ffh,0e5h,0ffh; Increments
        defb    0ffh,031h,007h
Word_0ff0:
        defb    00ch                                    ; Number of Words
        defw    0c882h                                  ; Base Address
        defb    0ffh,0ffh,0ffh,06dh,07fh,009h,007h,0f7h; Increments
        defb    007h,0dfh,0ffh
Word_1fe0:
        defb    009h                                    ; Number of Words
        defw    0c8cch                                  ; Base Address
        defb    07bh,092h,0ffh,0ffh,0ebh,0ffh,0ffh,0c4h; Increments
Word_3fc0:
        defb    010h                                    ; Number of Words
        defw    0c8afh                                  ; Base Address
        defb    0fch,002h,0fch,002h,0fch,002h,0fch,002h; Increments
        defb    0fch,002h,0fch,002h,00bh,0f3h,00bh
Word_7f80:
        defb    006h                                    ; Number of Words
        defw    0c887h                                  ; Base Address
        defb    0c0h,0ffh,0ffh,072h,0ffh                ; Increments
Word_ff00:
        defb    00bh                                    ; Number of Words
        defw    0c846h                                  ; Base Address
        defb    074h,06eh,03dh,052h,08eh,09fh,0feh,0ddh; Increments
        defb    0ffh,082h

; outer loop 1 writes Words with 9 consecutive 1's (8 middle loops)
Word_01ff:
        defb    011h                                    ; Number of Words
        defw    0c88ch                                  ; Base Address
        defb    062h,055h,020h,0e1h,070h,08eh,0dah,0a3h; Increments
        defb    07ah,05fh,09fh,05fh,0cch,02ah,028h,008h
Word_03fe:
        defb    00bh                                    ; Number of Words
        defw    0c862h                                  ; Base Address
        defb    0e5h,0ffh,0ffh,059h,0f7h,007h,0a5h,070h; Increments
        defb    0b6h,0d1h
```

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Word_07fc:
        defb    013h                                    ; Number of Words
        defw    0c8d1h                                  ; Base Address
        defb    002h,002h,001h,088h,037h,007h,02eh,090h; Increments
        defb    03fh,0a5h,08bh,0bfh,002h,0f9h,0ffh,0ffh; Increments
        defb    002h,002h
Word_0ff8:
        defb    008h                                    ; Number of Words
        defw    0cad1h                                  ; Base Address
        defb    076h,090h,0edh,09ah,02eh,0e8h,007h      ; Increments
Word_1ff0:
        defb    009h                                    ; Number of Words
        defw    0c9cch                                  ; Base Address
        defb    07bh,086h,078h,092h,0ebh,09fh,027h,0fch; Increments
Word_3fe0:
        defb    00bh                                    ; Number of Words
        defw    0c8ach                                  ; Base Address
        defb    022h,00bh,0f3h,0fch,0ffh,0fah,0ffh,0c7h; Increments
        defb    0ffh,01ch
Word_7fc0:
        defb    003h                                    ; Number of Words
        defw    0cdbbh                                  ; Base Address
        defb    0b4h,0f6h                               ; Increments
Word_ff80:
        defb    010h                                    ; Number of Words
        defw    0c82ah                                  ; Base Address
        defb    0c6h,054h,003h,0fdh,072h,038h,0f5h,0dch; Increments
        defb    07bh,030h,002h,0f8h,032h,0a6h,026h

; outer loop 2 writes Words with 10 consecutive 1's (7 middle loops)
Word_03ff:
        defb    00ch                                    ; Number of Words
        defw    0c866h                                  ; Base Address
        defb    028h,0fch,0bbh,0ffh,0deh,0fbh,0e4h,0bah; Increments
        defb    008h,0fah,007h
Word_07fe:
        defb    001h                                    ; Number of Words
        defw    0cf46h                                  ; Base Address
Word_0ffc:
        defb    001h                                    ; Number of Words
        defw    0ce46h                                  ; Base Address
Word_1ff8:
        defb    005h                                    ; Number of Words
        defw    0cacch                                  ; Base Address
        defb    0ffh,0fah,07eh,045h
Word_3ff0:
        defb    001h                                    ; Number of Words
        defw    0cc46h                                  ; Base Address
Word_7fe0:
        defb    001h                                    ; Number of Words
        defw    0cb46h                                  ; Base Address
Word_ffc0:
        defb    00ch                                    ; Number of Words
        defw    0c891h                                  ; Base Address
        defb    0b6h,0fdh,0e1h,09bh,07eh,085h,0fdh,0ffh; Increments
        defb    061h,079h,02dh

; outer loop 3 writes Words with 11 consecutive 1's (6 middle loops)
Word_07ff:
        defb    00ah                                    ; Number of Words
        defw    0c924h                                  ; Base Address
        defb    06ah,0fch,057h,0ech,0f5h,0fbh,008h,056h; Increments
        defb    0abh
Word_0ffe:
        defb    002h                                    ; Number of Words
        defw    0cbcch                                  ; Base Address
        defb    0bfh                                    ; Increments
Word_1ffc:
        defb    004h                                    ; Number of Words
        defw    0caedh                                  ; Base Address
        defb    0e1h,0bfh,0ddh                          ; Increments
Word_3ff8:
        defb    003h                                    ; Number of Words
        defw    0cbd1h                                  ; Base Address
        defb    095h,029h                               ; Increments
Word_7ff0:
        defb    001h                                    ; Number of Words
        defw    0c8e6h                                  ; Base Address
Word_ffe0:
        defb    00dh                                    ; Number of Words
        defw    0c8c3h                                  ; Base Address
        defb    0cdh,062h,0f3h,05ah,0e3h,0ffh,0a1h,040h; Increments
        defb    01eh,04ah,002h,059h

; outer loop 4 writes Words with 12 consecutive 1's (5 middle loops)
```

```
Word_0fff:
        defb    00dh                                    ; Number of Words
        defw    0ca8fh                                  ; Base Address
        defb    0fch,002h,097h,09bh,008h,002h,009h,0f5h; Increments
        defb    009h,02ah,062h,0bbh
Word_1ffe:
        defb    001h                                    ; Number of Words
        defw    0c8e6h                                  ; Base Address
Word_3ffc:
        defb    004h                                    ; Number of Words
        defw    0caf0h                                  ; Base Address
        defb    0f8h,0ddh,0a8h                          ; Increments
Word_7ff8:
        defb    002h                                    ; Number of Words
        defw    0c9c3h                                  ; Base Address
        defb    07fh                                    ; Increments
Word_fff0:
        defb    014h                                    ; Number of Words
        defw    0c894h                                  ; Base Address
        defb    002h,05ch,0fbh,0a0h,0ffh,022h,002h,052h; Increments
        defb    0a9h,002h,019h,009h,04dh,0a7h,009h,0edh
        defb    0aah,002h,05ch

; outer loop 5 writes Words with 13 consecutive 1's (4 middle loops)
Word_1fff:
        defb    00ch                                    ; Number of Words
        defw    0c843h                                  ; Base Address
        defb    0b5h,0eah,03fh,0e3h,0bah,040h,0c2h,0ffh; Increments
        defb    0ffh,011h,0ffh
Word_3ffe:
        defb    007h                                    ; Number of Words
        defw    0c943h                                  ; Base Address
        defb    0e4h,09ah,084h,0aah,07fh,002h           ; Increments
Word_7ffc:
        defb    007h                                    ; Number of Words
        defw    0c9edh                                  ; Base Address
        defb    0fah,07eh,0bfh,0ffh,0ffh,045h           ; Increments
Word_fff8:
        defb    017h                                    ; Number of Words
        defw    0c826h                                  ; Base Address
        defb    0d4h,098h,002h,01ch,002h,0dch,002h,01ch; Increments
        defb    002h,052h,0fdh,020h,0fdh,08ch,002h,01ch
        defb    0dfh,002h,01ch,006h,0edh,011h

; outer loop 6 writes Words with 14 consecutive 1's (3 middle loops)
Word_3fff:
        defb    02ah                                    ; Number of Words
        defw    0c828h                                  ; Base Address
        defb    06ah,002h,01ch,002h,03ch,09fh,002h,01ch; Increments
        defb    002h,03ch,005h,099h,002h,01ch,002h,03ch
        defb    005h,00ah,08eh,002h,01ch,002h,042h,0b9h
        defb    002h,01ch,09fh,002h,03ch,002h,01ch,09fh
        defb    002h,03ch,002h,01ch,09fh,002h,03ch,002h
        defb    01ch
Word_7ffe:
        defb    001h                                    ; Number of Words
        defw    0c8e6h                                  ; Base Address
Word_fffc:
        defb    017h                                    ; Number of Words
        defw    0c88dh                                  ; Base Address
        defb    026h,002h,031h,003h,038h,066h,06dh,091h; Increments
        defb    06dh,02dh,063h,006h,002h,063h,0d1h,006h
        defb    0f8h,0ffh,09fh,046h,002h,015h

; outer loop 7 writes Words with 15 consecutive 1's (2 middle loops)
Word_7fff:
        defb    006h                                    ; Number of Words
        defw    0c825h                                  ; Base Address
        defb    0beh,043h,0ffh,0fbh,0c0h                ; Increments
Word_fffe:
        defb    008h                                    ; Number of Words
        defw    0c9e9h                                  ; Base Address
        defb    03ch,002h,03dh,0e1h,0ddh,09dh,083h      ; Increments

; outer loop 8 writes Words with 16 consecutive 1's (1 middle loops)
Word_ffff:
        defb    029h                                    ; Number of Words
        defw    0c8e6h                                  ; Base Address
        defb    001h,05fh,09dh,001h,007h,037h,01fh,09dh; Increments
        defb    001h,01dh,001h,01dh,001h,01fh,09eh,001h
        defb    01ch,001h,01ch,001h,09dh,001h,05dh,001h
        defb    020h,07ch,001h,03eh,001h,01ch,001h,020h
        defb    027h,054h,001h,05dh,001h,020h,07ch,001h

Bytes:
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```
        defb    001h  ; Number of Bytes
        defw    0c845h; Address
        defb    0c3h  ; Byte
        defb    001h  ; Number of Bytes
        defw    0c887h; Address
        defb    003h  ; Byte
        defb    001h  ; Number of Bytes
        defw    0c987h; Address
        defb    001h  ; Byte
        defb    001h  ; Number of Bytes
        defw    0ce68h; Address
        defb    0e0h  ; Byte
        defb    001h  ; Number of Bytes
        defw    0cf67h; Address
        defb    007h  ; Byte
        defb    00ah  ; Number of Bytes
        defw    0d903h; Address
        defb    001h  ; Byte: blue    INK on black   PAPER
        defb    002h  ; Number of Bytes
        defw    0d927h; Address
        defb    035h  ; Byte: cyan    INK on yellow  PAPER
        defb    001h  ; Number of Bytes
        defw    0d947h; Address
        defb    015h  ; Byte: cyan    INK on red     PAPER
        defb    004h  ; Number of Bytes
        defw    0d948h; Address
        defb    002h  ; Byte: red     INK on black   PAPER
        defb    001h  ; Number of Bytes
        defw    0d98bh; Address
        defb    003h  ; Byte: magenta INK on black   PAPER
        defb    001h  ; Number of Bytes
        defw    0d9abh; Address
        defb    003h  ; Byte: magenta INK on black   PAPER
; missing loop
;       defb    001h  ; Number of Bytes
;       defw    0d9cbh; Address
;       defb    003h  ; Byte: magenta INK on black   PAPER

Colors:
        defb    007h  ;       white   INK on black   PAPER
        defb    00eh  ;       yellow  INK on blue    PAPER
        defb    015h  ;       cyan    INK on red     PAPER
        defb    01ch  ;       green   INK on magenta PAPER
        defb    023h  ;       magenta INK on green   PAPER
        defb    02ah  ;       red     INK on cyan    PAPER
        defb    031h  ;       blue    INK on yellow  PAPER
        defb    038h  ;       black   INK on white   PAPER

l0382h:
        defb              000h,000h,000h,000h,001h,000h,01ah,006h,080h,001h,007h,00dh,013h,019h;
        defb    005h,00bh,011h,017h,003h,009h,00fh,015h,002h,008h,00eh,014h,01ah,006h,00ch,012h;
        defb    018h,004h,00ah,010h,016h,001h,004h,007h,00ah,003h,006h,009h,002h,005h,008h;
l03afh:
        defb                                                                                  000h;
        defb    053h,059h,053h,020h,020h,020h,020h,020h,043h,04fh,04dh,0ddh,021h,000h,000h,011h;
        defb    000h,020h,03eh,0feh,037h,0cdh,0c2h,004h,0ddh,021h,000h,020h,011h,000h,020h,03eh;
        defb    0feh,037h,0cdh,0c2h,004h,0c3h,000h,000h;

l03d8h:
        call sub_05feh  ;03d8  cd fe 05   ;
        ld bc,00009h    ;03db  01 09 00   ;
        ld hl,l0382h    ;03de  21 82 03   ;
        ld de,0f00ch    ;03e1  11 0c f0   ;
        ldir            ;03e4  ed b0      ;
l03e6h:
        call sub_05e9h  ;03e6  cd e9 05   ;
        and 020h        ;03e9  e6 20      ;
        jr nz,l03fbh    ;03eb  20 0e      ;
        ld a,(0f00dh)   ;03ed  3a 0d f0   ;
        inc a           ;03f0  3c         ;
        ld (0f00dh),a   ;03f1  32 0d f0   ;
        cp 004h         ;03f4  fe 04      ;
        jr nz,l03e6h    ;03f6  20 ee      ;
        jp 00000h       ;03f8  c3 00 00   ;
l03fbh:
        call sub_05a8h  ;03fb  cd a8 05   ;
        ld hl,08000h    ;03fe  21 00 80   ;
        ld (0f008h),hl  ;0401  22 08 f0   ;
        ld bc,00a01h    ;0404  01 01 0a   ;
        call sub_04b3h  ;0407  cd b3 04   ;
        ld (0f020h),a   ;040a  32 20 f0   ;
        xor a           ;040d  af         ;
        ld (0f00eh),a   ;040e  32 0e f0   ;
        inc a           ;0411  3c         ;
        ld (0f010h),a   ;0412  32 10 f0   ;
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l0415h:
        call sub_0563h  ;0415  cd 63 05   ;
        ld c,008h       ;0418  0e 08      ;
        call sub_04eeh  ;041a  cd ee 04   ;
        jp nz,00000h    ;041d  c2 00 00   ;
        ld a,(08000h)   ;0420  3a 00 80   ;
        or a            ;0423  b7         ;
        jr z,l042eh     ;0424  28 08      ;
        cp 0e5h         ;0426  fe e5      ;
        jr z,l042eh     ;0428  28 04      ;
        ld a,002h       ;042a  3e 02      ;
        jr l042fh       ;042c  18 01      ;
l042eh:
        xor a           ;042e  af         ;
l042fh:
        ld (0f021h),a   ;042f  32 21 f0   ;
        call sub_0563h  ;0432  cd 63 05   ;
        call sub_04beh  ;0435  cd be 04   ;
        call sub_0509h  ;0438  cd 09 05   ;
        jp c,00000h     ;043b  da 00 00   ;
        ld de,08800h    ;043e  11 00 88   ;
        ld (0f008h),de  ;0441  ed 53 08 f0;
        ld de,00010h    ;0445  11 10 00   ;
        add hl,de       ;0448  19         ;
l0449h:
        ld a,(hl)       ;0449  7e         ;
        or a            ;044a  b7         ;
        jr z,l0490h     ;044b  28 43      ;
        push hl         ;044d  e5         ;
        ld h,000h       ;044e  26 00      ;
        ld l,a          ;0450  6f         ;
        call sub_0530h  ;0451  cd 30 05   ;
        ld a,(0f011h)   ;0454  3a 11 f0   ;
        or a            ;0457  b7         ;
        jr z,l045fh     ;0458  28 05      ;
        ld bc,00208h    ;045a  01 08 02   ;
        jr l0462h       ;045d  18 03      ;
l045fh:
        ld bc,00808h    ;045f  01 08 08   ;
l0462h:
        ld a,d          ;0462  7a         ;
        ld (0f00eh),a   ;0463  32 0e f0   ;
        push de         ;0466  d5         ;
        ld d,000h       ;0467  16 00      ;
        call sub_0559h  ;0469  cd 59 05   ;
        ld (0f010h),a   ;046c  32 10 f0   ;
        call sub_04eeh  ;046f  cd ee 04   ;
        jp nz,00000h    ;0472  c2 00 00   ;
        ld hl,(0f01eh)  ;0475  2a 1e f0   ;
        ld (0f008h),hl  ;0478  22 08 f0   ;
        ld hl,(0f01ch)  ;047b  2a 1c f0   ;
        inc hl          ;047e  23         ;
        pop de          ;047f  d1         ;
        inc e           ;0480  1c         ;
        ld a,e          ;0481  7b         ;
        cp l            ;0482  bd         ;
        jp c,l0489h     ;0483  da 89 04   ;
        inc d           ;0486  14         ;
        ld a,001h       ;0487  3e 01      ;
l0489h:
        ld e,a          ;0489  5f         ;
        djnz l0462h     ;048a  10 d6      ;
        pop hl          ;048c  e1         ;
        inc hl          ;048d  23         ;
        jr l0449h       ;048e  18 b9      ;
l0490h:
        ld hl,l04a5h    ;0490  21 a5 04   ;
        ld de,0e000h    ;0493  11 00 e0   ;
        ld bc,0000eh    ;0496  01 0e 00   ;
        ldir            ;0499  ed b0      ;
        ld hl,0e000h    ;049b  21 00 e0   ;
        ld a,040h       ;049e  3e 40      ;
        out (0feh),a    ;04a0  d3 fe      ;
        ld r,a          ;04a2  ed 4f      ;
        jp (hl)         ;04a4  e9         ;
l04a5h:
        ld hl,08800h    ;04a5  21 00 88   ;
        ld de,00100h    ;04a8  11 00 01   ;
        ld bc,05000h    ;04ab  01 00 50   ;
        ldir            ;04ae  ed b0      ;
        jp 00100h       ;04b0  c3 00 01   ;
sub_04b3h:
        ld a,b          ;04b3  78         ;
        ld (0f00ch),a   ;04b4  32 0c f0   ;
        ld hl,l06bah    ;04b7  21 ba 06   ;
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        call sub_0647h  ;04ba  cd 47 06   ;
        ret             ;04bd  c9         ;
sub_04beh:
        or a            ;04be  b7         ;
        jr z,l04c6h     ;04bf  28 05      ;
        ld bc,00408h    ;04c1  01 08 04   ;
        jr l04c9h       ;04c4  18 03      ;
l04c6h:
        ld bc,01008h    ;04c6  01 08 10   ;
l04c9h:
        ld a,(0f021h)   ;04c9  3a 21 f0   ;
        ld (0f00eh),a   ;04cc  32 0e f0   ;
        ld a,001h       ;04cf  3e 01      ;
l04d1h:
        push af         ;04d1  f5         ;
        ld e,a          ;04d2  5f         ;
        ld d,000h       ;04d3  16 00      ;
        call sub_0559h  ;04d5  cd 59 05   ;
        ld (0f010h),a   ;04d8  32 10 f0   ;
        call sub_04eeh  ;04db  cd ee 04   ;
        jr nz,l04ebh    ;04de  20 0b      ;
        ld hl,(0f01eh)  ;04e0  2a 1e f0   ;
        ld (0f008h),hl  ;04e3  22 08 f0   ;
        pop af          ;04e6  f1         ;
        inc a           ;04e7  3c         ;
        djnz l04d1h     ;04e8  10 e7      ;
        ret             ;04ea  c9         ;
l04ebh:
        jp 00000h       ;04eb  c3 00 00   ;
sub_04eeh:
        push bc         ;04ee  c5         ;
        call sub_05b9h  ;04ef  cd b9 05   ;
        pop bc          ;04f2  c1         ;
        ld e,00ah       ;04f3  1e 0a      ;
l04f5h:
        push bc         ;04f5  c5         ;
        push de         ;04f6  d5         ;
        ld hl,l06bah    ;04f7  21 ba 06   ;
        call sub_0647h  ;04fa  cd 47 06   ;
        pop de          ;04fd  d1         ;
        pop bc          ;04fe  c1         ;
        ret z           ;04ff  c8         ;
        dec e           ;0500  1d         ;
        jr nz,l04f5h    ;0501  20 f2      ;
        ld a,(0f015h)   ;0503  3a 15 f0   ;
        and 0c0h        ;0506  e6 c0      ;
        ret             ;0508  c9         ;
sub_0509h:
        ld hl,08000h    ;0509  21 00 80   ;
        ld de,l03afh    ;050c  11 af 03   ;
        ld b,040h       ;050f  06 40      ;
l0511h:
        call sub_051fh  ;0511  cd 1f 05   ;
        ret z           ;0514  c8         ;
        push de         ;0515  d5         ;
        ld de,0001fh+1  ;0516  11 20 00   ;
        add hl,de       ;0519  19         ;
        pop de          ;051a  d1         ;
        djnz l0511h     ;051b  10 f4      ;
        scf             ;051d  37         ;
        ret             ;051e  c9         ;
sub_051fh:
        push hl         ;051f  e5         ;
        push de         ;0520  d5         ;
        push bc         ;0521  c5         ;
        ld b,00ch       ;0522  06 0c      ;
l0524h:
        ld a,(de)       ;0524  1a         ;
        cp (hl)         ;0525  be         ;
        jr nz,l052ch    ;0526  20 04      ;
        inc hl          ;0528  23         ;
        inc de          ;0529  13         ;
        djnz l0524h     ;052a  10 f8      ;
l052ch:
        pop bc          ;052c  c1         ;
        pop de          ;052d  d1         ;
        pop hl          ;052e  e1         ;
        ret             ;052f  c9         ;
sub_0530h:
        ld bc,003ffh    ;0530  01 ff 03   ;
l0533h:
        sla l           ;0533  cb 25      ;
        rl h            ;0535  cb 14      ;
        djnz l0533h     ;0537  10 fa      ;
        inc hl          ;0539  23         ;
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        ld de,(0f01ch)  ;053a  ed 5b 1c f0;
l053eh:
        inc c           ;053e  0c         ;
        sbc hl,de       ;053f  ed 52      ;
        jr nc,l053eh    ;0541  30 fb      ;
        add hl,de       ;0543  19         ;
        ld a,(0f011h)   ;0544  3a 11 f0   ;
        or a            ;0547  b7         ;
        jr z,l0552h     ;0548  28 08      ;
        ld a,l          ;054a  7d         ;
        dec a           ;054b  3d         ;
        srl a           ;054c  cb 3f      ;
        srl a           ;054e  cb 3f      ;
        inc a           ;0550  3c         ;
        ld l,a          ;0551  6f         ;
l0552h:
        ld a,(0f021h)   ;0552  3a 21 f0   ;
        add a,c         ;0555  81         ;
        ld d,a          ;0556  57         ;
        ld e,l          ;0557  5d         ;
        ret             ;0558  c9         ;
sub_0559h:
        push ix         ;0559  dd e5      ;
        add ix,de       ;055b  dd 19      ;
        ld a,(ix+000h)  ;055d  dd 7e 00   ;
        pop ix          ;0560  dd e1      ;
        ret             ;0562  c9         ;
sub_0563h:
        xor a           ;0563  af         ;
        ld (0f01dh),a   ;0564  32 1d f0   ;
        ld a,(0f020h)   ;0567  3a 20 f0   ;
        or a            ;056a  b7         ;
        jr z,l058ch     ;056b  28 1f      ;
        ld a,046h       ;056d  3e 46      ;
        ld (0f00ch),a   ;056f  32 0c f0   ;
        ld a,00ah       ;0572  3e 0a      ;
        ld (0f012h),a   ;0574  32 12 f0   ;
        ld a,028h       ;0577  3e 28      ;
        ld (0f01ch),a   ;0579  32 1c f0   ;
        ld ix,003a4h    ;057c  dd 21 a4 03;
        ld hl,00002h    ;0580  21 02 00   ;
        ld (0f00ah),hl  ;0583  22 0a f0   ;
        ld a,002h       ;0586  3e 02      ;
        ld (0f011h),a   ;0588  32 11 f0   ;
        ret             ;058b  c9         ;
l058ch:
        ld a,006h       ;058c  3e 06      ;
        ld (0f00ch),a   ;058e  32 0c f0   ;
        ld a,01ah       ;0591  3e 1a      ;
        ld (0f012h),a   ;0593  32 12 f0   ;
        ld (0f01ch),a   ;0596  32 1c f0   ;
        ld ix,0038ah    ;0599  dd 21 8a 03;
        ld hl,08001h    ;059d  21 01 80   ;
        ld (0f00ah),hl  ;05a0  22 0a f0   ;
        xor a           ;05a3  af         ;
        ld (0f011h),a   ;05a4  32 11 f0   ;
        ret             ;05a7  c9         ;
sub_05a8h:
        call sub_05d2h  ;05a8  cd d2 05   ;
        ld a,007h       ;05ab  3e 07      ;
        out (0fdh),a    ;05ad  d3 fd      ;
        call sub_061ah  ;05af  cd 1a 06   ;
        ld a,(0f00dh)   ;05b2  3a 0d f0   ;
        out (0fdh),a    ;05b5  d3 fd      ;
        halt            ;05b7  76         ;
        ret             ;05b8  c9         ;
sub_05b9h:
        call sub_05d2h  ;05b9  cd d2 05   ;
        ld a,00fh       ;05bc  3e 0f      ;
        out (0fdh),a    ;05be  d3 fd      ;
        call sub_061ah  ;05c0  cd 1a 06   ;
        ld a,(0f00dh)   ;05c3  3a 0d f0   ;
        out (0fdh),a    ;05c6  d3 fd      ;
        call sub_061ah  ;05c8  cd 1a 06   ;
        ld a,(0f00eh)   ;05cb  3a 0e f0   ;
        out (0fdh),a    ;05ce  d3 fd      ;
        halt            ;05d0  76         ;
        ret             ;05d1  c9         ;
sub_05d2h:
        call sub_05e9h  ;05d2  cd e9 05   ;
        and 020h        ;05d5  e6 20      ;
        jr nz,l05dch    ;05d7  20 03      ;
        jp 00000h       ;05d9  c3 00 00   ;
l05dch:
        call sub_0620h  ;05dc  cd 20 06   ;
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        ld hl,l069ch    ;05df  21 9c 06   ;
        ld (0f000h),hl  ;05e2  22 00 f0   ;
        call sub_0614h  ;05e5  cd 14 06   ;
        ret             ;05e8  c9         ;
sub_05e9h:
        call sub_0614h  ;05e9  cd 14 06   ;
        ld a,004h       ;05ec  3e 04      ;
        out (0fdh),a    ;05ee  d3 fd      ;
        call sub_061ah  ;05f0  cd 1a 06   ;
        ld a,(0f00dh)   ;05f3  3a 0d f0   ;
        out (0fdh),a    ;05f6  d3 fd      ;
        call sub_061ah  ;05f8  cd 1a 06   ;
        in a,(0fdh)     ;05fb  db fd      ;
        ret             ;05fd  c9         ;
sub_05feh:
        call sub_0614h  ;05fe  cd 14 06   ;
        ld a,003h       ;0601  3e 03      ;
        out (0fdh),a    ;0603  d3 fd      ;
        call sub_061ah  ;0605  cd 1a 06   ;
        ld a,03fh       ;0608  3e 3f      ;
        out (0fdh),a    ;060a  d3 fd      ;
        call sub_061ah  ;060c  cd 1a 06   ;
        ld a,017h       ;060f  3e 17      ;
        out (0fdh),a    ;0611  d3 fd      ;
        ret             ;0613  c9         ;
sub_0614h:
        in a,(0f5h)     ;0614  db f5      ;
        and 01fh        ;0616  e6 1f      ;
        jr nz,sub_0614h ;0618  20 fa      ;
sub_061ah:
        in a,(0f5h)     ;061a  db f5      ;
        rlca            ;061c  07         ;
        jr nc,sub_061ah ;061d  30 fb      ;
        ret             ;061f  c9         ;
sub_0620h:
        di              ;0620  f3         ;
        im 2            ;0621  ed 5e      ;
        ld hl,0f000h    ;0623  21 00 f0   ;
        ld a,h          ;0626  7c         ;
        ld i,a          ;0627  ed 47      ;
        ld a,l          ;0629  7d         ;
        out (0e3h),a    ;062a  d3 e3      ;
        ld a,0dfh       ;062c  3e df      ;
        out (0e3h),a    ;062e  d3 e3      ;
        ld a,001h       ;0630  3e 01      ;
        out (0e3h),a    ;0632  d3 e3      ;
        ld hl,(0f00ah)  ;0634  2a 0a f0   ;
        ld a,05fh       ;0637  3e 5f      ;
        out (0ebh),a    ;0639  d3 eb      ;
        ld a,h          ;063b  7c         ;
        out (0ebh),a    ;063c  d3 eb      ;
        ld a,05fh       ;063e  3e 5f      ;
        out (0f3h),a    ;0640  d3 f3      ;
        ld a,l          ;0642  7d         ;
        out (0f3h),a    ;0643  d3 f3      ;
        ei              ;0645  fb         ;
        ret             ;0646  c9         ;
sub_0647h:
        ld (0f000h),hl  ;0647  22 00 f0   ;
        call sub_0620h  ;064a  cd 20 06   ;
        ld hl,(0f008h)  ;064d  2a 08 f0   ;
        ld de,0f00ch    ;0650  11 0c f0   ;
        call sub_0614h  ;0653  cd 14 06   ;
l0656h:
        call sub_061ah  ;0656  cd 1a 06   ;
        ld a,(de)       ;0659  1a         ;
        out (0fdh),a    ;065a  d3 fd      ;
        inc de          ;065c  13         ;
        dec c           ;065d  0d         ;
        jr nz,l0656h    ;065e  20 f6      ;
        call sub_061ah  ;0660  cd 1a 06   ;
        ld a,(de)       ;0663  1a         ;
        ld b,020h       ;0664  06 20      ;
        ld c,0fdh       ;0666  0e fd      ;
        out (0fdh),a    ;0668  d3 fd      ;
        halt            ;066a  76         ;
l066bh:
        in a,(0f5h)     ;066b  db f5      ;
        and b           ;066d  a0         ;
        jp nz,l066bh    ;066e  c2 6b 06   ;
        ld b,000h       ;0671  06 00      ;
l0673h:
        djnz l0673h     ;0673  10 fe      ;
        dec hl          ;0675  2b         ;
        ld (0f01eh),hl  ;0676  22 1e f0   ;
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        ld a,(hl)       ;0679  7e         ;
        ld hl,0f015h    ;067a  21 15 f0   ;
        ld (hl),a       ;067d  77         ;
        inc hl          ;067e  23         ;
l067fh:
        call sub_061ah  ;067f  cd 1a 06   ;
        in a,(0fdh)     ;0682  db fd      ;
        ld (hl),a       ;0684  77         ;
        inc hl          ;0685  23         ;
        ld b,000h       ;0686  06 00      ;
l0688h:
        djnz l0688h     ;0688  10 fe      ;
        in a,(0f5h)     ;068a  db f5      ;
        and 010h        ;068c  e6 10      ;
        jr nz,l067fh    ;068e  20 ef      ;
        ld hl,l069ch    ;0690  21 9c 06   ;
        ld (0f000h),hl  ;0693  22 00 f0   ;
        ld a,(0f015h)   ;0696  3a 15 f0   ;
        and 0c0h        ;0699  e6 c0      ;
        ret             ;069b  c9         ;
l069ch:
        push af         ;069c  f5         ;
        push hl         ;069d  e5         ;
        ld hl,0f015h    ;069e  21 15 f0   ;
        call sub_061ah  ;06a1  cd 1a 06   ;
        ld a,008h       ;06a4  3e 08      ;
        out (0fdh),a    ;06a6  d3 fd      ;
        call sub_061ah  ;06a8  cd 1a 06   ;
        in a,(0fdh)     ;06ab  db fd      ;
        ld (hl),a       ;06ad  77         ;
        inc hl          ;06ae  23         ;
        call sub_061ah  ;06af  cd 1a 06   ;
        in a,(0fdh)     ;06b2  db fd      ;
        ld (hl),a       ;06b4  77         ;
        pop hl          ;06b5  e1         ;
        pop af          ;06b6  f1         ;
        ei              ;06b7  fb         ;
        reti            ;06b8  ed 4d      ;
l06bah:
        ini             ;06ba  ed a2      ;
        inc b           ;06bc  04         ;
        ei              ;06bd  fb         ;
        reti            ;06be  ed 4d      ;
l06c0h:
        ld bc,02000h    ;06c0  01 00 20   ;
        ld de,08000h    ;06c3  11 00 80   ;
        ld hl,04000h    ;06c6  21 00 40   ;
        ldir            ;06c9  ed b0      ;
        ld bc,02000h    ;06cb  01 00 20   ;
        ld de,0e000h    ;06ce  11 00 e0   ;
        ld hl,06000h    ;06d1  21 00 60   ;
        ldir            ;06d4  ed b0      ;
l06d6h:
        ld hl,00000h    ;06d6  21 00 00   ;
        xor a           ;06d9  af         ;
        out (0feh),a    ;06da  d3 fe      ;
        ld r,a          ;06dc  ed 4f      ;
        jp (hl)         ;06de  e9         ;
l06dfh:
        ld sp,0c000h    ;06df  31 00 c0   ;
        scf             ;06e2  37         ;
        ld ix,08000h    ;06e3  dd 21 00 80;
        ld de,02000h    ;06e7  11 00 20   ;
        ld a,0feh       ;06ea  3e fe      ;
        call sub_073ah  ;06ec  cd 3a 07   ;
        jp nc,00000h    ;06ef  d2 00 00   ;
        ld ix,0e000h    ;06f2  dd 21 00 e0;
        ld de,02000h    ;06f6  11 00 20   ;
        scf             ;06f9  37         ;
        ld a,0feh       ;06fa  3e fe      ;
        call sub_073ah  ;06fc  cd 3a 07   ;
        jp nc,00000h    ;06ff  d2 00 00   ;
        jp l06d6h       ;0702  c3 d6 06   ;

; partial keyboard test                                                      ; cycles
l0705h:
        rra             ;0705  1f         ;  rotate right all, CY=or(col 0)          ;    4
        jp c,l0710h     ;0706  da 10 07   ;  jump if "SPACE,"Q","A","CS" not pressed ;   10
        ld a,002h       ;0709  3e 02      ;
l070bh:
        out (c),a       ;070b  ed 79      ;
        jp 006c0h       ;070d  c3 c0 06   ;
l0710h:
        rra             ;0710  1f         ;  rotate right all, CY=or(col 1)          ;    4
        jp c,KB_return  ;0711  da b7 00   ;  jump if "SS,"W","S","Z" not pressed     ;   10
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        ld a,005h       ;0714  3e 05      ;
        jp l070bh       ;0716  c3 0b 07   ;
l0719h:
        ld a,004h       ;0719  3e 04      ;
        out (c),a       ;071b  ed 79      ;
        jp l06c0h       ;071d  c3 c0 06   ;
        rlca            ;0720  07         ;
        jp c,l06c0h     ;0721  da c0 06   ;
        push ix         ;0724  dd e5      ;
        pop af          ;0726  f1         ;
        and 03fh        ;0727  e6 3f      ;
        srl a           ;0729  cb 3f      ;
        srl a           ;072b  cb 3f      ;
        srl a           ;072d  cb 3f      ;
        ld e,a          ;072f  5f         ;
        ld d,000h       ;0730  16 00      ;
        ld hl,0da8ch    ;0732  21 8c da   ;
        add hl,de       ;0735  19         ;
        set 7,(hl)      ;0736  cb fe      ;
l0738h:
        jr l0738h       ;0738  18 fe      ;
sub_073ah:
        inc d           ;073a  14         ;
        ex af,af'       ;073b  08         ;
        dec d           ;073c  15         ;
        di              ;073d  f3         ;
        ld a,00fh       ;073e  3e 0f      ;
        out (0feh),a    ;0740  d3 fe      ;
        in a,(0feh)     ;0742  db fe      ;
        rra             ;0744  1f         ;
        and 020h        ;0745  e6 20      ;
        or 002h         ;0747  f6 02      ;
        ld c,a          ;0749  4f         ;
        cp a            ;074a  bf         ;
l074bh:
        ret nz          ;074b  c0         ;
l074ch:
        call sub_07c7h  ;074c  cd c7 07   ;
        jr nc,l074bh    ;074f  30 fa      ;
        ld hl,l0415h    ;0751  21 15 04   ;
l0754h:
        djnz l0754h     ;0754  10 fe      ;
        dec hl          ;0756  2b         ;
        ld a,h          ;0757  7c         ;
        or l            ;0758  b5         ;
        jr nz,l0754h    ;0759  20 f9      ;
        call sub_07c3h  ;075b  cd c3 07   ;
        jr nc,l074bh    ;075e  30 eb      ;
l0760h:
        ld b,09ch       ;0760  06 9c      ;
        call sub_07c3h  ;0762  cd c3 07   ;
        jr nc,l074bh    ;0765  30 e4      ;
        ld a,0c6h       ;0767  3e c6      ;
        cp b            ;0769  b8         ;
        jr nc,l074ch    ;076a  30 e0      ;
        inc h           ;076c  24         ;
        jr nz,l0760h    ;076d  20 f1      ;
l076fh:
        ld b,0c9h       ;076f  06 c9      ;
        call sub_07c7h  ;0771  cd c7 07   ;
        jr nc,l074bh    ;0774  30 d5      ;
        ld a,b          ;0776  78         ;
        cp 0d4h         ;0777  fe d4      ;
        jr nc,l076fh    ;0779  30 f4      ;
        call sub_07c7h  ;077b  cd c7 07   ;
        ret nc          ;077e  d0         ;
        ld a,c          ;077f  79         ;
        xor 003h        ;0780  ee 03      ;
        ld c,a          ;0782  4f         ;
        ld h,000h       ;0783  26 00      ;
        ld b,0b0h       ;0785  06 b0      ;
        jr l07a8h       ;0787  18 1f      ;
l0789h:
        ex af,af'       ;0789  08         ;
        jr nz,l0793h    ;078a  20 07      ;
        jr nc,l079dh    ;078c  30 0f      ;
        ld (ix+000h),l  ;078e  dd 75 00   ;
        jr l07a2h       ;0791  18 0f      ;
l0793h:
        rl c            ;0793  cb 11      ;
        xor l           ;0795  ad         ;
        ret nz          ;0796  c0         ;
        ld a,c          ;0797  79         ;
        rra             ;0798  1f         ;
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        ld c,a          ;0799  4f         ;
        inc de          ;079a  13         ;
        jr l07a4h       ;079b  18 07      ;
l079dh:
        ld a,(ix+000h)  ;079d  dd 7e 00   ;
        xor l           ;07a0  ad         ;
        ret nz          ;07a1  c0         ;
l07a2h:
        inc ix          ;07a2  dd 23      ;
l07a4h:
        dec de          ;07a4  1b         ;
        ex af,af'       ;07a5  08         ;
        ld b,0b2h       ;07a6  06 b2      ;
l07a8h:
        ld l,001h       ;07a8  2e 01      ;
l07aah:
        call sub_07c3h  ;07aa  cd c3 07   ;
        ret nc          ;07ad  d0         ;
        ld a,0cbh       ;07ae  3e cb      ;
        cp b            ;07b0  b8         ;
        rl l            ;07b1  cb 15      ;
        ld b,0b0h       ;07b3  06 b0      ;
        jp nc,l07aah    ;07b5  d2 aa 07   ;
        ld a,h          ;07b8  7c         ;
        xor l           ;07b9  ad         ;
        ld h,a          ;07ba  67         ;
        ld a,d          ;07bb  7a         ;
        or e            ;07bc  b3         ;
        jr nz,l0789h    ;07bd  20 ca      ;
        ld a,h          ;07bf  7c         ;
        cp 001h         ;07c0  fe 01      ;
        ret             ;07c2  c9         ;
sub_07c3h:
        call sub_07c7h  ;07c3  cd c7 07   ;
        ret nc          ;07c6  d0         ;
sub_07c7h:
        ld a,016h       ;07c7  3e 16      ;
l07c9h:
        dec a           ;07c9  3d         ;
        jr nz,l07c9h    ;07ca  20 fd      ;
l07cch:
        and a           ;07cc  a7         ;
l07cdh:
        inc b           ;07cd  04         ;
        ret z           ;07ce  c8         ;
        ld a,07fh       ;07cf  3e 7f      ;
        in a,(0feh)     ;07d1  db fe      ;
        rra             ;07d3  1f         ;
        ret nc          ;07d4  d0         ;
        xor c           ;07d5  a9         ;
        and 020h        ;07d6  e6 20      ;
        jr z,l07cdh     ;07d8  28 f3      ;
        ld a,c          ;07da  79         ;
        cpl             ;07db  2f         ;
        ld c,a          ;07dc  4f         ;
        and 007h        ;07dd  e6 07      ;
        or 008h         ;07df  f6 08      ;
        out (0feh),a    ;07e1  d3 fe      ;
        scf             ;07e3  37         ;
        ret             ;07e4  c9         ;
        jr nc,$-26      ;07e5  30 e4      ;
        ld a,0c6h       ;07e7  3e c6      ;
        cp b            ;07e9  b8         ;
        jr nc,l07cch    ;07ea  30 e0      ;
        inc h           ;07ec  24         ;
        jr nz,$-13      ;07ed  20 f1      ;
l07efh:
        ld b,0c9h       ;07ef  06 c9      ;
        call sub_07c7h  ;07f1  cd c7 07   ;
        jr nc,$-41      ;07f4  30 d5      ;
        ld a,b          ;07f6  78
        cp 0d4h         ;07f7  fe d4      ;
        jr nc,l07efh    ;07f9  30 f4      ;
        call sub_07c7h  ;07fb  cd c7 07   ;
        ret nc          ;07fe  d0         ;
        ld a,c          ;07ff  79         ;

        end
```

Code 2. CoBra Logo Code